A Full Complexity Dichotomy for Complex-Valued Boolean Holant Problems

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Abstract

A Boolean Holant problem evaluates a graph partition function specified by local signatures at its vertices. We prove a complexity dichotomy for arbitrary finite sets of algebraic complex-valued signatures on general graphs: an explicit, decidable criterion characterizes the polynomial-time cases, and every remaining case is #P-hard under polynomial-time Turing reductions. The proof resolves the all-even case by constructing signatures with nonzero endpoints and all remaining support on the central layer, then classifying them according to half their arity. Binary transfer groups and matching-product reconstruction provide the main tools in the small- and large-height cases, respectively.

1 Introduction

Counting problems ask how many global configurations satisfy a collection of local constraints. Counting proper graph colourings, for example, sums over vertex colours subject to a constraint on each edge. In a weighted problem, each configuration contributes the product of its local weights, and their sum is a partition function. Although this sum has exponentially many terms, some choices of local rules permit efficient evaluation. A complexity dichotomy identifies all such choices and proves hardness for every remaining choice.

A Boolean Holant problem places local functions, called signatures, at the vertices of a graph. Each edge receives a bit, and the partition function sums the product of the vertex signatures over all edge assignments. We prove a dichotomy for arbitrary finite sets of algebraic complex-valued Boolean signatures on general graphs, with an explicit, decidable tractability criterion. The main case addressed here consists entirely of even-arity signatures, with neither a symmetry restriction nor a prescribed auxiliary binary signature.

1.1 From graph homomorphisms through counting CSP to Holant

Counting graph homomorphisms (#GH) asks how many maps from an input graph to a fixed target graph preserve adjacency. In the weighted version, a fixed matrix specifies the interaction between the labels at the two ends of each edge. Counting constraint satisfaction problems (#CSP) allow constraints of arbitrary arity, so one local function can constrain several variables at once. The weighted problem sums the product of the constraint values over all variable assignments.

Dichotomies for unweighted graph homomorphisms and Boolean counting CSP [DG00, CH96] established early classifications in these frameworks. Subsequent work extended the allowed domains and weights, from nonnegative weights to signed-real and complex weights, where cancellation must also be accounted for. The general-domain complex classification of Cai and Chen covers both weighted graph homomorphisms and counting CSP [CC17]. Table 1 records this development. A dash means that the case is covered by a more general entry, with no separate reference listed.

Table 1: Development of general-graph dichotomies for weighted graph homomorphisms and counting CSP.

Weight range	Symmetric #GH	#GH	Boolean-domain #CSP	General-domain #CSP
$\{0,1\}$	[DG00]	acyclic targets [DGP07]	[CH96]	[Bul13, DR13]
$\mathbb{Q}_{\geq 0}$	–	–	–	[BDG$^+$12]
$\mathbb{R}_{\geq 0}$	[BG05]	[CC19]	[DGJ09]	[CCL16]
$\mathbb{R}$	[GGJT10]	–	[BDG$^+$09]	–
$\mathbb{C}$	[CCL13]	[CC17]	[CLX14]	classification [CC17], decidability [LM26b]

Holant changes which local constraints are available. A counting-CSP instance can be represented by a bipartite graph with variable vertices and constraint vertices. At each variable vertex, an equality signature forces all occurrences of that variable to agree. In a Holant problem, these equality signatures need not be available: every vertex must use a signature from the prescribed set. The

signatures needed in a reduction must be obtained by gadgets or other justified reductions. This distinction brings problems such as counting perfect matchings into the classification and changes the constructions available for proving hardness.

Motivated by Valiant's holographic algorithms [Val08], the Holant framework was introduced in 2009 [CLX09]. A holographic transformation is an invertible simultaneous change of basis on the signatures and edge tensors that preserves their contraction. Such transformations relate apparently different counting problems and expose tractable signature sets.

On the Boolean domain, Holant classifications developed along two complementary directions: relaxing restrictions on symmetry and weights, and reducing the auxiliary signatures assumed available. Unary signatures are useful because they allow an input of a signature to be fixed or summed with a chosen weight. The auxiliary assumptions range from all unary signatures (Holant*), through four distinguished unaries (Holant$^+$), to the two unaries fixing a bit to 0 or 1 (Holant^c). Later work treated any signature set containing a nonzero odd-arity signature (Holant^{odd}). Table 2 separates these assumptions from restrictions on symmetry and weights, using the same dash convention as Table 1.

Table 2: Development of Boolean-domain Holant dichotomies on general graphs.

(a) Symmetric signature sets

Weight range	Holant*	Holant^c	Unrestricted Holant
$\mathbb{R}_{\geq 0}$	–	–	–
$\mathbb{R}$	–	[CLX09]	[HL16]
$\mathbb{C}$	[CLX09]	[CHL12]	[CGW16a]

(b) Arbitrary signature sets

Weight range	Holant*	Holant$^+$	Holant^c	Holant^{odd}	Unrestricted Holant
$\mathbb{R}_{\geq 0}$	–	–	–	–	[LW18]
$\mathbb{R}$	–	–	[CLX18]	[CFS20b]	[SC20]
$\mathbb{C}$	[CLX11]	[Bac17]	[Bac21, Bac25]	[MWXZ25, GSS26]	This paper

Without auxiliary unary signatures, dichotomies were obtained for symmetric complex-valued signatures [CGW16a], arbitrary nonnegative signatures [LW18], and arbitrary real-valued signatures [SC20].

In 2025, Meng, Wang, and Xia proved an $\mathsf{FP}^{\mathsf{NP}}$ versus #P-hard dichotomy for weighted Eulerian orientations [MWX25]. Meng, Wang, Xia, and Zheng combined this result with tensor decomposition to obtain the same complexity bounds for complex Boolean Holant with a nonzero odd-arity signature [MWXZ25]. In 2026, Guan, Shao, and Shi gave a polynomial-time algorithm based on linear programming for the former $\mathsf{FP}^{\mathsf{NP}}$ cases, yielding FP versus #P-hard dichotomies for both problems [GSS26].

These results direct attention to the all-even case. Every gadget built from even-arity signatures has even arity, since each internal edge consumes two inputs. Thus no such gadget supplies a unary signature. Binary signatures become the smallest positive-arity objects available for further constructions. Xia proposed a framework organized by the finite groups associated with

binary signatures, analyzing several branches in which all realizable nonzero quaternary signatures decompose [Xia26]. Liu and Meng then classified complex Boolean Holant with binary disequality available, for arbitrary finite sets of algebraic complex-valued signatures, and gave an explicit decidable tractability criterion [LM26a].

Our proof removes the prescribed-binary assumption in the all-even case. It combines the binary and quaternary dichotomies of Liu and Meng with a decomposition based on Hamming imbalance. This decomposition produces signatures of controlled support whose arity determines the remaining cases, as described in Sections 3 to 5. Together with the odd-arity dichotomy, it completes the unrestricted complex-valued entry in Table 2.

Independent work. Huang and Fu established a dichotomy for Holant problems with a single complex-valued quaternary signature [HF26]. Guan, Shao, and Tang proved a dichotomy for complex-valued Boolean Holant problems whose signature sets are closed under complex conjugation [GST26]. Zhou proved a dichotomy for finite sets of algebraic complex-valued Boolean signatures of even arity whose nonzero binary gadget signatures are nonsingular and whose projective matrix classes form the cyclic group C_2 of order two under series composition [Zho26].

We were also informed of the following independent work before the public release of the present paper. Jiayi Zheng informed us of unpublished work on the complex-valued Holant dichotomy in the case where the projective binary transfer classes form an odd dihedral group. Mingji Xia informed us of work concerning the twisted Klein-group case and the C_1 case, under the condition that all quaternary signatures are reducible. The results and proofs in the present paper were developed independently of these works.

1.2 Our result: the full dichotomy

In their odd-arity dichotomy, Meng, Wang, Xia, and Zheng [MWXZ25, conference version, Theorem 30; full version, Theorem 33] formulate condition (PC) as seven structural alternatives for signature sets. These specify support restrictions, tensor decompositions, or membership in prescribed function classes under a common change of basis. We write $\mathsf{Tract}(\mathcal{F})$ when $\mathcal{F}$ satisfies at least one of these seven alternatives, using the equivalent formulation in Section F.3. Each alternative concerns the whole set: in a transformability case, one basis must work simultaneously for the edge and every member of $\mathcal{F}$.

The algorithms for these seven forms, together with Guan, Shao, and Shi's FP improvement for the cases previously known to lie in $\mathsf{FP}^{\mathsf{NP}}$ [GSS26], give polynomial-time evaluation whenever this structural condition holds. Our contribution is to prove hardness outside these seven forms without the odd-arity assumption.

Theorem 1.1 (Full dichotomy)**.** *Let $\mathcal{F}$ be a finite set of algebraic complex-valued Boolean signatures, given by exact algebraic numbers. The predicate $\mathsf{Tract}(\mathcal{F})$ is decidable. If it holds, then $\mathrm{Holant}(\mathcal{F}) \in \mathsf{FP}$. Otherwise, $\mathrm{Holant}(\mathcal{F})$ is $\#\mathsf{P}$-hard under polynomial-time Turing reductions.*

For evaluation and hardness reductions, $\mathcal{F}$ is fixed and running time is measured in the encoded instance size. For recognition, $\mathcal{F}$ itself is the input. Exact algebraic data may be represented by minimal polynomials and isolating data for the chosen roots. Decidability of the criterion and evaluation in its tractable cases are supplied by [LM26a, Theorem 2.22]. We prove the remaining hardness direction. After the known cases are removed, a construction based on Hamming imbalance produces a signature with two nonzero endpoints and all other support on its central layer. We call this signature a mountain and classify the remaining problem according to its height.

The full paper follows the shared references. References with prefix F point to its detailed statements and proofs.

2 Known classifications and reductions

We first specify when an existing result completes a branch of the proof. A signature of arity r is a function $f : \{0,1\}^r \to \overline{\mathbb{Q}}$, where $\overline{\mathbb{Q}}$ is the field of algebraic complex numbers. Its arguments follow the fixed order of the incident edges. For a graph $G = (V, E)$ labelled by signatures f_v, its Holant is $\sum_{\sigma: E \to \{0,1\}} \prod_{v \in V} f_v(\sigma|_{E(v)})$. We allow parallel edges and self-loops. Zero signatures and nullary constants are handled directly, so subsequent sets contain nonzero positive-arity signatures.

For counting problems A, B, write $A \leq_{\mathrm{T}} B$ if a polynomial-time algorithm with oracle access to B computes A, and write $A \equiv_{\mathrm{T}} B$ when both directions hold. Put $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $K = 2^{-1/2} \left(\begin{smallmatrix} 1 & 1 \\ i & -i \end{smallmatrix}\right)$. For a matrix D, write $Df = D^{\otimes r} f$ and $D\mathcal{F} = \{Df : f \in \mathcal{F}\}$. Since $K^{\mathsf{T}} K = X$, the holographic transformation gives

$$\mathrm{Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \mathrm{Holant}(X \mid K^{-1}\mathcal{F}) = \text{K-Holant}(K^{-1}\mathcal{F}).$$

Here each edge joins opposite bits at its two ends. From now on, $\mathcal{F}$ denotes the current set in these K-coordinates. The structural condition is therefore $\mathsf{Tract}(K\mathcal{F})$. A gadget is a fixed network with ordered dangling edges, whose signature is obtained by summing over the internal bits. A directly realized signature can be substituted at every occurrence. More generally, a finite signature set S can be *simulated* from $\mathcal{F}$ if

$$\text{K-Holant}(\mathcal{F} \cup S) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}).$$

Simulation may use several oracle queries, as in interpolation, without a gadget realizing each signature of S. Known nonzero scalar factors are retained in these reductions.

Odd arity and Hamming layers. If $\mathcal{F}$ contains a nonzero odd-arity signature, the dichotomy of [MWXZ25], together with [GSS26], completes the classification. In particular, this applies whenever a reduction supplies a nonzero unary. We may therefore consider only even arities. For an arity-r input $\mathbf{x}$, define its *half-charge* by $\chi_r(\mathbf{x}) = \mathrm{wt}(\mathbf{x}) - r/2$, where wt is Hamming weight. Every internal X-edge removes one zero and one one, so half-charge is conserved: in a gadget it equals the boundary half-charge, and in a closed instance the total is zero. An *EO signature* is supported on the central layer $\mathrm{wt}(\mathbf{x}) = r/2$.

If all supported half-charges in the entire set are nonnegative, or all are nonpositive, only central entries contribute to a closed instance. The EO dichotomy [MWX25, GSS26] then classifies the problem. The case in which each signature is supported on one Hamming layer is also classified [MWX25, Corollary 68]. Its reduction appends fixed zero or one bits to balance each signature's charge. In each charge-balanced component these bits can be paired by X-edges, with a unique extension of every original assignment. The remaining set is *two-sided*, meaning both signs occur among its supported half-charges, and *multilayer*, meaning at least one member has support on different Hamming layers.

Factors, equality, and four variables. A tensor product uses disjoint variable blocks. A nonzero positive-arity signature is *irreducible* if it cannot be written as a tensor product of two positive-arity signatures, after a permutation of variables. The tensor-factor extraction theorem [MWXZ25, full version, Theorem 32 and Remark 39] allows the factors of a nonzero tensor product to be simulated, outside the common-one-sided case just handled. Initial factor extraction preserves

two-sidedness and the multilayer property. An odd factor invokes the odd-arity result. If all factors have arity at most two, tensor decomposition reduces evaluation to paths and cycles. Otherwise we start with irreducible even-arity signatures, at least one of arity greater than two.

Two further results apply to the whole current signature set. If binary equality I can be simulated, the dichotomy of [LM26a, Theorem 1.2] applies. The same is true for a nonsingular weighted equality $\mathrm{diag}(p, q)$: a common diagonal change of basis makes it I and preserves X up to a nonzero scalar. An *eight-vertex signature* is a quaternary signature supported on even Hamming weights. If an irreducible such signature has both endpoint values nonzero, [LM26a, Theorem 1.3] applies. A reducible one with these endpoint values has a nonsingular weighted-equality factor, so that case is classified as well. These consequences are stated in Theorem F.6.2 and Corollary F.6.29.

In the remaining hardness proof we assume $\neg\mathsf{Tract}(K\mathcal{F})$. This assumption persists when the set is enlarged by simulated signatures, since the seven structural alternatives are inherited by subsets. It also persists under common changes of basis in $GO(X) = \{D \in \mathrm{GL}_2(\overline{\mathbb{Q}}) : D^{\mathsf{T}} X D = \lambda X,\ \lambda \neq 0\}$. Thus each applicable classification above gives hardness, which transfers back along the reductions. Tensor reconstruction transfers any tractability conclusion for the initial factor set back to the original set (Section F.10.2).

3 Obtaining a mountain

The remaining signatures may have arbitrary arity and complex coefficients. We seek a signature with two nonzero endpoint values whose other support lies on its central Hamming layer. Such a signature retains both signs of half-charge, while its contractions have much more restricted support. For $d \geq 1$, a *mountain of height* d is an arity-$2d$ signature M satisfying

$$\mathrm{supp}(M) \subseteq HW^{=}_{2d} \cup \{\mathbf{0}^{2d}, \mathbf{1}^{2d}\}, \qquad M(\mathbf{0}^{2d})M(\mathbf{1}^{2d}) \neq 0,$$

where $HW^{=}_{2d}$ is the set of weight-d strings.

Lemma 3.1 (Mountain construction)**.** *Let $\mathcal{F}$ be a finite two-sided set of nonzero even-arity signatures, and let d be the gcd of all its nonzero supported half-charges. Then either a mountain M of height d can be simulated, or both pinning signatures $\Delta_0 = (1, 0)$ and $\Delta_1 = (0, 1)$ can be simulated. In the mountain case, every supported half-charge in $\mathcal{F} \cup \{M\}$ belongs to $d\mathbb{Z}$.*

Proof outline. Let C be the finite set of nonzero supported half-charges. Tensor products add half-charges, so the first task is to express both d and $-d$ as sums of elements of C with nonnegative multiplicities. The gcd is an integer linear combination $\sum_{c\in C} u_c c = d$, but some coefficients may be negative. For every $a, -b \in C$, with $a, b > 0$, the positive multiplicities $b/\gcd(a, b)$ and $a/\gcd(a, b)$ give total charge zero. Adding these zero-sum combinations over all positive–negative pairs gives $\mathbf{v} \in \mathbb{Z}^C_{>0}$ with $\sum_c v_c c = 0$. Consequently $\mathbf{u} + k\mathbf{v}$ and $-\mathbf{u} + k\mathbf{v}$ are nonnegative for sufficiently large k, while their total charges are d and $-d$.

Choose a supported input witnessing each charge used in these two tensor products. The endpoint-preserving self-loop lemma [MWXZ25, full version, Lemma 40] gives two arity-$2d$ gadget signatures P, N with $P(\mathbf{1}^{2d})N(\mathbf{0}^{2d}) \neq 0$. Charge conservation keeps every supported charge divisible by d. At this arity only $-d, 0, d$ are possible, so both signatures already have the required support. It remains to put the two nonzero endpoints into the same signature.

If either signature already has both endpoints, we are done. Otherwise $P(\mathbf{0}^{2d}) = N(\mathbf{1}^{2d}) = 0$. If both have nonzero central restrictions, permute supported central strings to $(\mathbf{0}^d, \mathbf{1}^d)$ in P and $(\mathbf{1}^d, \mathbf{0}^d)$ in N. Join their last d variables by X-edges and expose the first d variables of each. The

resulting signature Q satisfies

$$Q(\mathbf{0}^{2d}) = P(\mathbf{0}^d, \mathbf{1}^d)N(\mathbf{0}^{2d}),$$
$$Q(\mathbf{1}^{2d}) = P(\mathbf{1}^{2d})N(\mathbf{1}^d, \mathbf{0}^d).$$

Each endpoint has a unique nonzero extension, so cancellation is excluded. If just one central restriction is nonzero, use two copies of that signature and the pure endpoint of the other to force the connected bits. The same unique-extension argument gives both nonzero endpoints. Charge conservation retains the support restriction.

If both central restrictions are zero, P and N are nonzero scalar multiples of tensor powers of Δ_1 and Δ_0. Factor extraction supplies both pinnings. The original two-sided set remains present, so the one-sided exception is excluded. The full construction is Lemma F.5.2. □

The pinning outcome is classified by Section 2. We therefore retain the mountain and the whole-set divisibility condition. At height one the mountain is binary, so the available binary operations control the next steps. At larger heights, contractions below arity $2d$ have half-charge zero and hence are EO. This gives two different ways to constrain the remaining signatures.

4 Height one: binary transfers and contractions

A height-one mountain is a binary $B = \left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right)$ with $ae \neq 0$. If a nonzero singular binary can be realized, its unary factors invoke the odd-arity classification. Binary interpolation gives the same conclusion when a nonsingular binary transfer has infinite projective order (Lemma F.4.6). Here the *transfer* of a binary B is BX: series composition satisfies $(BXC)X = (BX)(CX)$. A projective class $[T]$ identifies matrices differing by a nonzero scalar, and its order is the least $m \geq 1$ with T^m scalar, if such an m exists.

In the remaining case the *complete transfer group* is

$$G(\mathcal{F}) = \{[BX] : B \neq 0 \text{ is an ordered binary directly realizable over } \mathcal{F}\}.$$

All these binaries are nonsingular and their transfers have finite order. The algebraicity of the fixed signature set makes $G(\mathcal{F})$ finite [LM26a, Lemma 5.2]. The group analysis in Theorem F.6.8 gives either a nonsingular weighted equality, which completes the classification, or a common $GO(X)$ change of basis after which the group is odd cyclic or odd dihedral. We now explain how these two remaining group forms constrain signatures of higher arity.

4.1 From restricted contractions to tensor factors

Fix one directly realized binary representative of each transfer class. Contract two variables of any current signature with such a representative, and extract the factors of the nonzero output. A new binary may enlarge the complete transfer group. A new nonbinary irreducible factor may lower the minimum such arity. Both changes supply new information and restart the analysis with an equivalent set. An odd factor or a binary covered by Section 2 completes the classification immediately. If neither occurs, recursive contractions give a finite set $\mathcal{F}^{\mathrm{cl}}$: every indicated contraction has been processed, and all its irreducible factors are retained. This closure is finite because a new nonbinary factor has arity at least two less than its parent (Lemma F.6.12).

Choose a nonbinary irreducible f of minimum arity n in this set. Every nonzero two-variable contraction of f must be a tensor product of binary factors whose transfer classes lie in the current group. We call a tensor product of binaries on the pairs of a perfect matching a *matching product*.

The difficulty is to recover information about f itself: different contractions could initially use different perfect matchings of the remaining variables.

Walsh coordinates make the group representatives diagonal or anti-diagonal. With $P = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, put $\widetilde{f} = (P^{-1})^{\otimes n} f$. Each binary factor of a nonzero contraction is supported on two complementary assignments, with the ratio of its ordered nonzero coefficients in $-\mu_N$. Here $\mu_N = \{s : s^N = 1\}$ and $N \geq 3$ is odd. Fix two contracted variables i, j, written first, and let $\mathbf{y}$ denote the remaining variables. After removing known nonzero scalars, the rotation classes in either group realize

$$q_s^+(\mathbf{y}) = \widetilde{f}(0, 0, \mathbf{y}) - s\widetilde{f}(1, 1, \mathbf{y}), \qquad s \in \mu_N.$$

The reflection classes in the dihedral group additionally realize

$$q_s^-(\mathbf{y}) = \widetilde{f}(0, 1, \mathbf{y}) - s\widetilde{f}(1, 0, \mathbf{y}), \qquad s \in \mu_N.$$

Each family has the form $q_s = A - sB$, where A, B are the indicated slices obtained by fixing i, j. We call it a *contraction pencil*. Thus the dihedral case permits comparisons between the equal-bit and opposite-bit restrictions, while the cyclic case has only the first family.

In the dihedral case, we distinguish $n = 4, 6$ from $n \geq 8$. The low-arity cases are treated directly. At arity four, coefficient analysis gives either a tensor factorization or a signature covered by the equality reductions in Section 2. At arity six, a finite support classification, including computer-assisted enumeration for the order-three group, followed by coefficient analysis forces a tensor factorization (Theorem F.6.19 and Appendix F.A.2.1). For $n \geq 8$, we instead recover a binary factor of f from the compatibility of its contractions. Although every nonzero contraction is already a product of binaries, its matching—which variables are paired—and the coefficients of its binary factors may initially depend on the contraction.

Fix a contracted pair i, j and one pencil $q_s = A - sB$. Its linear dependence on s links these otherwise separate factorizations. Comparing three phases shows that their matchings either coincide or differ only on four variables, with common binary factors elsewhere. In the latter case, closing the common external pairs leaves a six-variable proper contraction of f, which is itself a matching product. Oddness of N makes these closures nonzero. Its fixed matching cannot produce different residual pairings as the phase varies, excluding this possibility. Thus the matching of q_s is fixed. On that matching, the coefficients form a rank-one tensor; linear dependence on s then allows at most one binary factor to vary. Consequently,

$$q_s = H \otimes h_s,$$

where H is a fixed product of binaries and h_s occupies one remaining pair. Recovering A, B from two phases shows that H factors out of both slices. Applying this argument to the equal-bit and opposite-bit pencils gives factorizations of the two corresponding restrictions of $\widetilde{f}$. To compare them, we use contractions that are nonzero on both restrictions: their residual decompositions must come from the same matching product. This compatibility identifies a binary factor shared by the two restrictions, and hence by their sum $\widetilde{f}$. Inverting the Walsh coordinates gives a factorization of f, contradicting irreducibility (Proposition F.6.21 and Lemma F.6.22).

In the cyclic case, only the equal-bit pencils are available. We use them to prove that the minimum nonbinary arity is four, then handle that arity by binary attachments. For $n \geq 6$, the contraction supports consist of assignments equal on each matched pair. The pencil relation again fixes the matching as the phase varies. A pair whose entire pencil vanishes would force its two bits to be opposite on every supported input of $\widetilde{f}$; comparing this constraint with contractions on other pairs either gives incompatible matchings or a proper tensor factorization. Thus every pair has a

nonzero phase contraction. These comparisons also force $\widetilde{f}(\mathbf{0}^n) \neq 0$, so we normalize it to one. The weight-two coefficients now reveal which original variables must be paired. Put

$$W = \{\{p, q\} : \widetilde{f}(\mathbf{e}_p + \mathbf{e}_q) \neq 0\},$$

where $\mathbf{e}_p$ is the p-th unit vector. Any edge of W survives in the fixed matching of a pencil contracted outside that edge. Two adjacent edges would therefore have to belong to one matching, which is impossible. Moreover, each edge $pq \in W$ forces $x_p = x_q$ throughout the support of $\widetilde{f}$: a violating assignment could be preserved by a contraction outside p, q, in conflict with that contraction's matching.

If $W \neq \varnothing$, compatibility with contractions meeting its edges makes W a perfect matching. We may then regard $\widetilde{f}$ as a coefficient tensor $h(t_1, \ldots, t_{n/2})$, with one bit for each equal pair. Contracting variables from two pairs forces their bits to agree and gives a rank-one restriction of h. These overlapping restrictions determine a product decomposition of h, and hence of f. The six-variable case requires one further coefficient check using the fact that squaring permutes μ_N. If $W = \varnothing$, a different obstruction occurs: the constant coefficient of every contraction is one, while the rank-one identity on two residual pairs would equate a linear polynomial in s with a nonzero quadratic at all $N \geq 3$ phases. This is impossible. Both alternatives exclude $n \geq 6$. At arity four, binary attachments first align the coefficients on complementary Walsh inputs. The resulting forms give an equality reduction, a classified eight-vertex signature, or a tensor factorization, apart from one exceptional coefficient pattern. A two-copy construction in that last case yields a binary that is singular or has a transfer outside the allowed finite cyclic group. It therefore invokes an earlier classification or contradicts completeness of the group. This completes the cyclic branch (Theorem F.6.27 and Theorem F.6.31).

4.2 Completing the height-one classification

These arguments complete a fixed cyclic or dihedral case once the contraction procedure has stabilized. Either a known classification applies, or no nonbinary irreducible signature remains. In the latter case, the same Walsh matrix works for every signature: all binary factors belong to the product class $\mathscr{P}$, consisting of products of unary functions and equality or disequality constraints, and the edge becomes $P^{\mathsf{T}} X P = 2\,\mathrm{diag}(1, -1) \in \mathscr{P}$. Reconstructing the signatures from these factors gives one common $\mathscr{P}$-transformation of the whole set, satisfying one of the seven structural alternatives.

It remains to ensure that the restarts preceding this analysis terminate. A contraction may reveal a new binary transfer or a smaller nonbinary irreducible factor. Express every transfer group back in the original coordinates over one fixed number field. These groups are nested, and their orders have a uniform bound depending only on that field. Meanwhile, the minimum nonbinary irreducible arity never increases. Each strict update increases the former or decreases the latter. Between updates, the finite contraction recursion above terminates by arity descent. Thus a classification is eventually reached (Lemma F.6.11 and Theorem F.6.32). All changes retain the original signatures up to one common invertible transformation, so hardness or a common tractable basis transfers back to the initial set.

5 Higher heights: local contractions and reconstruction

Write the mountain as $M = a\delta_{\mathbf{0}^{2d}} + b\delta_{\mathbf{1}^{2d}} + Q$, where $ab \neq 0$, $\delta_{\mathbf{x}}$ is the indicator of the single assignment $\mathbf{x}$, and Q is its central restriction. Height two is already covered by the eight-vertex

classification in Section 2. For $d \geq 3$, the whole-set divisibility condition makes every gadget output of positive arity less than $2d$ EO: its half-charge lies in $d\mathbb{Z}$, but has absolute value less than d. In particular, an X self-loop gives $\partial_{ij} M = \partial_{ij} Q$, where $(\partial_{ij} f)(\mathbf{x}) = f(0,1,\mathbf{x}) + f(1,0,\mathbf{x})$ and the other variables retain their order.

We seek an EO output whose presence forces hardness. Height three uses a finite family of local constructions. At larger heights, constraints on all pair contractions reconstruct the central restriction, finishing heights at least five and leaving one exception at height four.

5.1 Which EO outputs suffice?

An EO output sits inside the current set, which also contains the non-EO mountain. We therefore need a conclusion about this entire set. Let $\mathcal{B}_{\mathrm{ARS}}$ consist of nonzero scalar multiples of tensor products, on oriented pairs of a perfect matching, of binaries $(0, z, \overline{z}, 0)$ with $z \neq 0$. The subscript refers to arrow-reversal symmetry, the identity $h(\mathbf{x}) = \overline{h(\overline{\mathbf{x}})}$, where $\overline{\mathbf{x}}$ is bitwise complement.

Lemma 5.1 (An EO output in the ambient set)**.** *Let $\mathcal{F}$ be a finite two-sided set of nonzero even-arity signatures with $\neg\mathsf{Tract}(K\mathcal{F})$. If $\mathcal{F}$ contains a nonzero positive-arity EO signature outside $\mathcal{B}_{\mathrm{ARS}}$, then* K-Holant$(\mathcal{F})$ *is $\#\mathsf{P}$-hard.*

The proof uses the constructions in the EO classifications [MWX25, CFS20a] to obtain either a signature classified by Section 2 or the asserted binary factorization. The first stage forces arrow-reversal symmetry up to a scalar; the second handles signatures with that symmetry that fail to factor. To use this argument in the present problem, each fixed gadget must remain valid in the ambient set, which also contains the non-EO mountain. Gadget substitution supplies this extension for direct constructions, and two-sidedness permits the separate factor-extraction reductions. The extension is proved in Section F.4.4. Thus, unless hardness is obtained, every nonzero low-arity EO output has the specified matching-product form. This turns a complexity classification into structural information about the mountain's contractions.

5.2 Height three and a two-copy construction

Let $\mathcal{B}_{\times}$ be the nonzero scalar multiples of matching products of nonsingular anti-diagonal binaries $\left(\begin{smallmatrix} 0 & p \\ q & 0 \end{smallmatrix}\right)$, with $pq \neq 0$. It contains $\mathcal{B}_{\mathrm{ARS}}$. A matching-product central restriction already suffices to finish the branch. If $Q \in \mathcal{B}_{\times}$, join two copies of the complete mountain so that their central matching edges and connecting edges form two odd-length alternating paths between four exposed variables. Central assignments supply four boundary patterns, with one bit equal to one on each exposed pair. The two complementary endpoint choices supply two additional patterns. Mixed endpoint–central assignments are excluded by an internal matching edge, and each supported boundary has a unique nonzero extension. Thus the quaternary EO output has all six weight-two entries nonzero, whereas a quaternary member of $\mathcal{B}_{\times}$ has only four supported strings. By Lemma 5.1 this gives hardness. The construction uses the whole mountain and works at every height $d \geq 3$ (Lemma F.8.2).

Height three is handled directly, without first reconstructing Q. Self-loops test its four-variable restrictions; to obtain stronger constraints, we also join two copies of the six-variable mountain along four pairs of variables, again leaving four variables exposed. A finite coefficient classification shows that if all fifteen self-loops and all 5400 labelled two-copy outputs are zero or in $\mathcal{B}_{\times}$, then Q is supported on at most one complementary pair. This support is too small to satisfy the same tests: if $Q \neq 0$, a self-loop joining positions with opposite bits gives a nonzero output supported on at most two strings; if $Q = 0$, a two-copy output has that property. Either contradicts the four-string support of a quaternary member of $\mathcal{B}_{\times}$. One of the tested outputs must therefore give

hardness by Lemma 5.1. The finite classification uses computer-assisted exhaustive enumeration and exact polynomial solver checks for arbitrary complex coefficients and every $ab \neq 0$. The polynomial infeasibility steps rely on Z3's `UNSAT` answers, as documented in Theorem F.8.5 and Appendix F.A.2.2.

5.3 Normalization and reconstruction at larger heights

For $d \geq 4$, we reconstruct the central restriction from its pair contractions. We use the reconstruction theorem of Shao and Cai [SC20, full version, Lemma 2.11] after making the central tensor real and complement-invariant and all its contractions products of X. The task is to obtain these simultaneous properties from our complex-valued mountain. If $Q = 0$, the mountain is a generalized equality, already classified by the auxiliary reductions (Lemma F.4.18). Otherwise, Lemma 5.1 makes every nonzero contraction a product of EO–ARS binaries. Factor extraction simulates these binaries. Attaching an appropriately oriented copy of a factor $B(z) = (0, z, \overline{z}, 0)$ to one of its variables turns that factor into $|z|^2 X$. The issue is to normalize all contractions of the same mountain, since an attachment can affect several of them. The resulting central restriction F must satisfy

$$\partial_e F = \lambda_e X_{P_e}, \qquad X_{P_e} := \bigotimes_{t \in P_e} X_t, \qquad \lambda_e \neq 0 \quad \left(e \in \binom{[2d]}{2} \right),$$

where P_e is a perfect matching of the remaining variables. To reach this form, first normalize one nonzero contraction. Contracting a disjoint pair in either order gives the same tensor, so the other contraction must also be nonzero, and any binary factor surviving the comparison must be proportional to X. These comparisons constrain the remaining factors sufficiently to choose further attachments that preserve the first normalized contraction. They then propagate the X-matching form to all pairs, including those whose original contractions might have vanished. This simultaneous normalization is proved in Lemma F.9.5. All attachments act on the complete mountain; its central restriction is the tensor analyzed here.

The scalars λ_e may still be complex. To recover symmetry of the central tensor, we need one common scalar normalization. Commuting contractions on disjoint pairs e, f compares λ_e and λ_f: closing a pair in an X-matching product contributes only a factor one or two, so their ratio is positive real. Since the graph of disjoint pairs is connected, all λ_e have the same phase. Dividing by one scalar makes every contraction real and complement-invariant. These properties also hold for the central tensor itself: its differences from its conjugate and from its bit-reversed tensor are EO and have all pair contractions zero, which forces both differences to vanish [CFS20a, Lemma 2.7]. This establishes the required symmetry from the contractions (Lemma F.9.6).

We can now apply reconstruction. At arity at least ten, it makes the central restriction itself an X-matching product. The two-copy construction above then gives hardness, completing every height $d \geq 5$. At arity eight, reconstruction needs an additional overlap: two distinct contracted pairs A, B must satisfy $A \cap B \neq \varnothing$ and their residual matchings share an edge, $P_A \cap P_B \neq \varnothing$. Such an overlap gives the same matching-product conclusion and finishes that height-four branch. The only remaining case has no such overlap.

5.4 The height-four exception

At height four, we first determine the remaining matching family, then recover the coefficients of the mountain. Without the preceding overlap, commutation of double contractions still requires the matchings to agree after deleting any two disjoint variable pairs in either order. A computer-assisted exhaustive classification of these compatible families identifies the eight variables with

$\mathbb{F}_2^3$: contracting a pair with difference $t \neq 0$ leaves the other three pairs $\{x, x+t\}$. This specifies the contraction matchings, but the coefficients of the central tensor remain to be recovered. A contraction coefficient is the sum of two entries whose supports differ by swapping the deleted bits. The prescribed matching determines which such sums vanish and which equal its nonzero scalar. The resulting linear relations force all coefficients outside the fourteen affine planes of $\mathbb{F}_2^3$ to vanish and all plane coefficients to agree (Lemma F.9.9 and Lemma F.9.10).

Consequently the mountain has the form $M^\circ = A\delta_{\mathbf{0}^8} + B\delta_{\mathbf{1}^8} + cR_8^{\mathrm{cen}}$, with $ABc \neq 0$, where R_8^{cen} indicates those fourteen planes. These fourteen weight-four strings, together with the two constant strings, are the truth tables of the affine functions on $\mathbb{F}_2^3$. Their set is the Reed–Muller code

$$RM(1,3) = \{(\alpha \cdot x + \beta)_{x \in \mathbb{F}_2^3} : \alpha \in \mathbb{F}_2^3,\ \beta \in \mathbb{F}_2\}.$$

Write RM_8 for its indicator. To obtain this signature up to scale, we must also control the endpoint coefficients A, B. Join two copies of M° along four coordinates that determine each affine function uniquely. The remaining coefficients distinguish endpoint contributions AB from central contributions c^2. One further closure produces a six-variable EO tensor whose coefficients cannot factor over a matching unless $AB = c^2$. Failure gives hardness by Lemma 5.1; equality permits a common diagonal change of basis in $GO(X)$ taking the mountain to $c\mathrm{RM}_8$ while preserving the whole-set charge condition (Lemma F.9.11).

We now simulate binary equality, which allows the known dichotomy in Section 2 to finish the branch. The Reed–Muller identity of [SC20, full version, Lemma 8.9] replaces three occurrences of I by one occurrence and one Reed–Muller vertex, up to a known nonzero scalar. Repetition leaves at most two occurrences, with only linear growth in the instance size. Thus it suffices to evaluate instances with one or two equality vertices. After removing these vertices, the remaining network is over the current set $\mathcal{F}$, whose supported half-charges all belong to $4\mathbb{Z}$. Its boundary tensor is therefore anti-diagonal on two variables and EO on four variables. One equality occurrence contributes zero, since it requires equal boundary bits. For two occurrences, only the weight-two entries matter, and on that support

$$I_{12}I_{34} = \frac{X_{13}X_{24} + X_{14}X_{23} - X_{12}X_{34}}{2}.$$

Here I_{ij} and X_{ij} act on the indicated pair of variables. The right-hand side uses only existing X-edges, so three oracle queries evaluate the two-occurrence instance. Adjacent equality vertices collapse directly, and equality self-loops have value zero, covering arbitrary instances. We have therefore proved K-Holant$(\mathcal{F}, I) \leq_{\mathrm{T}}$ K-Holant$(\mathcal{F})$, and the binary-equality dichotomy completes the classification. The full version proves the stronger simulation of all four Bell signatures in Lemma F.9.13.

6 Completing the dichotomy

Proof of Theorem 1.1. Decidability and polynomial-time evaluation follow from [LM26a, Theorem 2.22]. Suppose none of the seven structural alternatives holds. By Section 2, it remains to treat a two-sided, multilayer set of irreducible even-arity signatures in K-coordinates. Lemma 3.1 either simulates pinnings, invoking the odd-arity classification, or supplies a mountain of height d and the whole-set charge condition. Section 4 handles $d = 1$, and Section 5 handles $d \geq 2$. The simulations and common changes of basis preserve failure of the structural condition, so the applicable dichotomies give hardness. Composing the reductions with factor reconstruction and the initial holographic transformation proves the theorem (Section F.10). □

Appendix F.A specifies the finite classifications used in the odd-dihedral, height-three, and height-four arguments, including their coverage and computational dependencies. It distinguishes enumeration and solver checks used in the proofs from supplementary computations.

AI methodology statement

We used OpenAI's GPT-5.6 Sol to assist with mathematical exploration and proof development throughout this paper, including exact symbolic calculations and finite-case checks. All companion verification code was generated by OpenAI Codex and subsequently reviewed by the authors.

We used OpenAI's GPT-6 Astra to improve the writing and presentation and to assist with the Lean formalization. We completed the Lean formalization of the proof. The authors take full responsibility for the correctness and originality of the paper.

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Full paper

F.1 Introduction

Counting problems ask how many global configurations satisfy a collection of local constraints. For example, counting proper colourings of a graph means counting assignments of colours to vertices such that adjacent vertices receive different colours. In a weighted version, each local configuration contributes a weight, and the weight of a global configuration is the product of its local weights. Summing these products over all configurations gives a partition function. Ordinary counting is the special case in which the local weights are zero or one.

A short description of the local rules can therefore specify a sum with exponentially many terms. Some rules allow this sum to be evaluated efficiently without enumerating the configurations, whereas others lead to computationally hard counting problems. A complexity dichotomy identifies all tractable choices of local rules within a given framework and proves hardness for every remaining choice. Holant provides a graph-based framework for studying this boundary.

A Boolean Holant problem evaluates a graph partition function specified by local functions, called signatures, at its vertices. Each edge receives a bit, and the partition function sums the product of the vertex signatures over all edge assignments. The classification problem asks which fixed signature sets permit polynomial-time evaluation and which give #P-hard problems.

We prove a dichotomy for arbitrary finite sets of algebraic complex-valued Boolean signatures on general graphs, with an explicit, decidable tractability criterion. The main case addressed here consists entirely of even-arity signatures, with neither a symmetry restriction nor a prescribed auxiliary binary signature. Its resolution completes a line of classification results extending from graph homomorphisms through counting constraint satisfaction problems to Holant problems.

F.1.1 From graph homomorphisms through counting CSP to Holant

Counting graph homomorphisms (#GH) asks how many maps from an input graph to a fixed target graph preserve adjacency. In the weighted version, a fixed matrix specifies the interaction between the labels at the two ends of each edge. Counting constraint satisfaction problems (#CSP) allow constraints of arbitrary arity, so one local function can constrain several variables at once. The weighted problem sums the product of the constraint values over all variable assignments.

Dichotomies for unweighted graph homomorphisms and Boolean counting CSP [DG00, CH96] established early classifications in these frameworks. Subsequent work extended the allowed domains and weights, from nonnegative weights to signed-real and complex weights, where cancellation must also be accounted for. The general-domain complex classification of Cai and Chen covers both weighted graph homomorphisms and counting CSP [CC17]. Table F.1 records this development. A dash means that the case is covered by a more general entry, with no separate reference listed.

Holant changes which local constraints are available. A counting-CSP instance can be represented by a bipartite graph with variable vertices and constraint vertices. At each variable vertex, an equality signature forces all occurrences of that variable to agree. In a Holant problem, these equality signatures need not be available: every vertex must use a signature from the prescribed set. The signatures needed in a reduction must be obtained by gadgets or other justified reductions. This distinction brings problems such as counting perfect matchings into the classification and changes the constructions available for proving hardness.

Motivated by Valiant's holographic algorithms [Val08], the Holant framework was introduced in 2009 [CLX09]. A holographic transformation is an invertible simultaneous change of basis on the

Table F.1: Development of general-graph dichotomies for weighted graph homomorphisms and counting CSP.

Weight range	Symmetric #GH	#GH	Boolean-domain #CSP	General-domain #CSP
$\{0,1\}$	[DG00]	acyclic targets [DGP07]	[CH96]	[Bul13, DR13]
$\mathbb{Q}_{\geq 0}$	–	–	–	[BDG$^+$12]
$\mathbb{R}_{\geq 0}$	[BG05]	[CC19]	[DGJ09]	[CCL16]
$\mathbb{R}$	[GGJT10]	–	[BDG$^+$09]	–
$\mathbb{C}$	[CCL13]	[CC17]	[CLX14]	classification [CC17], decidability [LM26b]

signatures and edge tensors that preserves their contraction. Such transformations relate apparently different counting problems and expose tractable signature sets.

On the Boolean domain, Holant classifications developed along two complementary directions: relaxing restrictions on symmetry and weights, and reducing the auxiliary signatures assumed available. Unary signatures are useful because they allow an input of a signature to be fixed or summed with a chosen weight. The auxiliary assumptions range from all unary signatures (Holant*), through four distinguished unaries (Holant$^+$), to the two unaries fixing a bit to 0 or 1 (Holant^c). Later work treated any signature set containing a nonzero odd-arity signature (Holant^{odd}). Table F.2 separates these assumptions from restrictions on symmetry and weights, using the same dash convention as Table F.1.

Table F.2: Development of Boolean-domain Holant dichotomies on general graphs.

(a) Symmetric signature sets

Weight range	Holant*	Holant^c	Unrestricted Holant
$\mathbb{R}_{\geq 0}$	–	–	–
$\mathbb{R}$	–	[CLX09]	[HL16]
$\mathbb{C}$	[CLX09]	[CHL12]	[CGW16a]

(b) Arbitrary signature sets

Weight range	Holant*	Holant$^+$	Holant^c	Holant^{odd}	Unrestricted Holant
$\mathbb{R}_{\geq 0}$	–	–	–	–	[LW18]
$\mathbb{R}$	–	–	[CLX18]	[CFS20b]	[SC20]
$\mathbb{C}$	[CLX11]	[Bac17]	[Bac21, Bac25]	[MWXZ25, GSS26]	This paper

Without auxiliary unary signatures, dichotomies were obtained for symmetric complex-valued signatures [CGW16a], arbitrary nonnegative signatures [LW18], and arbitrary real-valued signatures [SC20].

In 2025, Meng, Wang, and Xia proved an $\mathsf{FP}^{\mathsf{NP}}$ versus #P-hard dichotomy for weighted Eulerian orientations [MWX25]. Meng, Wang, Xia, and Zheng combined this result with tensor decomposition to obtain the same complexity bounds for complex Boolean Holant with a nonzero

odd-arity signature [MWXZ25]. In 2026, Guan, Shao, and Shi gave a polynomial-time algorithm based on linear programming for the former $\mathsf{FP}^{\mathsf{NP}}$ cases, yielding FP versus #P-hard dichotomies for both problems [GSS26].

These results direct attention to the all-even case. Every gadget built from even-arity signatures has even arity, since each internal edge consumes two inputs. Thus no such gadget supplies a unary signature. Binary signatures become the smallest positive-arity objects available for further constructions. Xia proposed a framework organized by the finite groups associated with binary signatures, analyzing several branches in which all realizable nonzero quaternary signatures decompose [Xia26]. Liu and Meng then classified complex Boolean Holant with binary disequality available, for arbitrary finite sets of algebraic complex-valued signatures, and gave an explicit decidable tractability criterion [LM26a].

Our proof removes the prescribed-binary assumption in the all-even case. It combines the binary and quaternary dichotomies of Liu and Meng with a decomposition based on Hamming imbalance. This decomposition produces signatures of controlled support whose arity determines the remaining cases, as described in Section F.1.3. Together with the odd-arity dichotomy, it completes the unrestricted complex-valued entry in Table F.2.

Independent work. Huang and Fu established a dichotomy for Holant problems with a single complex-valued quaternary signature [HF26]. Guan, Shao, and Tang proved a dichotomy for complex-valued Boolean Holant problems whose signature sets are closed under complex conjugation [GST26]. Zhou proved a dichotomy for finite sets of algebraic complex-valued Boolean signatures of even arity whose nonzero binary gadget signatures are nonsingular and whose projective matrix classes form the cyclic group C_2 of order two under series composition [Zho26].

We were also informed of the following independent work before the public release of the present paper. Jiayi Zheng informed us of unpublished work on the complex-valued Holant dichotomy in the case where the projective binary transfer classes form an odd dihedral group. Mingji Xia informed us of work concerning the twisted Klein-group case and the C_1 case, under the condition that all quaternary signatures are reducible. The results and proofs in the present paper were developed independently of these works.

F.1.2 Our result: the full dichotomy

The predicate $\mathsf{Tract}(\mathcal{F})$, defined in Section F.3, is an equivalent formulation of the seven conditions in the odd-arity dichotomy of Meng, Wang, Xia, and Zheng [MWXZ25, conference version, Theorem 30; full version, Theorem 33]. Our result applies this criterion without an odd-arity assumption. Each condition concerns the signature set as a whole. In each transformability case, one basis must work simultaneously for the edge and every member of $\mathcal{F}$.

Theorem F.1.1 (Full dichotomy theorem)**.** *Let $\mathcal{F}$ be a finite set of algebraic complex-valued Boolean signatures, given in the exact algebraic representation fixed in Section F.2.4. The predicate* $\mathsf{Tract}(\mathcal{F})$ *is decidable, and the following complexity classification holds:*

$$\begin{aligned} \mathsf{Tract}(\mathcal{F}) &\implies \text{Holant}(\mathcal{F}) \in \mathsf{FP}, \\ \neg\mathsf{Tract}(\mathcal{F}) &\implies \text{Holant}(\mathcal{F}) \textit{ is } \#\mathsf{P}\textit{-hard} \end{aligned}$$

under polynomial-time Turing reductions.

For evaluation and hardness reductions, $\mathcal{F}$ is fixed and running time is measured in the encoded instance size. For recognition, $\mathcal{F}$ itself is the input. The proof of Theorem F.1.1 is completed in Section F.10.2, where Theorem F.10.3 is transferred back to ordinary coordinates.

F.1.3 Proof organization

We first apply the known odd-arity dichotomy. For the all-even case, we reduce to known classifications or construct a *mountain*, whose height determines the remaining analysis. An input $\mathbf{x}$ to an arity-r signature has *half-charge* $\chi_r(\mathbf{x}) = \mathrm{wt}(\mathbf{x}) - r/2$, where $\mathrm{wt}(\mathbf{x})$ is its number of 1-bits. In the K-basis, edges carry disequality, so the vertex charges in a closed instance sum to zero. If all supported charges are nonnegative, or all are nonpositive, only the central Hamming layers contribute, and the weighted Eulerian-orientation classification applies. That classification also handles sets in which each signature is supported on a single Hamming layer, meaning inputs of one fixed weight.

The remaining set is *multilayer*: some signature has nonzero entries at different Hamming weights. It is also *two-sided*: positive and negative half-charges both occur, possibly in different signatures. We extract tensor factors and handle the cases covered by known classifications. For the remaining set, let d be the gcd of its nonzero supported half-charges. The construction in Lemma F.5.2 first uses tensor products and self-loops to obtain two signatures, one nonzero at its all-zero input and the other at its all-one input. We combine these endpoint properties by further connections, or obtain unary pins and apply the odd-arity dichotomy. In the first case, the output is a *mountain of height* d: an arity-$2d$ signature with both endpoint coefficients nonzero and all other support on the weight-d layer. Every supported half-charge in the current set remains divisible by d, a condition used in the higher-height arguments.

At height one, the mountain is binary. We use interpolation and finite projective groups to analyze binary gadgets. Binary equality is handled by Liu and Meng's dichotomy [LM26a, Theorem 1.2]. The remaining groups are odd cyclic or odd dihedral. Contraction analysis and a terminating sequence of signature additions settle these two cases.

Height two is resolved by the eight-vertex dichotomy and factor extraction. Height three uses self-loop and two-copy constructions and computer-assisted exhaustive enumeration.

For the remaining cases at heights at least four, we normalize the mountain and reconstruct its central restriction from its self-loop contractions. At heights at least five, the recovered matching structure yields a hard two-copy gadget. Height four has an exceptional Reed–Muller structure, which we handle by a reduction adding binary equality. Some base cases in the odd-dihedral and higher-height arguments depend on exact enumeration and solver checks.

Sections F.2–F.5 present the basic conventions, tractability criterion, auxiliary reductions, and mountain construction, respectively. Sections F.6–F.9 treat the height cases, Section F.10 completes the main proof, and Section F.11 concludes. Appendix F.A records the finite calculations and their verification.

F.1.4 Related work

The historical progression above concerns exact counting on general graphs. Further work studies recognition of the tractable cases, larger domains, specific vertex models, and planar restrictions.

The last cell of Table F.1 distinguishes a classification for each fixed signature set from an algorithm that decides whether the tractability criterion holds. Cai and Chen's 2017 theorem expressed tractability through three conditions on an unbounded generated family of partial sums [CC17]. Liu and Meng subsequently gave a total exact algorithm deciding these conditions from a finite set of algebraic complex-valued constraints. They proved that Block Orthogonality implies the other two conditions and also decided the corresponding degree-multiple counting-CSP criterion [LM26b].

For Holant on larger domains, classifications are available under further restrictions on the

signatures. A domain-three theorem treated one symmetric ternary signature with all unary signatures available [CLX13]. Its journal version appeared in 2026 [CLX26]. Subsequent classifications covered special edge-colouring signatures and restricted domain-three and domain-four settings [CGW16b, LFC25]. In 2025, Cai and Ihm proved a dichotomy for real-valued symmetric signatures on domain three with all unary signatures available, allowing an arbitrary finite signature set [CI25].

More specific Boolean models include a single ternary signature on 3-regular graphs, the six-vertex and eight-vertex models, and weighted Eulerian orientations with arrow-reversal symmetry [YHF24, CFX18, CF23, CFS20a].

Restricting the input graph to be planar can make additional counting problems tractable. For Boolean counting CSP, Cai, Lu, and Xia classified real-valued symmetric constraints [CLX17]. Guo and Williams extended the classification to symmetric constraints with complex weights [GW20], and Cai and Fu removed the symmetry assumption [CF22]. In this planar model, the incidence graph is embedded in the plane and the argument order follows the cyclic order around each constraint vertex. Every case that is hard on general graphs but tractable in this planar setting admits a holographic reduction to matchgates. These reductions evaluate the partition function through weighted perfect-matching sums computed by the FKT algorithm.

Graph homomorphism problems are a subclass of counting CSP, so these Boolean-domain results also classify planar graph homomorphisms with complex symmetric weight matrices. For larger domains, Cai and Maran proved a dichotomy for real symmetric 3×3 matrices in 2023 [CM23]. Their 2024 preprint classified nonnegative real symmetric 4×4 matrices of full rank [CM24]. In a 2026 preprint, Cai, Maran, and Young obtained a dichotomy for nonnegative real symmetric matrices of arbitrary finite size whose diagonal entries are pairwise distinct [CMY26].

For planar Holant with symmetric complex-valued Boolean signatures, Cai, Fu, Guo, and Williams gave a classification that includes tractable cases beyond holographic reductions to the FKT algorithm [CFGW22]. In 2026, Cai, Fan, Shao, and Tang classified the eight-vertex model into three cases: tractable on general graphs, hard on general graphs but tractable on planar graphs, and hard even on planar graphs [CFST26].

F.2 Preliminaries

F.2.1 Basic notation

We use $\mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$ for the rationals, reals, and complex numbers, and write

$$\overline{\mathbb{Q}} := \{\alpha \in \mathbb{C} : p(\alpha) = 0 \text{ for some } 0 \neq p \in \mathbb{Q}[t]\}, \qquad \mathbb{F}_2 := \mathbb{Z}/2\mathbb{Z}$$

for the algebraic complex numbers and the two-element field. For a field F, put $F^\times = F \setminus \{0\}$. For a finite set V, F^V denotes the functions $V \to F$. We write $[m] = \{1, \dots, m\}$, with $[0] = \varnothing$, and $\mathrm{Mat}_{m\times n}(F)$ for matrices over F, abbreviated to $\mathrm{Mat}_n(F)$ when $m = n$. Transposes are denoted by M^T and do not include complex conjugation. The group of invertible matrices is $\mathrm{GL}_n(F)$, and I_n is its identity. For a vector space V over F, tensor products use the convention $V^{\otimes 0} = F$, with empty product 1.

Vector symbols are bold, while scalar coordinates and variable or vertex labels are not. For a Boolean vector $\mathbf{x} \in \mathbb{F}_2^V$, put $\mathrm{wt}(\mathbf{x}) = |\{v : x_v = 1\}|$. A Hamming layer fixes $\mathrm{wt}(\mathbf{x})$, and parity means $\mathrm{wt}(\mathbf{x})$ mod 2. All binary-vector arithmetic is over $\mathbb{F}_2$. We use $\oplus$ for coordinatewise addition and $\bar{\mathbf{x}}$ for bitwise complement. For $\mathbf{y} \in \mathbb{F}_2^V$, the dot product $\mathbf{x} \cdot \mathbf{y} = \sum_{v\in V} x_v y_v$ is taken in $\mathbb{F}_2$. Binary strings on ordered index sets are ordered lexicographically with $0 < 1$. For $A \subseteq V$, $\mathbf{x}_{\widehat{A}}$ denotes the restriction of $\mathbf{x}$ to $V \setminus A$, in the inherited order when V is ordered. In particular, $\mathbf{x}_{\widehat{pq}}$ deletes positions p, q.

F.2.2 Signatures and Holant instances

For an integer $r \geq 0$, an arity-r Boolean signature is a function $f : \{0,1\}^r \to \overline{\mathbb{Q}}$. Its truth table, listed as a column vector in lexicographic order, identifies it with a tensor in $(\overline{\mathbb{Q}}^2)^{\otimes r}$. Its variables have labelled positions whose order is part of the signature. We write $\mathrm{arity}(f) = r$ and $\mathrm{supp}(f) = \{\mathbf{x} : f(\mathbf{x}) \neq 0\}$, and use $\mathbf{1}[\mathsf{P}]$ for the indicator of a proposition P. A signature set is *all-even* when each member has even arity. The terms nullary, unary, binary, ternary, and quaternary mean arity $0, 1, 2, 3, 4$, respectively. For an ordered binary signature B, we identify B with the matrix $(B(a,b))_{a,b\in\mathbb{F}_2}$, with rows indexed by a and columns by b.

The symbols $\mathbf{0}^r$ and $\mathbf{1}^r$ denote the all-zero and all-one strings. At $r = 0$, both denote the unique empty string. More generally, $\mathbf{0}^V, \mathbf{1}^V \in \mathbb{F}_2^V$ are the constant vectors on a finite index set V. The values $f(\mathbf{0}^r)$ and $f(\mathbf{1}^r)$ are the two *endpoint coefficients*. For $T \subseteq [r]$, $\mathbf{1}_T \in \{0,1\}^r$ is the incidence vector of T, so that $\mathbf{1}_\varnothing = \mathbf{0}^r$. We write $\mathbf{e}_j = \mathbf{1}_{\{j\}}$ for the j-th standard basis vector when the ambient arity is clear.

Graphs may have parallel edges and loops. Each loop contributes two to $\deg_G(v)$. A perfect matching of a finite set V is a partition into pairs. The mate of a vertex is the other endpoint of its matching edge. We write K_n for the complete graph and $\binom{V}{k}$ for the k-element subsets of V.

For a signature set $\mathcal{F}$, a signature grid over $\mathcal{F}$ is a finite multigraph $G = (V, E)$, together with a signature $f_v \in \mathcal{F}$ satisfying $\mathrm{arity}(f_v) = \deg_G(v)$ and an ordering of the incident edge-ends at every vertex. Let $\mathbf{x}_v$ list the local variables at v in this order. An edge assignment $\sigma : E \to \{0,1\}$ induces the string $\mathbf{x}_v|_\sigma$ by assigning each variable the bit on its edge. The two ends of a loop receive the same edge bit. For the resulting grid Ω,

$$\mathrm{Holant}_\Omega = \sum_{\sigma : E \to \{0,1\}} \prod_{v \in V} f_v(\mathbf{x}_v|_\sigma).$$

The problem $\mathrm{Holant}(\mathcal{F})$ asks for this value on an input grid over $\mathcal{F}$. Throughout, comma-separated signature or signature-set arguments mean adding their union. For example, $\mathrm{Holant}(\mathcal{F}, a, b)$ abbreviates $\mathrm{Holant}(\mathcal{F} \cup \{a, b\})$.

We use the unary pinnings

$$\Delta_0 = [1,0]^{\mathsf{T}}, \qquad \Delta_1 = [0,1]^{\mathsf{T}},$$

and the binary equality and disequality tensors

$$I = I_2 = (=_2) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad X = (\neq_2) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

We also fix the sign and skew matrices

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Y = ZX = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

More generally, for $r \geq 1$, let

$$(=_r)(x_1, \ldots, x_r) = \mathbf{1}[x_1 = \cdots = x_r].$$

For signature sets $\mathcal{G}$ and $\mathcal{F}$, the bipartite problem $\mathrm{Holant}(\mathcal{G} \mid \mathcal{F})$ is defined on a bipartite graph $H = (U, V, E)$: vertices in the left part U are assigned signatures from $\mathcal{G}$, and vertices in the right part V are assigned signatures from $\mathcal{F}$.

Subdividing every ordinary edge by an $=_2$-vertex preserves the partition value, as does suppressing these degree-two equality vertices. Under this correspondence between instances, we identify

$$\mathrm{Holant}(\mathcal{F}) = \mathrm{Holant}(=_2 \mid \mathcal{F}).$$

For an integer $r \geq 0$ and $\mathbf{a} \in \{0,1\}^r$, define the *indicator signature* $\delta_{\mathbf{a}}$ by

$$\delta_{\mathbf{a}}(\mathbf{x}) = \mathbf{1}[\mathbf{x} = \mathbf{a}].$$

F.2.3 Zero and nullary preprocessing

An instance containing an identically zero signature has value zero. A nullary signature is a scalar: delete its vertex and multiply that scalar back into the final value. The empty residual instance has value 1. After this preprocessing, we assume that the input signature set $\mathcal{F}$ contains neither zero nor nullary signatures. Intermediate contractions may still be zero or nullary and are treated as specified in each argument.

F.2.4 Complexity and exact-input conventions

We treat counting problems as functions from finite encoded instances to exact values. We use *tractable* to mean membership in FP, the functions computable exactly in deterministic polynomial time. The class $\#\mathsf{P}$ consists of functions counting accepting paths of nondeterministic polynomial-time Turing machines. A possibly complex-valued counting problem is *hard* if it is $\#\mathsf{P}$-hard under polynomial-time Turing reductions.

For counting problems A, B, write $A \leq_{\mathrm{T}} B$ if an algorithm computes A using polynomially many exact evaluations of B on instances of polynomial size, with polynomial time for all other work. This time bound includes any known nonzero algebraic rescaling and exact algebraic postprocessing. Write $A \equiv_{\mathrm{T}} B$ when reductions exist in both directions. The notation $\mathsf{FP}^{\mathsf{NP}}$ permits an NP decision oracle.

We fix an *exact algebraic encoding* of $\alpha \in \overline{\mathbb{Q}}$ by its monic irreducible polynomial over $\mathbb{Q}$ and an isolating rectangle in $\mathbb{C}$: its vertices have rational real and imaginary parts, and it contains α and no other complex root of the polynomial. Equality, zero tests, conjugation, and field arithmetic are exact. A *number field* is a finite extension $L/\mathbb{Q}$, and $[L : \mathbb{Q}]$ is its dimension as a $\mathbb{Q}$-vector space.

For evaluation of $\mathrm{Holant}(\mathcal{F})$, the signature set $\mathcal{F}$ is fixed, and running time is measured in the encoded instance size. For recognition, the input is a finite list of signatures given by exact algebraic entries. No polynomial-time bound in the length of this list is asserted.

F.2.5 Tensor operations and projective notation

For an integer $r \geq 0$, let S_r denote the symmetric group on $[r]$. For an arity-r signature f and $\tau \in S_r$, $(\tau f)(\mathbf{x}) = f(x_{\tau^{-1}(1)}, \ldots, x_{\tau^{-1}(r)})$. For signatures $f : \mathbb{F}_2^r \to \overline{\mathbb{Q}}$ and $g : \mathbb{F}_2^s \to \overline{\mathbb{Q}}$ on disjoint ordered variables, and $\mathbf{x} \in \mathbb{F}_2^r, \mathbf{y} \in \mathbb{F}_2^s$, define $(f \otimes g)(\mathbf{x}, \mathbf{y}) = f(\mathbf{x})g(\mathbf{y})$. The signature f is *symmetric* when $\tau f = f$ for every $\tau \in S_r$, equivalently when $f(\mathbf{x})$ depends only on $\mathrm{wt}(\mathbf{x})$.

For $\mathbf{b} \in \mathbb{F}_2^r$, the bit-flip operation sends f to $f^{\oplus \mathbf{b}}(\mathbf{x}) = f(\mathbf{x} \oplus \mathbf{b})$. Bit-flip/variable-permutation orbits use these flips and permutations of argument positions. A nonzero tensor has rank one when it is a product of one vector on each of its specified factors. For two factors this is matrix rank one. We say a divides f, with cofactor b, when $f = a \otimes b$ up to variable permutation and nonzero scalar redistribution. Every tensor is taken to divide the zero tensor. A proper tensor factor has positive arity strictly smaller than its parent.

A nonzero positive-arity signature is *irreducible* if, up to variable permutation and nonzero scalar, it cannot be written as $a \otimes b$ with both a and b of positive arity. A nonzero positive-arity signature that is not irreducible is *reducible*. Factorization into irreducibles is unique up to these operations and factor order [CFS20a, full version, Lemma 2.13]. For any class $\mathscr{C}$ of signatures, its tensor closure $\langle\mathscr{C}\rangle$ consists of tensor products of members of $\mathscr{C}$, up to variable permutation and nonzero scalar. By convention, it also contains zero and the arity-zero empty product.

For an integer $r \geq 1$ and $\mathbf{a} \in \{0,1\}^r$, a *complementary-pair signature* (CP) is a signature

$$c_0\delta_{\mathbf{a}} + c_1\delta_{\bar{\mathbf{a}}}, \qquad c_0, c_1 \in \overline{\mathbb{Q}}^{\times}.$$

When $\mathbf{a} = \mathbf{0}^r$, it is a *generalized equality* (GE). As a tensor identity, individual bit flips turn a CP signature into a GE signature. The notation $\mathrm{diag}(p, q)$ denotes the 2-by-2 diagonal matrix with diagonal entries p, q. For $t \in \overline{\mathbb{Q}}^{\times}$, put $D(t) := \mathrm{diag}(1, t)$. For $p, q \in \overline{\mathbb{Q}}$, in arity two, $p\delta_{00} + q\delta_{11} = \mathrm{diag}(p, q)$ is a *weighted binary equality*, and $p\delta_{01} + q\delta_{10}$ is a *weighted binary disequality*. Either is nonsingular when $pq \neq 0$.

The notation $\operatorname{span} S$ denotes linear span. An affine subspace is a translate $\mathbf{a} + L$ of a linear subspace. Its translation space is L, and an affine plane has dimension two.

For a nonzero vector $\mathbf{v}$ over a field F, write $[\mathbf{v}] = F^{\times}\mathbf{v}$ for its projective class. We likewise write $[A] = F^{\times}A$ for a nonzero tensor or matrix A. In particular, $\mathrm{PGL}_n(F) = \mathrm{GL}_n(F)/(F^{\times}I_n)$. For tensors or matrices A, B over $\overline{\mathbb{Q}}$, write $A \doteq B$ when $A = \lambda B$ for some $\lambda \in \overline{\mathbb{Q}}^{\times}$. In gadget constructions we suppress overall nonzero algebraic factors. Their contribution to a Holant value can be removed by exact algebraic postprocessing. This convention does not suppress relative coefficients in sums of signatures. Angle brackets denote generated subgroup in a group context and tensor closure for a signature class.

F.2.6 Holographic transformations and the K-basis

In $\mathrm{Holant}(\mathcal{G} \mid \mathcal{F})$, represent each left-side signature g of arity r as a row vector and each right-side signature f of arity s as a column vector, both in lexicographic order. For a fixed $M \in \mathrm{GL}_2(\overline{\mathbb{Q}})$, replace them by

$$gM^{\otimes r}, \qquad (M^{-1})^{\otimes s} f,$$

respectively. The matrices on the two ends of each edge cancel by $MM^{-1} = I$, so this replacement preserves the value of every instance. Write $gM := gM^{\otimes r}$ for the left-side action and $Mf := M^{\otimes s}f$ for the right-side action, and put

$$\mathcal{G}M := \{gM : g \in \mathcal{G}\}, \qquad M\mathcal{F} := \{Mf : f \in \mathcal{F}\}.$$

The value-preserving transformation in both directions gives

$$\mathrm{Holant}(\mathcal{G} \mid \mathcal{F}) \equiv_{\mathrm{T}} \mathrm{Holant}(\mathcal{G}M \mid M^{-1}\mathcal{F}).$$

For ordinary Holant, take $\mathcal{G} = \{=_2\}$. The transformed row vector $(1,0,0,1)M^{\otimes 2}$ represents the binary signature with matrix $M^{\mathsf{T}}M$, since its (a,b)-entry is $\sum_{x\in\mathbb{F}_2} M_{xa}M_{xb}$. Thus

$$\mathrm{Holant}(\mathcal{F}) = \mathrm{Holant}(=_2 \mid \mathcal{F}) \equiv_{\mathrm{T}} \mathrm{Holant}(M^{\mathsf{T}}M \mid M^{-1}\mathcal{F}).$$

Fix the K-coordinate matrix

$$K = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \qquad K^{-1} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}. \tag{F.1}$$

Since $K^{\mathsf{T}}K = X$, putting $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$ gives

$$\text{Holant}(\mathcal{F}) \equiv_{\text{T}} \text{Holant}(X \mid \widehat{\mathcal{F}}) =: \text{K-Holant}(\widehat{\mathcal{F}}). \tag{F.2}$$

For an original-coordinate signature f, we likewise write $\widehat{f} = K^{-1}f$. The hat records passage to K-coordinates. Once we work with a signature set already in K-coordinates, we suppress these hats and denote its member signatures simply by f. Unless stated otherwise, signatures and signature sets appearing in a K-Holant problem are understood to be in K-coordinates, with edge X.

Write $Z_X(\Omega)$ for the partition function of a closed grid whose edges carry X. To describe simultaneous changes of basis that preserve this edge up to scalar, define the general orthogonal similitude group of X by

$$GO(X) = \{D \in \text{GL}_2(\overline{\mathbb{Q}}) : D^{\mathsf{T}}XD = \lambda X \text{ for some } \lambda \in \overline{\mathbb{Q}}^{\times}\}.$$

If $E(\Omega)$ is the edge set, $D\Omega$ applies D at every vertex, and $D^{\mathsf{T}}XD = \lambda X$, then

$$Z_X(D\Omega) = \lambda^{|E(\Omega)|} Z_X(\Omega). \tag{F.3}$$

The known edge-count scalar gives a polynomial-time equivalence. This operation transforms the whole signature set. It does not by itself realize D as a local gadget.

F.2.7 Gadgets and reductions

A gadget is a finite network over the allowed signatures, with ordered, labelled dangling edges. Its signature is obtained by summing over the internal variables, including the edge signatures in the product. Edges carry equality in ordinary coordinates and disequality in K-coordinates. A *direct* construction uses one fixed gadget. Below we work in K-coordinates.

A slice fixes some arguments of a tensor. Implementing it by pinning requires the corresponding pinning signatures. Interpolation gives an oracle reduction by solving a linear system from exact evaluations. Formal tensor operations make a signature available only when a gadget or a stated reduction supplies it. Whole-set changes of basis act on the problem as described in Section F.2.6.

Using bit flips in a gadget or reduction requires available signatures or a stated whole-set transformation. Algebraic support analysis may also use the relabellings defined above.

Equivalence convention. Let $\mathcal{F}$ be the current finite signature set. Whenever a fixed gadget construction, polynomial-time interpolation, or a cited reduction supplies a finite set S such that

$$\text{K-Holant}(\mathcal{F} \cup S) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}), \tag{F.4}$$

the reverse reduction is immediate by restriction. We call the signatures in S *available by reduction over* $\mathcal{F}$. This does not assert membership in the original set or direct gadget realizability. When an argument replaces the current set by an equivalent set and retains the notation $\mathcal{F}$, that replacement is stated explicitly. The same convention applies to ordinary Holant when $\text{Holant}(\mathcal{F}\cup S) \leq_{\text{T}} \text{Holant}(\mathcal{F})$. A use that depends on local topology or variable order still requires an explicit direct realization over the stated signature set. Overall nonzero scalars are handled as in Section F.2.5. These are the gadget and interpolation conventions of [LM26a, Section 2.5] and [MWXZ25, Section 2.3].

If a direct X-edge gadget over $\mathcal{F}$ realizes h, replacing each vertex carrying h by that gadget gives

$$\text{K-Holant}(\mathcal{F} \cup \{h\}) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}). \tag{F.5}$$

A fixed gadget has constant size. If it realizes ρh, with $\rho \in \overline{\mathbb{Q}}^{\times}$, an instance with N occurrences is rescaled by ρ^{-N}. Gadget replacement is thus one case of Equation (F.4).

Let f have arity r. For an ordered pair of distinct variables $p = (i, j)$, a binary kernel $C : \mathbb{F}_2^2 \to \overline{\mathbb{Q}}$, and a residual string $\mathbf{z}$ indexed by $[r] \setminus \{i, j\}$ in increasing variable order, let $\mathbf{z}^{i \leftarrow a, j \leftarrow b}$ be the length-r string obtained by inserting a, b at positions i, j. Define the contraction of two variables by

$$(\partial_p^C f)(\mathbf{z}) = \sum_{a,b \in \mathbb{F}_2} C(a, b) f(\mathbf{z}^{i \leftarrow a, j \leftarrow b}).$$

We also write $\partial_{ij}^C f = \partial_{(i,j)}^C f$.

In the notation of Shao–Cai [SC20, full version, Section 2.4], their $\widehat{\partial}_{ij} \widehat{f}$ and $\partial_{ij} f$ are, respectively, our $\partial_{(i,j)}^X \widehat{f}$ and $\partial_{(i,j)}^I f$.

Let an ordered binary signature B be directly realized over the current signature set. Connecting its two inputs through X-edges to variables i, j of f, in that order, gives the contraction kernel $C = XBX$, with $C(a, b) = B(1 - a, 1 - b)$. Swapping the inputs of B gives the kernel $XB^{\mathsf{T}}X$.

We also write $f_{ij}^{ab}(\mathbf{z}) = f(\mathbf{z}^{i \leftarrow a, j \leftarrow b})$ for the residual slice. If an ordered binary B is placed on the ordered variable pair $e = (u, v)$, we write B_e.

For tensors A, B on the same ordered variables, define the bilinear coefficient pairing

$$\langle A, B \rangle = \sum_{\mathbf{x}} A(\mathbf{x}) B(\mathbf{x}),$$

without conjugation. For binary matrices this is the Frobenius pairing $\langle U, V \rangle = \operatorname{tr}(U^{\mathsf{T}} V)$. Two tensors are *orthogonal* for this pairing when it is zero.

For any $T \in \mathrm{Mat}_2(\overline{\mathbb{Q}})$, define

$$T^{\#} := XT^{\mathsf{T}}X.$$

For an ordered binary signature B, its *transfer* is $T_B := BX$. Swapping the two inputs of B transposes its matrix, so the resulting transfer is

$$T_{B^{\mathsf{T}}} = B^{\mathsf{T}}X = X(T_B)^{\mathsf{T}}X = (T_B)^{\#}.$$

Joining the second input of B to the first input of an ordered binary signature C through an X-edge gives BXC. Hence

$$T_{BXC} = (BXC)X = (BX)(CX) = T_B T_C.$$

For all $S, T \in \mathrm{Mat}_2(\overline{\mathbb{Q}})$, we have $(ST)^{\#} = T^{\#}S^{\#}$ and $(T^{\#})^{\#} = T$. For $T \in \mathrm{GL}_2(\overline{\mathbb{Q}})$, the *projective order* of $[T]$ is the least $m \geq 1$ such that $T^m = \lambda I$ for some $\lambda \in \overline{\mathbb{Q}}^{\times}$, and is infinite if no such m exists.

F.3 The Tractability Criterion

The criterion in Theorem F.1.1 is a disjunction of seven conditions on the whole signature set. The first two use support restrictions and EO signatures, the next two use tensor decompositions, and the last three require a common change of basis for the edge and all vertex signatures. We define these conditions and then state their decidability and polynomial-time evaluation result. The hardness direction is proved in Section F.10.2.

F.3.1 Charge and Hamming layers

All charge terminology in this subsection is in K-coordinates. For an integer $r \geq 0$, define the doubled-charge function $q_r : \mathbb{F}_2^r \to \mathbb{Z}$ by

$$q_r(\mathbf{x}) = 2\,\mathrm{wt}(\mathbf{x}) - r.$$

For an even integer r, we also use the integer half-charge

$$\chi_r(\mathbf{x}) = \frac{q_r(\mathbf{x})}{2} = \mathrm{wt}(\mathbf{x}) - \frac{r}{2}.$$

Put

$$HW_r^{\geq} = \{\mathbf{x} \in \mathbb{F}_2^r : q_r(\mathbf{x}) \geq 0\}, \quad HW_r^{\leq} = \{\mathbf{x} \in \mathbb{F}_2^r : q_r(\mathbf{x}) \leq 0\}, \quad HW_r^{=} = \{\mathbf{x} \in \mathbb{F}_2^r : q_r(\mathbf{x}) = 0\},$$

and define $HW_r^{>}, HW_r^{<}$ by strict inequalities. We omit the subscript when the arity is clear. A signature has one of these properties when its support lies in the corresponding set. A signature is *single-layer* when its support lies in one Hamming layer. A signature set is single-layer when each member is so supported, with member-dependent layers allowed, and is *multilayer* otherwise. A signature set is *common one-sided* when all members are $HW^{\geq}$ or all are $HW^{\leq}$. Otherwise it is *two-sided*, equivalently both charge signs occur among support strings of its members. Here EO abbreviates Eulerian orientation. An arity-$2d$ *EO signature* is supported in $HW_{2d}^{=}$. This central support condition is the local balance constraint for an Eulerian orientation.

For a closed X-edge grid Ω with vertex set $V(\Omega)$, every edge contributes one 0-bit and one 1-bit. Hence each contributing assignment $\mathbf{x}$, with induced local string $\mathbf{x}_v$ at vertex v, satisfies

$$\sum_{v \in V(\Omega)} q_{\mathrm{arity}(f_v)}(\mathbf{x}_v) = 0.$$

For a gadget with dangling edges, the same cancellation occurs on internal X-edges: the sum of local charges equals the charge of the external string. This is the charge conservation used in contractions. In a closed instance over a common one-sided set, all local charges have the same weak sign and sum to zero. Hence every contributing assignment uses only central-layer entries. Setting all other entries to zero therefore preserves the Holant value.

Remark F.3.1 (Notation relative to prior work). We follow [MWXZ25, Sections 2.1–2.3] in writing $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$, $HW^{\geq}$, $HW^{\leq}$, $HW^{=}$, and *EO signature*, as well as their equality and disequality notation $=_r$ and $\neq_2$.

F.3.2 The standard signature classes

Zero signatures are included whenever their arity is permitted by the class. The affine class $\mathscr{A}$ consists of the zero signature and the nonzero arity-r signatures

$$f(\mathbf{x}) = \lambda\, \mathbf{1}[A\mathbf{x} = \mathbf{b}]\, i^{Q(\mathbf{x})}, \tag{F.6}$$

where $r, m \geq 0$ are integers, $\lambda \in \overline{\mathbb{Q}}^{\times}$, $A \in \mathrm{Mat}_{m \times r}(\mathbb{F}_2)$, $\mathbf{b} \in \mathbb{F}_2^m$, and $Q : \mathbb{F}_2^r \to \mathbb{Z}_4$, where $\mathbb{Z}_4 = \mathbb{Z}/4\mathbb{Z}$, is represented by a multilinear polynomial over $\mathbb{Z}_4$ of degree at most two whose mixed coefficients are even. Equivalently, the phase is a product of fourth-root phases of affine linear forms interpreted as $\{0,1\}$-valued integers. Here a *mixed coefficient* is the coefficient of $x_i x_j$ for $i \neq j$, and an affine linear form over $\mathbb{F}_2$ is $a_0 + \sum_{j=1}^{r} a_j x_j$ with all coefficients in $\mathbb{F}_2$.

The product class $\mathscr{P}$ consists of zero and all signatures of the form $\lambda \prod_{j=1}^{m} h_j$, where $m \geq 0$ is an integer, $0 \neq \lambda \in \overline{\mathbb{Q}}$, and every factor h_j is a unary, binary equality, or binary disequality constraint applied to specified signature variables. This is a pointwise product of constraints that may share variables, whereas tensor products use disjoint variables. Empty products and repeated variables are allowed. An inconsistent product denotes the zero signature.

Put $\zeta_8 := e^{\pi i/4} = (1+i)/\sqrt{2}$. The local-affine class $\mathscr{L}$ consists of the signatures f such that

$$\left(\bigotimes_{j=1}^{\operatorname{arity}(f)} D(\zeta_8)^{\alpha_j} \right) f \in \mathscr{A} \qquad \text{for every } \boldsymbol{\alpha} \in \operatorname{supp}(f).$$

Each support assignment $\boldsymbol{\alpha}$ specifies a phase change at every variable, and the condition tests all of these choices on the original support.

Let $\mathscr{T}$ be the set of all unary and binary signatures, and let

$$\mathscr{M} = \{f : f \text{ is a Boolean signature and } f(\mathbf{x}) = 0 \text{ for every } \mathbf{x} \text{ with } \operatorname{wt}(\mathbf{x}) > 1\}.$$

Definition F.3.2 (Transformability)**.** Let $\mathscr{C}$ be closed under nonzero scalars and variable permutations. A finite signature set $\mathcal{F}$ is $\mathscr{C}$-transformable if one $M \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ satisfies

$$M^{\mathsf{T}} M \in \mathscr{C}, \qquad M^{-1}\mathcal{F} \subseteq \mathscr{C}.$$

Equivalently, for $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$, one matrix $U = K^{-1}M$ satisfies

$$U^{\mathsf{T}} X U \in \mathscr{C}, \qquad U^{-1}\widehat{\mathcal{F}} \subseteq \mathscr{C}.$$

Thus one matrix must satisfy the edge condition and the conditions for all signatures. The tractability criterion uses this definition for $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$.

The equivalence in Definition F.3.2 follows directly from $K^{\mathsf{T}}K = X$. For $U = K^{-1}M$,

$$U^{\mathsf{T}} X U = M^{\mathsf{T}} M, \qquad U^{-1}\widehat{\mathcal{F}} = M^{-1}\mathcal{F}.$$

Conversely, $M = KU$, which proves the equivalence.

F.3.3 The Eulerian-orientation predicate

The first two tractable cases use two different operations on signatures in K-coordinates. The central restriction keeps the arity and retains only the zero-charge entries. For an arity-r signature g, define

$$g|_{\mathrm{EO}}(\mathbf{x}) = g(\mathbf{x})\mathbf{1}[q_r(\mathbf{x}) = 0].$$

It is zero for odd r, and zero restrictions are deleted. For a signature set $\mathcal{F}$, set

$$\mathcal{F}|_{\mathrm{EO}} = \{g|_{\mathrm{EO}} : g \in \mathcal{F},\ g|_{\mathrm{EO}} \neq 0\}.$$

When applying the central restriction to a closed instance over a common one-sided set, an occurrence with zero central restriction makes the instance value zero. The deletion above is a convention for the signature set.

Let $g \neq 0$ have arity k and be supported on the single Hamming layer of weight d. To balance its charge while retaining its coefficients, its *EO completion* appends pinning signatures in the following order:

$$g_{\to \mathrm{EO}} = \begin{cases} g \otimes \Delta_0^{\otimes(2d-k)}, & 2d \geq k, \\ g \otimes \Delta_1^{\otimes(k-2d)}, & 2d < k. \end{cases} \tag{F.7}$$

The original variables precede the added unaries. A zeroth power is the nullary unit. The completed signature has even arity and central support, hence is EO. For a signature set of nonzero single-layer signatures, write

$$\mathcal{F}_{\to\mathrm{EO}} = \{g_{\to\mathrm{EO}} : g \in \mathcal{F}\}.$$

In the evaluation reduction, a component whose fixed local charges do not sum to zero contributes zero. Otherwise, the appended zero and one variables are paired by X-edges within each charge-balanced component, giving a unique extension of each original assignment [MWX25, Section 6.2, proof of Corollary 68]. This construction uses the explicit completion tensors in an auxiliary EO instance. It requires no assumption that the input signature set realizes the added pinnings.

For arity-$2d$ EO signature g, an oriented perfect pairing $\pi = ((a_1, b_1), \ldots, (a_d, b_d))$ has unordered pairs partitioning $[2d]$. For $\mathbf{z} \in \mathbb{F}_2^d$, let $\mathbf{y} \in \mathbb{F}_2^{2d}$ be the unique string with $y_{a_j} = z_j$ and $y_{b_j} = 1 - z_j$. The pairing induces

$$g^\pi(z_1, \ldots, z_d) = g(\mathbf{y}).$$

Definition F.3.3 (The predicate $\mathsf{EOTract}$)**.** For a finite EO signature set $\mathcal{F}$, $\mathsf{EOTract}(\mathcal{F})$ holds if there exist a sign $\varepsilon \in \{+1, -1\}$ and a class $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}\}$, both shared by the entire signature set, such that:

1. for every $g \in \mathcal{F}$ and all $\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma} \in \mathrm{supp}(g)$, if $\boldsymbol{\delta} = \boldsymbol{\alpha} \oplus \boldsymbol{\beta} \oplus \boldsymbol{\gamma}$, then $\boldsymbol{\delta} \in \mathrm{supp}(g)$ or $\varepsilon q_{\mathrm{arity}(g)}(\boldsymbol{\delta}) > 0$.
2. for every $g \in \mathcal{F}$ and every oriented perfect pairing π of its variables, $g^\pi \in \mathscr{C}$.

In particular, $\mathsf{EOTract}(\varnothing)$ holds.

The common-sign condition follows [MWX25, Theorem 26]. The compressed signature g^π expresses the opposite-pair restriction used to define the affine and product-type EO classes in [MWX24, Definition 22]. To compare the two forms, reorder the original variables as $(a_1, \ldots, a_d, b_1, \ldots, b_d)$, with assignments $(\mathbf{z}, \mathbf{w})$. The full restriction, extended by zero away from opposite pairs, is

$$R_\pi g(\mathbf{z}, \mathbf{w}) := g(\mathbf{z}, \mathbf{w}) \prod_{j=1}^{d} X(z_j, w_j) = g^\pi(\mathbf{z}) \prod_{j=1}^{d} X(z_j, w_j).$$

For $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}\}$, adding these disequality factors gives $g^\pi \in \mathscr{C} \Rightarrow R_\pi g \in \mathscr{C}$. Conversely, $\sum_{\mathbf{w} \in \mathbb{F}_2^d} R_\pi g(\mathbf{z}, \mathbf{w}) = g^\pi(\mathbf{z})$: each factor $X(z_j, w_j)$ admits exactly one value of w_j. The affine and product normal forms are preserved under this elimination, so $R_\pi g \in \mathscr{C} \Rightarrow g^\pi \in \mathscr{C}$, as detailed in [LM26a, Section 2.10.2]. Reversing a pair flips the corresponding compressed variable, and reordering pairs permutes the compressed variables. Both operations preserve $\mathscr{A}$ and $\mathscr{P}$, so using oriented pairings retains the source condition and its single shared class.

F.3.4 The tractability predicate

For an arbitrary finite signature set, evaluate Tract after the zero and nullary preprocessing of Section F.2.3. For a preprocessed signature set $\mathcal{F}$, put $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$. Define $\mathsf{Tract}(\mathcal{F})$ to be the disjunction of the following seven alternatives, which may overlap. The first two test the K-coordinate set $\widehat{\mathcal{F}}$. The remaining five are stated for the original-coordinate set $\mathcal{F}$. Transformability and the EO condition are defined in Definitions F.3.2 and F.3.3, respectively.

(1) The signature set $\widehat{\mathcal{F}}$ is common one-sided, that is,

$$[\forall \widehat{f} \in \widehat{\mathcal{F}},\ \mathrm{supp}(\widehat{f}) \subseteq HW^{\geq}_{\mathrm{arity}(\widehat{f})}] \quad \text{or} \quad [\forall \widehat{f} \in \widehat{\mathcal{F}},\ \mathrm{supp}(\widehat{f}) \subseteq HW^{\leq}_{\mathrm{arity}(\widehat{f})}],$$

and $\mathsf{EOTract}(\widehat{\mathcal{F}}|_{\mathrm{EO}})$ holds.

(2) Every $\widehat{f} \in \widehat{\mathcal{F}}$ is supported on one Hamming weight d_f. Across the signature set, some support string has negative charge and some (possibly other) support string has positive charge, and $\mathsf{EOTract}(\widehat{\mathcal{F}}_{\to \mathrm{EO}})$ holds, with the completion operation defined in Equation (F.7).

(3) $\mathcal{F} \subseteq \langle \mathscr{T} \rangle$.

(4) $\mathcal{F} \subseteq \langle K\mathscr{M} \rangle$, or $\mathcal{F} \subseteq \langle KX\mathscr{M} \rangle$.

(5) $\mathcal{F}$ is $\mathscr{A}$-transformable.

(6) $\mathcal{F}$ is $\mathscr{P}$-transformable.

(7) $\mathcal{F}$ is $\mathscr{L}$-transformable.

In Item (4), one of the two inclusions must hold for the entire signature set.

Remark F.3.4. For a nonzero positive-arity signature f,

$$f \in \langle K\mathscr{M} \rangle \iff \text{every irreducible factor of } K^{-1}f \text{ lies in } \mathscr{M},$$
$$f \in \langle KX\mathscr{M} \rangle \iff \text{every irreducible factor of } XK^{-1}f \text{ lies in } \mathscr{M}.$$

Zero and nullary signatures are covered by the tensor-closure conventions of Section F.2.5 and by the preprocessing in Section F.2.3.

F.3.5 Decidability and polynomial-time evaluation

Theorem F.3.5 (Recognition and evaluation)**.** *For every finite algebraic signature set $\mathcal{F}$, the predicate* $\mathsf{Tract}(\mathcal{F})$*, with the preprocessing convention of Section F.2.3, is decidable. If it holds, then* $\mathrm{Holant}(\mathcal{F}) \in \mathsf{FP}$.

Proof. This is [LM26a, Theorem 2.22]. □

F.4 Auxiliary Reductions

The height arguments use the reductions in this section to turn particular gadget outputs into a classification of the whole signature set. Factor extraction and binary interpolation supply additional signatures, while access to a nonzero unary gives a dichotomy. The EO and equality results then handle the corresponding outputs of the height constructions. Availability by reduction is understood as in Equation (F.4).

F.4.1 Factor Extraction and Preservation of Tractability

The support alternatives of Section F.3.4 determine when factors may be added to the current signature set. We first state the EO classification needed for the common-one-sided exception, then record factor extraction and preservation of the tractability criterion under the reductions.

Theorem F.4.1 (Weighted Eulerian orientations [MWX25, Theorem 26], [GSS26, Theorem 1.1])**.** *For every finite algebraic EO signature set $\mathcal{F}$, if* $\mathsf{EOTract}(\mathcal{F})$ *holds, then* $\text{K-Holant}(\mathcal{F}) \in \mathsf{FP}$. *Otherwise* $\text{K-Holant}(\mathcal{F})$ *is* $\#\mathsf{P}$*-hard.*

Theorem F.4.2 (Tensor-factor extraction [MWXZ25, full version, Theorem 32 and Remark 39; conference version, Theorem 29])**.** *Let $\mathcal{F}$ be a finite signature set and let $h = a \otimes b$ be a nonzero tensor product. Unless $\mathcal{F} \cup \{h\}$ is common one-sided,*

$$\text{K-Holant}(\mathcal{F}, a, b) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}, h).$$

In the common-one-sided exception, K-Holant$(\mathcal{F}, h)$ is in FP when $\mathsf{EOTract}((\mathcal{F} \cup \{h\})|_{\mathrm{EO}})$ holds and is #P-hard otherwise, by charge conservation and Theorem F.4.1.

F.4.1.1 Irreducible factors

The common-one-sided case is already classified by Theorem F.4.1. In the remaining case, Theorem F.4.2 permits tensor-factor extraction, so the analysis of individual signatures reduces to their irreducible factors. The iterations below also retain previously constructed signatures. For those arguments, we use the following finite closure.

For a preprocessed finite signature set $\mathcal{F}$, choose an exact irreducible factorization of every member and let $\Pi(\mathcal{F})$ be the finite set of all irreducible factors, with their ordered variable blocks. We call $\mathcal{F}$ *factor-closed* when $\Pi(\mathcal{F}) \subseteq \mathcal{F}$.

Lemma F.4.3 (Factor closure)**.** *If $\mathcal{F}$ is not common one-sided, then*

$$\text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F} \cup \Pi(\mathcal{F})). \tag{F.8}$$

Proof. Iterate Theorem F.4.2, retaining the original set at each step so that the common-one-sided exception cannot occur. This is the argument of [LM26a, Lemma 2.23]. □

Repeated factorization terminates because every proper factor has smaller arity than its parent. Each original signature can be reconstructed as the tensor product of its factors, with the recorded scalar and variable order. Substituting these products in an instance therefore gives

$$\text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \text{K-Holant}(\Pi(\mathcal{F}))$$

outside the common-one-sided case. Thus initial preprocessing may replace the signature set by its irreducible factors, still denoted by $\mathcal{F}$. This assumption applies to the initial set: later gadget outputs may be reducible and must be processed again.

Some later iterations retain both earlier signatures and their factors. When such an iteration requires factor closure, we instead use $\mathcal{F} \cup \Pi(\mathcal{F})$, retaining $\mathcal{F}$ for the current set after this explicit replacement.

F.4.1.2 Taking subsets and changing basis

The next two lemmas transfer the tractability criterion through taking subsets and through a common change of basis.

Lemma F.4.4 (Preservation under taking subsets)**.** *If $\mathcal{F} \subseteq \mathcal{F}'$ are preprocessed signature sets and $\mathsf{Tract}(K\mathcal{F}')$ holds, then $\mathsf{Tract}(K\mathcal{F})$ holds.*

Proof. This is [LM26a, Lemma 2.24]. □

Lemma F.4.5 ($GO(X)$-invariance of tractability)**.** *Let $\mathcal{F}$ be a finite preprocessed signature set and let $D \in GO(X)$. Then*

$$\mathsf{Tract}(K\mathcal{F}) \quad \Longleftrightarrow \quad \mathsf{Tract}\big(K(D\mathcal{F})\big).$$

Proof. This is [LM26a, Lemma 2.25]. □

F.4.2 Binary Interpolation

For an ordered binary B, the transfer $T_B = BX$ records composition along X-edges, as in Section F.2.7. The lemma below has two ways to obtain a unary: the singular case uses factor extraction, while the infinite-order case first supplies a rank-one binary by interpolation. Both conclusions are reductions over the original signature set.

Lemma F.4.6 (Binary interpolation)**.** *Let $\mathcal{F}$ be a finite algebraic signature set whose supported doubled charges include both signs, let $0 \neq B \in \mathcal{F}$ be ordered binary, with transfer $T_B = BX$.*

1. *If $B = u \otimes v$ is singular, where $u, v \neq 0$, then*
$$\text{K-Holant}(\mathcal{F}, u, v) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$
2. *If T_B is nonsingular and has infinite projective order, there is a nonzero rank-one binary H such that*
$$\text{K-Holant}(\mathcal{F}, H) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$

Proof. This is [LM26a, Lemma 3.1]. □

F.4.3 Single-occurrence Reductions and Nonzero Unaries

Theorem F.4.7 (Odd-arity Holant [MWXZ25, full version, Theorem 33; conference version, Theorem 30])**.** *Let $\mathcal{F}$ be a finite preprocessed algebraic signature set containing an odd-arity signature. Then* Holant$(\mathcal{F}) \in \mathsf{FP}$ *if* $\mathsf{Tract}(\mathcal{F})$ *holds, and is* $\#\mathsf{P}$*-hard otherwise.*

Its first two alternatives were originally in $\mathsf{FP}^{\mathsf{NP}}$. The stated FP conclusion follows from [GSS26, Theorem 1.1 and Corollary 1.2].

The single-occurrence replication argument below originated in an expanded version of [MWXZ25] that is not publicly available. The version we use is also stated in [LM26a, Lemma 3.3]. We reproduce it with its ordered variable connections explicit.

If a nonzero odd-arity signature is realized or made available by a reduction, this dichotomy settles that case. The same holds in K-coordinates by Equation (F.2). After zero and nullary preprocessing, the remaining analysis may therefore assume that every signature in $\mathcal{F}$ has even arity. In particular, there are no unaries. Write

$$\Delta = \Delta_0 \otimes \Delta_1,$$

with the displayed order of its two variables. For a signature g, the superscripts $g^{=1}$ and $g^{\leq 1}$ mean that an oracle instance contains, respectively, exactly one or at most one g-vertex.

Lemma F.4.8 (Reduction with at most one Δ-vertex)**.** *For every fixed finite algebraic signature set $\mathcal{F}$,*
$$\text{K-Holant}(\mathcal{F}, \Delta^{\leq 1}) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$

Proof. This is [LM26a, Lemma 3.2]. □

The next lemma turns one-occurrence access to g into unrestricted access to h. Each copy of a fixed gadget propagates the boundary signature g and exposes an output block with signature h. A fixed nonzero evaluation then closes the retained end of the chain, as illustrated in Figure F.1. The lemma applies separately to ordinary Holant and to K-Holant.

Lemma F.4.9 (Single-occurrence replication)**.** *Let $\mathcal{F}$ be a fixed finite algebraic signature set, and let g and h be nonzero signatures of arities r and s, respectively. For ordinary Holant, all connections below use $=_2$. Suppose that*

1. $\mathrm{Holant}(\mathcal{F}, g^{=1}) \le_{\mathrm{T}} \mathrm{Holant}(\mathcal{F})$.
2. *there is a fixed gadget C over $\mathcal{F}$ whose dangling edges are the three disjoint ordered blocks*

$$A = (a_1, \dots, a_r), \qquad P = (p_1, \dots, p_r), \qquad Y = (y_1, \dots, y_s).$$

Connecting one g-vertex coordinatewise to A realizes

$$\rho\, g(P) h(Y)$$

on P, Y, for a known $\rho \in \overline{\mathbb{Q}}^{\times}$. Moreover, for consecutive copies C_i, C_{i+1}, the block P_i can be connected coordinatewise to A_{i+1}, with no other connections. Thus a chain of k copies, beginning with one g-vertex, contains exactly that one g-vertex and realizes

$$\rho^k g(P_k) \prod_{i=1}^{k} h(Y_i),$$

where the blocks $Y_1, \dots, Y_k$ are pairwise disjoint.

3. *there is a fixed instance J over $\mathcal{F} \cup \{g\}$, containing exactly one g-vertex, whose exactly computable partition value $Z(J)$ is nonzero.*

Then

$$\mathrm{Holant}(\mathcal{F}, h) \le_{\mathrm{T}} \mathrm{Holant}(\mathcal{F}).$$

For K-Holant, if all connections use X and assumption 1 is $\text{K-Holant}(\mathcal{F}, g^{=1}) \le_{\mathrm{T}} \text{K-Holant}(\mathcal{F})$*, then the same gadget and nonzero-evaluation hypotheses give*

$$\text{K-Holant}(\mathcal{F}, h) \le_{\mathrm{T}} \text{K-Holant}(\mathcal{F}).$$

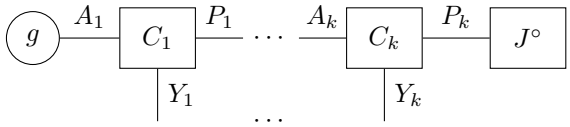
Proof. This is [LM26a, Lemma 3.3]. An input with no h-vertex is already an instance over $\mathcal{F}$. Let Ω be an input with $k \ge 1$ occurrences of h. Delete those vertices, retain their incident edges as k ordered boundary blocks, and attach the blocks $Y_1, \dots, Y_k$ of the displayed chain coordinatewise in the corresponding orders. Let J° be the network obtained by deleting the unique g-vertex of J, and join its boundary coordinatewise to P_k in the variable order inherited from that removed vertex. Denote the resulting instance by Ω'. It contains exactly one g-vertex, and the two tensor identities give

$$Z(\Omega') = \rho^k Z(J) Z(\Omega).$$

Divide by the known nonzero scalar $\rho^k Z(J)$ and apply assumption 1. The gadgets C and J and the scalars ρ and $Z(J)$ are fixed. The replacement uses k copies of C, so its size grows linearly with the number of replaced vertices, and the scalar recovery uses exact algebraic arithmetic. □

Lemma F.4.10 (Dichotomy with a nonzero unary)**.** *Let $\mathcal{F}$ be a finite algebraic signature set, and let $u \neq 0$ be unary. If $u \in \mathcal{F}$, then* $\mathrm{Holant}(\mathcal{F})$ *is* #P*-hard unless* $\mathsf{Tract}(\mathcal{F})$ *holds, and* $\text{K-Holant}(\mathcal{F})$ *is* #P*-hard unless* $\mathsf{Tract}(K\mathcal{F})$ *holds.*

Proof. The ordinary-coordinate assertion is the unary case of Theorem F.4.7. For the K-coordinate assertion, apply the same theorem after the transformation in Equation (F.2), which preserves the arity and nonzeroness of the unary signature. This is the argument recorded in [LM26a, Lemma 3.4]. □



ordered boundary blocks of the removed h-vertices

Figure F.1: The replacement in Lemma F.4.9. Each horizontal connection represents r edges in the specified variable order. Each Y_i represents s edges attached to the original boundary of one removed h-vertex. The network contains only the one displayed g-vertex. Here J° is obtained by removing the unique g-vertex from J.

Corollary F.4.11 (Higher equality supplies unaries)**.** *Let $d \geq 4$, and let $\mathcal{F}$ be a finite algebraic signature set containing $(=_d)$. Then*

$$\text{Holant}(\mathcal{F} \cup \{\Delta_0, \Delta_1\}) \leq_{\text{T}} \text{Holant}(\mathcal{F}),$$
$$\text{K-Holant}(\mathcal{F} \cup \{\Delta_0, \Delta_1\}) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$

Proof. Both reductions follow from [LM26a, Corollary 3.6]. Restriction supplies the reverse reductions. □

Together with Lemma F.4.10, these reductions settle the complexity of the original problem in either setting.

F.4.4 Hardness from an Eulerian-orientation Signature

The height constructions produce EO signatures inside a set that may also contain non-EO signatures. We show that, under the assumptions stated before Theorem F.4.13, an EO signature outside the matching-product class defined below forces hardness.

All gadget connections in this subsection use the edge X.

Arrow-reversal symmetry (ARS) means $h(\mathbf{x}) = \overline{h(\bar{\mathbf{x}})}$ for every input $\mathbf{x}$. An EO signature h has *projective arrow-reversal symmetry* (projective ARS) if, for some $c \in \overline{\mathbb{Q}}^\times$,

$$(c^{-1}h)(\mathbf{x}) = \overline{(c^{-1}h)(\bar{\mathbf{x}})} \qquad \text{for every } \mathbf{x} \in \{0,1\}^{\text{arity}(h)}.$$

Here $\bar{\mathbf{x}}$ is bitwise complement and the overline on the coefficient is complex conjugation. An EO–ARS binary is both EO and ARS without rescaling.

Let M be a perfect matching of a finite variable set V. After choosing an orientation of every edge, write $\vec{M}$ for the resulting set of ordered pairs. A nonzero *matching product* on V is a function $f : \mathbb{F}_2^V \to \overline{\mathbb{Q}}$ of the form

$$f(\mathbf{x}) = \lambda \prod_{(u,v)\in\vec{M}} b_{uv}(x_u, x_v), \qquad \lambda \in \overline{\mathbb{Q}}^\times, \quad b_{uv} : \mathbb{F}_2^2 \to \overline{\mathbb{Q}}.$$

Let $\mathcal{B}_{\text{ARS}}$ consist of the nonzero scalar multiples of products on arbitrary perfect matchings, with an orientation chosen for each matched pair, of binary EO–ARS signatures

$$(0, a, \overline{a}, 0), \qquad a \in \overline{\mathbb{Q}}^\times. \tag{F.9}$$

We write $(\neq_4) = \delta_{0011} + \delta_{1100}$. Nonzero scalars and permutations of the output variables are suppressed according to the equivalence convention. The required matching-product factorization constrains both the support and the coefficients.

We first record how fixed direct gadgets over a subset apply to the ambient signature set.

Lemma F.4.12 (Replacing vertices by fixed gadgets in a larger signature set)**.** *Let $\mathcal{F}_0 \subseteq \mathcal{F}$ be finite algebraic signature sets in K-coordinates. Suppose that a finite sequence of fixed gadget constructions starting from $\mathcal{F}_0$ adds a finite set S using only fixed direct X-gadgets. Then the same derivation is valid over $\mathcal{F}$, and*

$$\text{K-Holant}(\mathcal{F} \cup S) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$

If, at some stage, the problem K-Holant$(\mathcal{F}_0')$*, with $\mathcal{F}_0' \subseteq \mathcal{F}_0 \cup S$, is $\#\mathsf{P}$-hard, then* K-Holant$(\mathcal{F})$ *is $\#\mathsf{P}$-hard.*

Proof. Each direct gadget is substituted independently at every occurrence of its output signature, leaving vertices labelled from $\mathcal{F} \setminus \mathcal{F}_0$ unchanged. Induction over the finite sequence of fixed gadget constructions proves the displayed reduction. For the final claim, restriction and that reduction give

$$\text{K-Holant}(\mathcal{F}_0') \leq_{\text{T}} \text{K-Holant}(\mathcal{F} \cup S) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$

The lemma applies to the fixed direct gadgets in a literature proof. Tensor-factor extraction is supplied separately by Theorem F.4.2. □

The common-one-sided case and the tractable cases have already been handled. For the following theorem and Lemmas F.4.14 to F.4.16, fix a finite preprocessed algebraic set $\mathcal{F}$ of even-arity signatures in K-coordinates satisfying:

(i) Among the values $\chi_{\text{arity}(f)}(\mathbf{x})$, with $f \in \mathcal{F}$ and $\mathbf{x} \in \text{supp}(f)$, at least one is positive and at least one is negative.

(ii) $\neg\mathsf{Tract}(K\mathcal{F})$.

Finite enlargements retain witnesses of both charge signs, so (i) persists. By Lemma F.4.4, (ii) also persists under enlargement. A nonzero unary then gives hardness by Lemma F.4.10.

Theorem F.4.13 (Hardness from an EO signature outside $\mathcal{B}_{\text{ARS}}$)**.** *Assume (i)–(ii). Let $h \in \mathcal{F}$ be an EO signature. Suppose there are no $\lambda \in \overline{\mathbb{Q}}^{\times}$, perfect matching P_h of its variables together with an orientation (i, j) of each matched pair, and coefficients $a_{ij} \in \overline{\mathbb{Q}}^{\times}$ such that*

$$h = \lambda \bigotimes_{(i,j)\in P_h} (0, a_{ij}, \overline{a_{ij}}, 0)_{ij}.$$

Then K-Holant$(\mathcal{F})$ *is $\#\mathsf{P}$-hard.*

The proof first obtains projective ARS or hardness, then proves hardness when the signature satisfying projective ARS lies outside $\mathcal{B}_{\text{ARS}}$.

Lemma F.4.14 (Quaternary EO CP signatures supply a unary)**.** *Under the assumptions preceding Theorem F.4.13, suppose $D \in \mathcal{F}$, where*

$$D = p\delta_{\mathbf{s}} + q\delta_{\bar{\mathbf{s}}}, \qquad \mathbf{s} \in \{0,1\}^4, \quad \text{wt}(\mathbf{s}) = 2, \quad p, q \in \overline{\mathbb{Q}}^{\times}.$$

Then there is a nonzero unary u such that

$$\text{K-Holant}(\mathcal{F}, u) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$

Proof. First reduce D to the unweighted quaternary CP signature. After a variable permutation take $\mathbf{s} = 0101$, and connect the same last two variable positions of two copies through X-edges. The resulting ordered quaternary is

$$\begin{aligned} Q(x_1, x_2, y_1, y_2) &= \sum_{u,v \in \mathbb{F}_2} D(x_1, x_2, u, v) D(y_1, y_2, 1-u, 1-v) \\ &= pq(\delta_{0110} + \delta_{1001}). \end{aligned}$$

Thus, in output order (x_1, y_2, x_2, y_1), the output is the nonzero scalar multiple $pq\,(\neq_4)$. Hence $(\neq_4)$ is directly realizable over $\mathcal{F}$, up to the displayed nonzero scalar.

Put $E = (\neq_4)$. Joining the fourth variables of $E(a_1, a_2, a_3, u)$ and $E(b_1, b_2, b_3, v)$ through $X(u, v)$ realizes, in output order $(a_1, a_2, b_3, a_3, b_1, b_2)$, the six-ary generalized disequality

$$\delta_{000111} + \delta_{111000}.$$

Attach the Δ_0-variable of one Δ-vertex through X to a_1, and its Δ_1-variable through X to a_3. This selects the string 111000. In remaining-variable order (b_1, a_2, b_2, b_3), the output is $\delta_{0101} = \Delta \otimes \Delta$. In the notation of Lemma F.4.9, delete the unique Δ-vertex to obtain a gadget with input block $A = (a_1, a_3)$ and output blocks $P = (b_1, a_2)$, $Y = (b_2, b_3)$. Connecting a Δ-vertex to A realizes $\Delta(P)\Delta(Y)$. Connect P of each copy to A of the next, in the displayed orders, with no other connections. A chain of k copies therefore uses exactly one Δ-vertex and realizes $\Delta(P_k) \prod_{i=1}^{k} \Delta(Y_i)$ on pairwise disjoint output blocks. Closing a single Δ-vertex by X has value one. The reductions in Lemmas F.4.8 and F.4.9 therefore give

$$\text{K-Holant}(\mathcal{F}, \Delta) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$

Now use $\Delta = \Delta_0 \otimes \Delta_1$. Since both charge signs occur, the common-one-sided exception in Theorem F.4.2 is impossible. Factor extraction and the preceding reduction give

$$\text{K-Holant}(\mathcal{F}, \Delta_0, \Delta_1) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}). \qquad \square$$

Write $\mu_m = \{\omega \in \overline{\mathbb{Q}} : \omega^m = 1\}$ for $m \geq 1$.

Lemma F.4.15 (Projective ARS or hardness)**.** *Under the assumptions preceding Theorem F.4.13, let $h \in \mathcal{F}$ be an EO signature. Then either there is a $c \in \overline{\mathbb{Q}}^{\times}$ such that*

$$(c^{-1}h)(\mathbf{x}) = \overline{(c^{-1}h)(\bar{\mathbf{x}})} \qquad (\mathbf{x} \in \{0,1\}^{\text{arity}(h)}),$$

or K-Holant$(\mathcal{F})$ *is* #P*-hard.*

Proof. We first show that projective ARS can be witnessed by an algebraic scalar. Suppose an algebraic signature h_0 is ARS after multiplication by an arbitrary nonzero complex scalar. Then, for some $\rho \in \mathbb{C}^{\times}$,

$$h_0(\mathbf{x}) = \rho\, \overline{h_0(\bar{\mathbf{x}})}.$$

Choose $\mathbf{x}$ with $h_0(\mathbf{x}) \neq 0$. The same identity implies $h_0(\bar{\mathbf{x}}) \neq 0$, and hence

$$\rho = \frac{h_0(\mathbf{x})}{\overline{h_0(\bar{\mathbf{x}})}} \in \overline{\mathbb{Q}}^{\times}, \qquad \rho\overline{\rho} = 1.$$

Set $c = 1 + \rho$ if $\rho \neq -1$, and set $c = i$ if $\rho = -1$. Then $c \in \overline{\mathbb{Q}}^{\times}$ and $c/\overline{c} = \rho$, so $c^{-1}h_0$ is ARS with the required algebraic scalar.

We next handle the binary case. Write $h = (0, a, b, 0)$. If $ab = 0$, then h is a nonzero singular binary, so Lemma F.4.6 and factor extraction give a nonzero unary. By $\neg\mathsf{Tract}(K\mathcal{F})$ and Lemma F.4.10, this is hard. Suppose $ab \neq 0$, and put

$$T_h = hX = \mathrm{diag}(a, b).$$

If T_h has infinite projective order, Lemma F.4.6 first gives a rank-one binary. Applying its singular case and factor extraction again gives a unary and hence hardness. Otherwise a/b is a root of unity, so $|a| = |b|$. With $\rho = a/\bar{b}$, one has $h(\mathbf{x}) = \rho\overline{h(\bar{\mathbf{x}})}$ for all $\mathbf{x}$, and the preceding scalar argument proves the required projective ARS conclusion.

Assume now that $\mathrm{arity}(h) \geq 4$. Let L be the number field generated over $\mathbb{Q}$ by all entries of $\mathcal{F}$ and their complex conjugates. It is a finite extension of $\mathbb{Q}$. Every output of a fixed direct X-gadget over signatures already generated from h has entries in L, and every nonzero ratio used to normalize such an output also lies in L.

Run the generating process of [MWX25, arXiv v2, Definition 29] starting from $\{h\}$, with X as its initial binary. All numbered references to this paper below use the same version. Each self-loop and each binary path in that definition is a fixed direct X-gadget. We therefore add its output to $\mathcal{F}$ by Lemma F.4.12. Zero outputs are discarded. If a generated nonzero binary is singular, the singular case of Lemma F.4.6 gives unary factors. If it is nonsingular but its transfer has infinite projective order, the other case of that lemma gives a rank-one binary and then unary factors. Either event is hard by $\neg\mathsf{Tract}(K\mathcal{F})$ and Lemma F.4.10. The original witnesses of both charge signs exclude the factor-extraction exception at every stage.

Otherwise every generated nonzero binary is nonsingular and, up to a nonzero scalar, has the form

$$B_r = \begin{pmatrix} 0 & 1 \\ r & 0 \end{pmatrix}, \qquad r \in L^\times, \qquad B_r X = \mathrm{diag}(1, r),$$

where r is a root of unity. Let R_i be the parameter set represented by the family $B_i(h)$ at stage i. The path clause in Definition 29 makes each nonempty R_i the subgroup generated by the root-of-unity parameters available at that stage. Repeated self-loop descent [MWX25, Lemma 39] supplies nonemptiness. To make the monotonicity explicit, write the stage operation as Φ. Its self-loop and path construction is monotone under inclusion of its input parameter set. Finite order and the positive-power path clause give $1 \in R_1$, so $R_0 = \{1\} \subseteq R_1$. Inductively, $R_i = \Phi(R_{i-1}) \subseteq \Phi(R_i) = R_{i+1}$. Equality at one stage forces equality thereafter by the same operation. Every R_i lies in the group of roots of unity of L, which is finite. Consequently this increasing chain stabilizes at some finite stage i_0.

For a unique $k \geq 1$, the stabilized binary family is

$$B(h) = B_{i_0}(h) = \{B_\omega : \omega \in \mu_k\}.$$

Thus every gadget construction invoked below is finite, as required by Lemma F.4.12.

If $k \geq 3$, [MWX25, Lemmas 40–42] show that h is ARS up to a nonzero scalar. The opening scalar argument completes this case.

Suppose $k = 2$, and write $Y = B_{-1} = (0, 1, -1, 0)$. Both X and Y are available in this construction, and Y is realized by a direct gadget. The inductive argument in the proof of [MWX25, Lemma 47] uses fixed X- or Y-self-loops. At arity four, the proofs of [MWX25, Lemmas 45–46], before invoking their general problem reduction, construct either an EO quaternary q for which the singleton problem is hard by [CFX18, Theorem 3.1], or a nonzero quaternary CP signature

$$D = p\delta_{\mathbf{s}} + q'\delta_{\bar{\mathbf{s}}}, \qquad \mathbf{s} \in \mathbb{F}_2^4, \quad \mathrm{wt}(\mathbf{s}) = 2, \quad pq' \neq 0.$$

The two-copy subcase realizes $\neq_4$ itself. All these signatures arise by the fixed self-loops and two-copy gadgets just identified, so Lemma F.4.12 adds them relative to $\mathcal{F}$. The hard quaternary gives hardness by restriction. The quaternary CP signature gives a unary by Lemma F.4.14, and then hardness by Lemma F.4.10. In the remaining case, the proof of Lemma 47 gives ARS up to a nonzero scalar, to which the opening argument applies.

Finally suppose $k = 1$. Follow the proof of [MWX25, Lemma 48], and let $C(h)$ denote its set of nonzero quaternary signatures obtained by successive X-self-loops. Every member of $C(h)$ is therefore available by Lemma F.4.12. Moreover, every binary generated from such a signature is also generated from h. Hence its generated binary set is contained in $B(h) = \{X\}$, and Lemma 39 makes that set nonempty. It is therefore exactly $\{X\}$. If a member of $C(h)$ lies outside $\{X^{\otimes 2}, Y^{\otimes 2}\}$, the proof of Lemma 45 reaches exactly one of the hard quaternaries or quaternary CP signatures described in the preceding paragraph. If $Y^{\otimes 2} \in C(h)$, add this signature and apply Theorem F.4.2. The presence of both charge signs excludes its exceptional case, so Y is available. The proof of Lemma 47 then applies because $B(h) \subseteq \{X, Y\}$ and both binaries are available. Here factor extraction supplies the reduction otherwise obtained from [MWX25, Lemma 38]. The case $C(h) = \varnothing$ is impossible for nonzero h by Lemma 39. In the remaining case $C(h) = \{X^{\otimes 2}\}$, the reorganizing argument in Lemma 48, together with [MWX25, Lemma 43], makes h a scalar multiple of a real complement-invariant signature, hence projectively ARS. Its real and imaginary parts enter only the vector identities in that argument. The opening scalar argument again makes the scalar algebraic. □

Lemma F.4.16 (Hardness outside $\mathcal{B}_{\mathrm{ARS}}$ under projective ARS)**.** *Under the assumptions preceding Theorem F.4.13, let $h \in \mathcal{F}$ be an EO signature for which there is a $c \in \overline{\mathbb{Q}}^{\times}$ satisfying*

$$(c^{-1}h)(\mathbf{x}) = \overline{(c^{-1}h)(\bar{\mathbf{x}})} \qquad (\mathbf{x} \in \{0,1\}^{\mathrm{arity}(h)}).$$

If there are no $\lambda \in \overline{\mathbb{Q}}^{\times}$, perfect matching P_h together with an orientation (i, j) of each matched pair, and $a_{ij} \in \overline{\mathbb{Q}}^{\times}$ such that

$$h = \lambda \bigotimes_{(i,j)\in P_h} (0, a_{ij}, \overline{a_{ij}}, 0)_{ij},$$

then K-Holant($\mathcal{F}$) *is* #P*-hard.*

Proof. Put $h' = c^{-1}h$, which is algebraic, EO, and ARS. By uniqueness of irreducible factorization and the ARS factorization lemma [CFS20a, Lemmas 2.13–2.14], choose an irreducible factorization of h', up to an overall scalar, into factors $g_1, \ldots, g_m$ on disjoint variable blocks, each satisfying ARS exactly. The coordinate restrictions called *pinnings* in the proof of [CFS20a, Lemma 2.14] are slices used to prove tensor-divisibility identities. They do not supply direct pinning gadgets.

We next choose algebraic representatives and restore exact ARS. For each j, choose $\boldsymbol{\alpha}_j \in \mathrm{supp}(g_j)$, let $\boldsymbol{\alpha} = \boldsymbol{\alpha}_1 \cdots \boldsymbol{\alpha}_m$, and define on the variable block of g_j

$$q_j(\mathbf{z}) = \frac{h'(\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_{j-1}, \mathbf{z}, \boldsymbol{\alpha}_{j+1}, \ldots, \boldsymbol{\alpha}_m)}{h'(\boldsymbol{\alpha})}.$$

The chosen support assignments make the denominator nonzero, and $q_j = g_j / g_j(\boldsymbol{\alpha}_j)$. Hence every q_j is algebraic, EO, irreducible, and projectively ARS, and

$$h' = h'(\boldsymbol{\alpha}) \bigotimes_{j=1}^{m} q_j.$$

For each q_j, the algebraic-scalar argument in Lemma F.4.15 gives $c_j \in \overline{\mathbb{Q}}^{\times}$ such that $p_j = c_j^{-1} q_j$ satisfies ARS exactly. Thus

$$h' = h'(\boldsymbol{\alpha}) \Big(\prod_{j=1}^{m} c_j \Big) \bigotimes_{j=1}^{m} p_j,$$

with an algebraic overall scalar and algebraic EO–ARS irreducible factors p_j. If every irreducible factor were binary, we would have $h \in \mathcal{B}_{\mathrm{ARS}}$, contrary to the hypothesis. Thus there is a nonbinary irreducible EO–ARS factor $f \notin \mathcal{B}_{\mathrm{ARS}}$. Iterated use of Theorem F.4.2 adds f to the current set. Since both charge signs occur, its common-one-sided exception is impossible. Whenever the cited descent subsequently extracts another factor, we repeat the displayed ratio construction and rescale by a nonzero algebraic scalar to obtain an ARS representative. Thus each signature used in the descent is a member of the current set or an extracted tensor factor.

Apply the proof of [CFS20a, Lemmas 3.2 and 4.1–4.7; Theorem 4.8] to the EO–ARS signature f. The signature-producing operations are local: the tensor connections in Lemma 3.2 and the contractions in Lemma 4.6 are fixed direct X-gadgets, while the selected-variable attachments of the fixed binary $iY = (0, i, -i, 0)$ in Lemmas 4.5–4.6 are direct gadgets after that binary factor has been extracted. These direct gadgets can be substituted over $\mathcal{F}$ by Lemma F.4.12. Every factor-extraction step is supplied separately by Theorem F.4.2, whose exception is again excluded by the presence of both charge signs. Sections 3–4 of the cited proof use no polynomial interpolation.

Concretely, Lemma 3.2 produces a direct EO quaternary whose singleton problem is hard, directly realizes $\neq_4$, or leaves the case treated in Lemmas 4.1–4.7. Using Lemmas 4.1–4.4, the descent in Lemmas 4.5–4.6 preserves nonmembership in $\mathcal{B}_{\mathrm{ARS}}$. Lemma 4.7 completes the base arities $4, 6, 8$. Hence the descent adds either a hard quaternary or $\neq_4$ by the reductions just described. The first gives hardness by restriction. The second gives a unary by Lemma F.4.14 and then hardness by Lemma F.4.10. □

Proof of Theorem F.4.13. Apply Lemma F.4.15. Either K-Holant($\mathcal{F}$) is #P-hard, or h is projectively ARS. In the latter case, $h \notin \mathcal{B}_{\mathrm{ARS}}$, so Lemma F.4.16 gives hardness. □

F.4.5 Walsh Coordinates

The equality reductions below and the height-one analysis use the same Walsh basis to describe tensor coefficients. For an arity-r signature f, with the column-vector convention of Section F.2.2, we fix the unnormalized Walsh matrix and the Walsh coordinates

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \widetilde{f} = (P^{-1})^{\otimes r} f. \tag{F.10}$$

Thus inversion is $f = P^{\otimes r} \widetilde{f}$. In coordinates,

$$f(\mathbf{x}) = \sum_{\mathbf{y} \in \mathbb{F}_2^r} (-1)^{\mathbf{x} \cdot \mathbf{y}} \widetilde{f}(\mathbf{y}).$$

Consequently, if $\widetilde{f}$ has even-parity support then f is *complement-invariant*, meaning $f(\bar{\mathbf{x}}) = f(\mathbf{x})$. If $\widetilde{f}$ has odd-parity support, then $f(\bar{\mathbf{x}}) = -f(\mathbf{x})$. We write $\mathbf{1}_{\mathrm{even}}$ for the even-parity indicator tensor, with its arity inferred from context.

For an ordered binary B, its Walsh transfer matrix is

$$P^{-1} T_B P = P^{-1}(BX)P = 2\widetilde{B}Z, \qquad \widetilde{B} = P^{-1} B (P^{-1})^{\mathsf{T}},$$

since $P^{\mathsf{T}}XP = 2Z$. Its Walsh transfer class is therefore $[P^{-1}T_BP] = [\widetilde{B}Z]$ when B is nonsingular.

These coordinates describe the same tensor contractions with edge kernel $2Z$, whose factor 2 is retained in the gadget calculations. This use of Walsh coordinates is distinct from a global transformation of the problem.

F.4.6 Equalities and Complementary-Pair Signatures

Three lemmas handle the equality signatures needed below. The first concerns a quaternary that becomes $=_4$ under a common transformation taking the edge to equality. The second is a generalized equality already available in K-coordinates. The third starts from a tensor factor with two complementary support strings in Walsh coordinates. Its proof reduces to the first form or obtains a nonzero unary by interpolation.

Lemma F.4.17 (Dichotomy from a transformed quaternary equality)**.** *Let $\mathcal{F}$ be a finite algebraic signature set containing an ordered quaternary signature g. Suppose there are*

$$U \in \mathrm{GL}_2(\overline{\mathbb{Q}}), \qquad \kappa, \lambda \in \overline{\mathbb{Q}}^{\times},$$

such that

$$(U^{-1})^{\mathsf{T}} X U^{-1} = \kappa(=_2), \qquad U^{\otimes 4} g = \lambda(=_4).$$

Then either K-Holant($\mathcal{F}$) *is* #P*-hard or* $\mathsf{Tract}(K\mathcal{F})$ *holds.*

Proof. This is [LM26a, Lemma 3.8]. □

Lemma F.4.18 (Dichotomy with a generalized equality)**.** *Let $\mathcal{F}$ be a finite algebraic signature set containing*

$$G_q = c_0\delta_{\mathbf{0}^q} + c_1\delta_{\mathbf{1}^q}, \qquad c_0, c_1 \in \overline{\mathbb{Q}}^{\times}, \qquad q \geq 4 \text{ an integer.}$$

Then either K-Holant($\mathcal{F}$) *is* #P*-hard or* $\mathsf{Tract}(K\mathcal{F})$ *holds.*

Proof. This is [LM26a, Lemma 3.9]. □

Lemma F.4.19 (Dichotomy from a CP factor in Walsh coordinates)**.** *Let $\mathcal{F}$ be a finite algebraic signature set whose supported doubled charges include both signs. Suppose a finite equivalent enlargement of $\mathcal{F}$ contains a nonzero signature with a tensor factor h of even arity $2k \geq 4$ with*

$$\widetilde{h} = (P^{-1})^{\otimes 2k} h = c_0\delta_{\mathbf{a}} + c_1\delta_{\bar{\mathbf{a}}}, \qquad \mathbf{a} \in \mathbb{F}_2^{2k}, \quad c_0c_1 \neq 0.$$

Then K-Holant($\mathcal{F}$) *is* #P*-hard or* $\mathsf{Tract}(K\mathcal{F})$ *holds. If c_1/c_0 is a root of unity, a finite equivalent enlargement contains a quaternary g satisfying the two transformation conditions of Lemma F.4.17 with $U = K$.*

Proof. Add the specified signature and then its factor h using Equation (F.4) and Theorem F.4.2. The supported charges of both signs exclude the common-one-sided exception. All subsequent gadgets are K-gadgets. In Walsh coordinates an edge is $P^{\mathsf{T}}XP = 2Z$.

Choose two positions at which $\mathbf{a}$ has the same bit. Relabel the two support strings, and their coefficients, so that this bit is zero in $\mathbf{a}$. Take two copies of h, leave these two positions exposed in each copy, and join all other corresponding positions between the copies. Write J for the joined positions. The diagonal edges allow only the two equally oriented pairs of support strings. Since $|J| = 2k - 2$ is even, their signs agree. Thus the quaternary gadget output q has

$$\widetilde{q} = \lambda(\delta_{0000} + \rho\delta_{1111}), \quad \rho = (c_1/c_0)^2, \quad \lambda = 2^{2k-2}(-1)^{\mathrm{wt}(\mathbf{a}|_J)}c_0^2 \neq 0.$$

Closing two variables of q gives a binary B with

$$\widetilde{B} = 2\lambda\,\mathrm{diag}(1,-\rho), \qquad P^{-1}(BX)P = 4\lambda\,\mathrm{diag}(1,\rho).$$

The multiplicative order of ρ now determines which reduction applies. If it is infinite, Lemma F.4.6 supplies a rank-one binary. Factor extraction supplies a nonzero unary, and Lemma F.4.10 gives the conclusion.

Otherwise let ℓ be the order of ρ. A chain of $\ell-1$ copies of B has Walsh transfer class $[\mathrm{diag}(1,\rho^{-1})]$. When $\ell = 1$ use the empty chain. Attaching this chain to one variable of q gives a gadget output g with $\widetilde{g} = c(\delta_{0000} + \delta_{1111})$, $c \neq 0$. Walsh inversion gives

$$g = 2c\mathbf{1}_{\mathrm{even}}, \qquad K^{\otimes 4} g = 4c(=_4), \qquad (K^{-1})^{\mathsf{T}} X K^{-1} = I.$$

These are the conditions of Lemma F.4.17 with $U = K$, so that lemma applies. The equivalent enlargements transfer hardness back to $\mathcal{F}$. Tractability passes to the original subset by Lemma F.4.4. □

F.5 Obtaining a Mountain

Starting from an even-arity signature set with supported half-charges of both signs, we seek either both pinning signatures or a signature with two nonzero endpoints and all remaining support on the central EO layer. The latter is a mountain, defined below. Its height will be the gcd of the nonzero supported half-charges of the entire input set. This choice also retains the divisibility condition on the input signatures needed in the later height analysis.

Definition F.5.1 (Mountain)**.** For an integer $d \geq 1$, an arity-$2d$ signature M in K-coordinates is a mountain of height d if

$$\mathrm{supp}(M) \subseteq HW^{=}_{2d} \cup \{\mathbf{0}^{2d}, \mathbf{1}^{2d}\} \qquad \text{and} \qquad M(\mathbf{0}^{2d})M(\mathbf{1}^{2d}) \neq 0.$$

Equivalently,

$$\{\chi_{2d}(\mathbf{x}) : M(\mathbf{x}) \neq 0\} \subseteq \{-d, 0, d\}, \qquad \{-d, d\} \subseteq \{\chi_{2d}(\mathbf{x}) : M(\mathbf{x}) \neq 0\}.$$

Thus the height is both half the arity and the absolute integer half-charge of either endpoint. No restriction is placed on the coefficients on the central Hamming layer.

Lemma F.5.2 (Producing the gcd-height signature or both pinning signatures)**.** *Let $\mathcal{F}$ be a finite preprocessed set of even-arity algebraic signatures. Assume that among the supported half-charges*

$$\chi_{\mathrm{arity}(f)}(\mathbf{x}) = \mathrm{wt}(\mathbf{x}) - \frac{\mathrm{arity}(f)}{2}, \qquad f \in \mathcal{F}, \quad f(\mathbf{x}) \neq 0,$$

there is at least one positive value and at least one negative value. Put

$$d = \gcd\left\{\left|\chi_{\mathrm{arity}(f)}(\mathbf{x})\right| : f \in \mathcal{F},\ f(\mathbf{x}) \neq 0,\ \chi_{\mathrm{arity}(f)}(\mathbf{x}) \neq 0\right\}.$$

Then $d \geq 1$, every supported half-charge $\chi_{\mathrm{arity}(f)}(\mathbf{x})$ lies in $d\mathbb{Z}$, and at least one of the following reductions holds.

1. *There is an arity-$2d$ signature M such that*

$$\text{K-Holant}(\mathcal{F}, M) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}),$$

$$M(\mathbf{0}^{2d})M(\mathbf{1}^{2d}) \neq 0, \qquad M(\mathbf{x}) = 0 \quad \textit{if } \mathrm{wt}(\mathbf{x}) \notin \{0, d, 2d\}.$$

2. Both pinning signatures are available by reduction:

$$\text{K-Holant}(\mathcal{F}, \Delta_0, \Delta_1) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$

Proof. Let

$$C = \{\chi_{\text{arity}(f)}(\mathbf{x}) : f \in \mathcal{F},\ f(\mathbf{x}) \neq 0,\ \chi_{\text{arity}(f)}(\mathbf{x}) \neq 0\}$$

be the finite set of nonzero supported half-charges of the input set. The gcd assertion follows from the definition of d. The next argument expresses d and $-d$ as sums of elements of C with nonnegative multiplicities. Tensor products add half-charges, but their factor multiplicities cannot be negative. Because C contains both signs, take every pair $a, -b \in C$ with $a, b > 0$. If $g = \gcd(a, b)$, assigning multiplicity b/g to a and a/g to $-b$ gives total half-charge zero. Summing these assignments over all positive–negative pairs gives a strictly positive multiplicity for each element of C, since each participates in at least one pair. We therefore obtain a vector $\mathbf{v} \in \mathbb{Z}_{>0}^{C}$ satisfying

$$\sum_{c \in C} v_c c = 0.$$

Bézout's identity, iterated over the finite set C, supplies an integer vector $\mathbf{u} \in \mathbb{Z}^C$ with $\sum_{c \in C} u_c c = d$ [Apo76, Section 1.3, Theorem 1.2]. The entries of $\mathbf{u}$ need not be nonnegative, but every entry of $\mathbf{v}$ is positive. Consequently, for all sufficiently large nonnegative integers k, both vectors

$$\mathbf{n}^+ = \mathbf{u} + k\mathbf{v}, \qquad \mathbf{n}^- = -\mathbf{u} + k\mathbf{v}$$

are nonnegative integer vectors, while their total half-charges remain d and $-d$, respectively. Thus $\mathbf{n}^+$ and $\mathbf{n}^-$ are the required multiplicity lists.

For each $c \in C$, choose a signature $f_c \in \mathcal{F}$ and an input $\mathbf{x}_c \in \text{supp}(f_c)$ with $\chi_{\text{arity}(f_c)}(\mathbf{x}_c) = c$. Form the two tensor products using n_c^+ and n_c^- copies of f_c, respectively. Concatenating the chosen inputs in these copies gives a nonzero product of signature values, with half-charge d in the first tensor product and $-d$ in the second. Every supported half-charge of either tensor product is divisible by d, since the same is true of each input signature.

For these two supported inputs, the excess of ones over zeros, or of zeros over ones, is $2d$. The endpoint-preserving self-loop lemma [MWXZ25, full version, Lemma 40; conference version, Lemma 37] therefore supplies sequences of X self-loops whose outputs P, N have arity $2d$ and satisfy $P(\mathbf{1}^{2d})N(\mathbf{0}^{2d}) \neq 0$. Both outputs are realized by fixed gadgets over $\mathcal{F}$. Each X self-loop removes one zero and one one from every contributing input, so the supported half-charges remain divisible by d. At arity $2d$, they lie between $-d$ and d, giving

$$\text{supp}(P) \cup \text{supp}(N) \subseteq HW_{2d}^{=} \cup \{\mathbf{0}^{2d}, \mathbf{1}^{2d}\}.$$

If either P or N has both endpoints, take that signature as the required M. We may therefore assume

$$P(\mathbf{0}^{2d}) = 0, \qquad N(\mathbf{1}^{2d}) = 0.$$

Apart from the two nonzero endpoints already identified, all supported inputs have weight d. Their central restrictions are $P|_{\text{EO}}$ and $N|_{\text{EO}}$.

If both central restrictions are nonzero, permute chosen central strings to $(\mathbf{0}^d, \mathbf{1}^d)$ in P and $(\mathbf{1}^d, \mathbf{0}^d)$ in N. Expose the first d variables of each signature and join their last d variables in the same order by X edges. Call the resulting arity-$2d$ signature Q. At the all-zero boundary, the support restriction and $P(\mathbf{0}^{2d}) = 0$ force the last block of P to be $\mathbf{1}^d$, and hence the last block of N

to be $\mathbf{0}^d$. At the all-one boundary, $N(\mathbf{1}^{2d}) = 0$ similarly forces the last block of N to be $\mathbf{0}^d$, and that of P to be $\mathbf{1}^d$. Thus each endpoint sum has exactly one nonzero term, and

$$Q(\mathbf{0}^{2d}) = P(\mathbf{0}^d, \mathbf{1}^d)N(\mathbf{0}^{2d}), \qquad Q(\mathbf{1}^{2d}) = P(\mathbf{1}^{2d})N(\mathbf{1}^d, \mathbf{0}^d),$$

both nonzero.

If only P has a nonzero central restriction, write $N = n\delta_{\mathbf{0}^{2d}}$ for $n \in \overline{\mathbb{Q}}^\times$, and permute a supported central string of P to $(\mathbf{0}^d, \mathbf{1}^d)$. Join the two d-variable blocks of N to the last d variables of two copies of P, again by X edges, leaving the first d variables of both copies exposed. The support of N forces its variables to zero and the connected variables of both copies of P to one. The endpoint extensions are therefore unique, with values $nP(\mathbf{0}^d, \mathbf{1}^d)^2$ and $nP(\mathbf{1}^{2d})^2$, both nonzero. If only N has a nonzero central restriction, use the same construction with the roles of P, N and of zero and one interchanged. In either construction, X charge conservation preserves divisibility by d. Since the output has arity $2d$, its supported half-charges belong to $\{-d, 0, d\}$. Take that output as M.

It remains to consider the case in which both central restrictions are zero. Then $P = p\Delta_1^{\otimes 2d}$ and $N = n\Delta_0^{\otimes 2d}$ for $p, n \in \overline{\mathbb{Q}}^\times$. The set $\mathcal{F} \cup \{P, N\}$ retains the original witnesses of positive and negative half-charge, so the common-one-sided exception in Theorem F.4.2 cannot occur. Applying that theorem to the pure tensor powers, with nonzero scalars handled by preprocessing, and substituting the gadgets for P, N proves the second reduction. □

In the pinning outcome of Lemma F.5.2, if $\neg\mathsf{Tract}(K\mathcal{F})$, then Lemmas F.4.4 and F.4.10 and the displayed reduction give #P-hardness of K-Holant($\mathcal{F}$). Thus the remaining analysis uses the mountain outcome.

Remark F.5.3. The pinning case in Lemma F.5.2 is necessary. Consider $\{\mathrm{EXACT}_3^4, \mathrm{EXACT}_1^4\}$, where $\mathrm{EXACT}_j^4(\mathbf{x}) = \mathbf{1}[\mathrm{wt}(\mathbf{x}) = j]$. These signatures have single supported half-charges 1 and -1, respectively. In a fixed gadget built directly from them using X edges, charge conservation fixes the half-charge of every supported boundary input: it is the number of EXACT_3^4-vertices minus the number of EXACT_1^4-vertices. Such a gadget cannot have both endpoints of a mountain. The lemma instead gives both pinning signatures through factor extraction.

F.6 Mountain Height One

A height-one mountain is an ordered binary signature whose two diagonal entries are nonzero. We prove the dichotomy for any finite algebraic signature set containing one. First, a nonzero unary settles the case by the odd-arity dichotomy. A nonzero singular binary, or a nonsingular binary whose transfer has infinite projective order, leads to the same conclusion by Lemma F.4.6.

In the remaining case, the projective transfers of all nonzero directly realizable binaries form a finite group. By Theorem F.6.8, we either obtain a nonsingular weighted equality or normalize this group to an odd cyclic or odd dihedral form. A weighted equality can be normalized to $I = (=_2)$ by a global diagonal transformation in $GO(X)$, so Theorem F.6.2 settles that branch.

For the two remaining group forms, contractions and factor extraction either give hardness, identify a common tractable basis, or yield a strict update: the transfer group grows in the original coordinates, or the minimum nonbinary irreducible arity decreases. A uniform bound over the initial number field makes these updates finite. We establish the common contraction tools, treat the dihedral and cyclic cases separately, and then combine them with the termination argument.

Theorem F.6.1 (Dichotomy with a height-one mountain)**.** *Let $\mathcal{F}$ be a finite algebraic signature set containing a height-one mountain. Then*

$$\mathsf{Tract}(K\mathcal{F}) \qquad \textit{or} \qquad \text{K-Holant}(\mathcal{F}) \textit{ is \#P-hard.}$$

We prove this theorem at the end of the section, after establishing the finite-group alternatives and the termination of their updates.

F.6.1 Binary equality

Binary equality gives the following classification, due to Liu and Meng [LM26a].

Theorem F.6.2 (Binary-equality K-Holant dichotomy)**.** *Let $\mathcal{F}$ be a finite algebraic signature set containing $I = (=_2)$. If $\mathsf{Tract}(K\mathcal{F})$ holds, then* K-Holant$(\mathcal{F}) \in \mathsf{FP}$*. Otherwise* K-Holant$(\mathcal{F})$ *is $\#\mathsf{P}$-hard. The predicate $\mathsf{Tract}(K\mathcal{F})$ is decidable.*

Proof. [LM26a, Theorem 1.2] is stated with a reduced four-clause predicate. Since $I \in \mathcal{F}$, its coordinate image contains $Z = K^{\otimes 2} I \in K\mathcal{F}$. Hence the binary-anchor lemma [LM26a, Lemma 2.13] identifies that reduced predicate with $\mathsf{Tract}(K\mathcal{F})$. The displayed formulation follows. □

F.6.2 Finite Transfer Groups

Series composition turns binary gadgets into matrix products. Passing to projective classes removes the nonzero scalars already recorded in the reductions. We use finite-group structure to find a weighted equality or restrict the remaining transfers to cyclic and dihedral forms. The edge and sign matrices remain fixed throughout this classification.

We use $H \le G$ for a subgroup, $H \triangleleft G$ for a normal subgroup, and $[G : H]$ for its index. An involution has order two. Our finite-group notation is

$$C_m, \qquad D_{2m} = \langle r, s \mid r^m = s^2 = 1,\ srs = r^{-1} \rangle, \qquad V_4 = C_2 \times C_2,$$

where C_m is cyclic of order m and D_{2m} has order $2m$. The symmetric and alternating groups are S_m and A_m. For $S \subseteq G$, the centralizer is $C_G(S) = \{g \in G : gs = sg \text{ for every } s \in S\}$. If ζ is a primitive m-th root of unity, then $[\mathbb{Q}(\zeta) : \mathbb{Q}] = \varphi(m)$, where φ is Euler's totient.

F.6.2.1 Ordered Bell signatures

The ordered Bell signatures are $\mathcal{B} = \{I, X, Y, Z\}$. With truth-table order $00, 01, 10, 11$, they are

$$\begin{aligned} I &= (1,0,0,1), \quad Z = (1,0,0,-1), \\ X &= (0,1,1,0), \quad Y = (0,1,-1,0). \end{aligned} \tag{F.11}$$

Our real skew matrix Y differs from the conventional complex Pauli Y-matrix by a nonzero scalar.

Remark F.6.3 (Bell notation in Shao–Cai). The tuple in Equation (F.11) is Shao–Cai's $(=_2^+, =_2^-, \neq_2^+, \neq_2^-)$ [SC20, Section 2]. $Y^{\mathsf{T}} = -Y$. Their holographic matrix named Z is our K, whereas our Z is only the diagonal sign matrix.

Recall from Section F.2.7 that $T_B = BX$, $T_{BXC} = T_B T_C$, and $T_{B^{\mathsf{T}}} = (T_B)^{\#}$. For a general $T \in \mathrm{Mat}_2(\overline{\mathbb{Q}})$, the operation defined in Section F.2.7 satisfies

$$T^{\#} = XT^{\mathsf{T}}X, \qquad T^{\#} = (\det T) Z T^{-1} Z \quad \text{if } T \text{ is nonsingular.} \tag{F.12}$$

Remark F.6.4 (Relation to binary-signature classifications). Lemma F.4.6 handles nonzero singular binaries and nonsingular binaries whose transfer $T_B = BX$ has infinite projective order, leaving the finite-order case. This is the K-Holant form of the binary classification used by Cai–Lu–Xia [CLX09, CLX18]. The four ordered Bell signatures in $\mathcal{B}$ represent the projective Bell classes of Shao–Cai [SC20]. Beyond the binary classification, we need the complete transfer group and uniform control of its higher-arity contractions.

Lemma F.6.5 (Bounded torsion)**.** *Fix a number field L, and let G be the set of projective transfers of all directly realizable nonsingular ordered binary gadgets over a fixed signature set with entries in L. If $G \neq \varnothing$ and every element of G has finite projective order, then G is a finite subgroup of $\mathrm{PGL}_2(L)$.*

Proof. This is [LM26a, Lemma 5.2]. □

For a current signature set $\mathcal{F}$, suppose that every nonzero directly realizable ordered binary B is nonsingular and that $[BX]$ has finite order. Series composition and Lemma F.6.5 then give the finite group

$$G(\mathcal{F}) = \{[BX] : B \neq 0 \text{ is an ordered binary directly realizable over } \mathcal{F}\}. \tag{F.13}$$

We call this the *complete transfer group*, since it includes the transfer classes of all directly realizable nonsingular binaries over $\mathcal{F}$.

We next classify this finite group. Changes of basis and comparisons between successive groups are handled in Section F.6.3.

Proposition F.6.6 (Finite projective binary groups)**.** *Every finite subgroup of $\mathrm{PGL}_2(\mathbb{C})$ is conjugate to exactly one of the following types:*

$$C_m, \qquad D_{2m}, \qquad A_4, \qquad S_4, \qquad A_5,$$

where $m \geq 1$ for C_m, $m \geq 2$ for D_{2m}, $|C_m| = m$, $|D_{2m}| = 2m$, and $|A_4| = 12$, $|S_4| = 24$, $|A_5| = 60$.

Proof. This is the classical conjugacy classification [Bea83, Section 5.1], also stated in [LM26a, Proposition 2.1]. □

Lemma F.6.7 (Finite projective-group facts)**.** *The following properties will be used below.*

1. *In*
$$D_{2m} = \langle r, s \mid r^m = s^2 = 1,\ srs = r^{-1} \rangle,$$
the rotations $\langle r \rangle \cong C_m$ form a normal index-two subgroup, and every element outside it is a reflection and an involution. If m is odd, the rotation group is the unique index-two subgroup. If m is even, the only nonidentity rotational involution is the central element $r^{m/2}$.

2. *The group A_4 has no index-two subgroup, and the centralizer of an involution is V_4.*

3. *The unique index-two subgroup of S_4 is A_4. Every involution is a transposition or a double transposition. Precisely the double transpositions belong to A_4.*

4. *The group A_5 is simple and has no index-two subgroup. The centralizer of each involution is V_4.*

5. *The projective centralizer of the distinguished involution is*
$$C_{\mathrm{PGL}_2(\mathbb{C})}([Z]) = \left\{ \left[\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \right] : a, b \in \mathbb{C},\ ab \neq 0 \right\} \sqcup \left\{ \left[\begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix} \right] : a, b \in \mathbb{C},\ ab \neq 0 \right\}.$$
Thus $[Z]$ is the only nonidentity diagonal involution, while every nonsingular anti-diagonal class is an involution.

Proof. The first four items follow from the cyclic and dihedral presentations and the standard permutation models [DM96]. The projective centralizer calculation in the last item is given in [LM26a, Lemma 2.2]. □

The classification identifies the abstract group type. To use it in K-Holant, we must also preserve the edge through a change of basis in $GO(X)$. The proof below keeps the distinguished classes $[X]$ and $[Z]$ fixed and constructs the required normalization explicitly.

Let $G \le \mathrm{PGL}_2(\overline{\mathbb{Q}})$ be finite, and suppose each class has a directly realized ordered-binary representative. Choose one nonsingular ordered binary B_g for each class g, normalized by $R_g = B_g X$, $[R_g] = g$, and $R_1 = I$, where $1 = [I]$. The matrix of every representative of class g is ρB_g for a unique nonzero scalar ρ. Fix one direct gadget for each of these finitely many representatives and substitute it whenever that representative is used. Displayed projective identities are read up to their known nonzero scalars. A scalar is shown explicitly whenever its zero/nonzero status or a coefficient ratio is used.

Theorem F.6.8 (Weighted equality or an odd cyclic or dihedral group)**.** *Let $\mathcal{F}$ be a finite algebraic signature set containing the height-one mountain*

$$B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad ad \neq 0,$$

and suppose every nonzero directly realizable binary over $\mathcal{F}$ is nonsingular and its transfer has finite projective order. Then one of the following holds.

1. *A nonsingular weighted equality $E = \mathrm{diag}(p, q)$, with $pq \neq 0$, is directly realizable over $\mathcal{F}$, up to a known nonzero scalar.*

2. *There exist an odd $N \ge 3$ and $D \in GO(X)$ such that the global normalization $\mathcal{F} \mapsto D\mathcal{F}$ puts $G(D\mathcal{F})$ in exactly one of the forms*

$$\{[R(s)] : s \in \mu_N\}, \qquad \{[R(s)], [ZR(s)] : s \in \mu_N\}, \qquad R(s) = \frac{1}{2}\begin{pmatrix} 1+s & 1-s \\ 1-s & 1+s \end{pmatrix}.$$

   Every displayed projective class has a fixed directly realizable ordered-binary representative over $D\mathcal{F}$.

For $s \in \overline{\mathbb{Q}}$, the displayed matrix family satisfies $P^{-1}R(s)P = \mathrm{diag}(1, s)$.

Proof. Put $G = G(\mathcal{F})$ and

$$H = \langle G, [Z] \rangle .$$

All gadget entries lie in the number field generated by the entries of $\mathcal{F}$, so Lemma F.6.5 makes G finite. If $[T] \in G$, swapping the two variables and Equation (F.12) give $[ZT^{-1}Z] \in G$. Since inversion permutes G, it follows that $[Z]G[Z] = G$. Consequently

$$G \triangleleft H, \qquad [H : G] \le 2,$$

and $H = G$ or $H = G \sqcup [Z]G$, so H is finite.

For the distinguished mountain,

$$T = BX = \begin{pmatrix} b & a \\ d & c \end{pmatrix}$$

has class $[T] \in G$. Since $a, d \neq 0$, both off-diagonal entries of T are nonzero. An anti-diagonal transfer will give the weighted equality we seek. Suppose $[S] \in G$ has an anti-diagonal representative

$$S = \begin{pmatrix} 0 & p \\ q & 0 \end{pmatrix}, \qquad pq \neq 0.$$

By the definition of G, there is a directly realizable ordered binary B_S and a scalar $\rho \neq 0$ such that $B_S X = \rho S$. Hence

$$B_S = \rho S X = \rho \operatorname{diag}(p, q),$$

which is the nonsingular weighted equality in outcome 1.

We now use Proposition F.6.6 and Lemma F.6.7. Their use here is only group-theoretic: the distinguished class is the fixed $[Z]$, and no conjugating matrix from the abstract classification is applied to the signature set.

If H is cyclic, it is abelian, so $[T]$ commutes with $[Z]$. The projective centralizer of $[Z]$ consists of the diagonal and anti-diagonal classes. The two nonzero off-diagonal entries of T exclude the diagonal case. Thus $[T] \in G$ is anti-diagonal and gives the weighted equality.

Suppose $H \cong A_4$. Since A_4 has no index-two subgroup, the index bound forces $G = H$. The centralizer of $[Z]$ in H is a V_4, so it contains an involution $u \neq 1, [Z]$. The class u belongs to G and commutes with $[Z]$. Because $[Z]$ is the only nonidentity diagonal involution, u is anti-diagonal. The same argument applies when $H \cong A_5$: simplicity gives $G = H$, and the centralizer of an involution is again V_4.

Suppose $H \cong S_4$. If $G = H$ and $[Z]$ corresponds to a transposition, choose the disjoint transposition. If $G = H$ and $[Z]$ corresponds to a double transposition, choose either of the other two double transpositions. In both cases the chosen involution lies in G, commutes with $[Z]$, and differs from it, so it is anti-diagonal. If $G \neq H$, then the unique index-two subgroup gives $G = A_4$. Here $[Z] \notin G$ is an involution outside A_4, hence a transposition. If z' is the disjoint transposition, then $u = [Z]z'$ is a double transposition in $A_4 = G$. It commutes with $[Z]$, differs from it, and is therefore anti-diagonal.

It remains to consider

$$H = D_{2m} = \left\langle r, z : r^m = z^2 = 1,\ zrz = r^{-1} \right\rangle,$$

where the distinguished reflection is denoted by $z = [Z]$ when $[Z]$ is a reflection. If m is even and $[Z] = r^{m/2}$ is the central half-turn, then $[T] \in G$ commutes with $[Z]$. Its two nonzero off-diagonal entries again make it anti-diagonal. Suppose instead that $[Z]$ is a reflection and m is even. The two involutions

$$r^{m/2}, \qquad [Z]r^{m/2}$$

commute with $[Z]$ and differ from it. If $G = H$, both belong to G. If $[H : G] = 2$, then $[Z] \notin G$, and the two displayed elements lie in opposite cosets of G. Hence exactly one belongs to G. In either event G contains an involution different from $[Z]$ that commutes with $[Z]$, and this involution is anti-diagonal.

The sole unresolved case is therefore

$$H \cong D_{2N}, \qquad N \geq 3 \text{ odd}, \qquad [Z] \text{ a reflection},$$

with $G = C_N$ or $G = D_{2N}$. Indeed, for odd N the rotation subgroup is the unique index-two subgroup. Thus a proper G is the rotation subgroup, while otherwise $G = H$.

Choose a rotation class $[R]$ of exact order N, and write

$$R = \begin{pmatrix} u & v \\ w & x \end{pmatrix}.$$

The dihedral relation gives $[ZRZ] = [R^{-1}]$, while

$$ZRZ = \begin{pmatrix} u & -v \\ -w & x \end{pmatrix}, \qquad R^{-1} \doteq \begin{pmatrix} x & -v \\ -w & u \end{pmatrix}.$$

The entries v, w cannot both vanish: otherwise $[R] = [R^{-1}]$, so $[R]^2 = 1$, contrary to the odd order $N \geq 3$. Since at least one of v, w is nonzero, comparison in the preceding projective equality forces the projective scalar to be 1, and hence $u = x$. Exactly one of v, w cannot vanish. In that case nonsingularity gives $u = x \neq 0$, and a projective rescaling makes R a nonidentity unipotent matrix. Its powers are pairwise distinct in characteristic zero. Therefore

$$R = \begin{pmatrix} \alpha & \beta \\ \gamma & \alpha \end{pmatrix}, \qquad \beta\gamma \neq 0.$$

Choose $q \in \overline{\mathbb{Q}}^{\times}$ with $q^2 = \gamma/\beta$, and put

$$D = \mathrm{diag}(q, 1).$$

Then

$$D^{\mathsf{T}} X D = qX,$$

so $D \in GO(X)$. Moreover $DZD^{-1} = Z$. With $\delta = q\beta = q^{-1}\gamma$,

$$DRD^{-1} = \begin{pmatrix} \alpha & \delta \\ \delta & \alpha \end{pmatrix}.$$

In Walsh coordinates,

$$P^{-1}(DRD^{-1})P = \mathrm{diag}(\alpha + \delta, \alpha - \delta).$$

Both eigenvalues are nonzero because R is nonsingular. Put

$$s_0 = \frac{\alpha - \delta}{\alpha + \delta}.$$

The exact projective order of $[R]$ implies that s_0 is a primitive N-th root of unity, and

$$DRD^{-1} = (\alpha + \delta)R(s_0), \qquad R(s) = \frac{1}{2}\begin{pmatrix} 1+s & 1-s \\ 1-s & 1+s \end{pmatrix}.$$

Since $R(s)R(t) = R(st)$, the normalized rotation classes are precisely

$$\{[R(s)] : s \in \mu_N\}.$$

The diagonal matrix D fixes $[Z]$. Thus, when $G = C_N$, these are all normalized classes, and, when $G = D_{2N}$, the remaining classes are precisely

$$\{[ZR(s)] : s \in \mu_N\}.$$

To identify the normalized group exactly, we compare gadget outputs in both directions. If B_0 is a directly realizable ordered binary over $\mathcal{F}$, the global change of basis gives the boundary signature B_0^D over $D\mathcal{F}$ with

$$B_0^D \doteq DB_0D^{\mathsf{T}}, \qquad (DB_0D^{\mathsf{T}})X = qD(B_0X)D^{-1}.$$

Hence $T_{B_0^D} \doteq DT_{B_0}D^{-1}$. Applying the same argument with $D^{-1} \in GO(X)$ gives the converse, so

$$G(D\mathcal{F}) = [D]G(\mathcal{F})[D]^{-1}.$$

Every displayed class therefore has an ordered binary representative realizable over $D\mathcal{F}$, with the normalization $D \in GO(X)$. □

F.6.3 Invariants and Finite Updates

A later contraction or factor extraction can enlarge the transfer group, so the group classification may need to be applied again. To prove that this process terminates, we compare all groups in the original coordinates over one fixed number field. We also retain the original height-one mountain under the accumulated change of basis. The following lemma records these invariants and transfers the final classification back to the initial signature set.

Lemma F.6.9 (Invariants under adding signatures and changing basis)**.** *Let $\mathcal{F}_0$ be a finite preprocessed algebraic signature set, and let L_0 be a number field containing its entries, their conjugates, and every algebraic constant fixed before the construction. Consider a sequence $(A_i, \mathcal{F}_i)$, in which every $\mathcal{F}_i$ is a finite preprocessed algebraic signature set, starting from the given $\mathcal{F}_0$ with $A_0 = I$, in which each step is one of the following.*

1. *Add signatures under the following conditions: set $A_{i+1} = A_i$, choose a finite $\mathcal{F}_{i+1} \supseteq \mathcal{F}_i$ with*

$$\text{K-Holant}(\mathcal{F}_{i+1}) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}_i),$$

   and require every newly added positive-arity signature h to satisfy

$$h \doteq A_i^{\otimes \mathrm{arity}(h)} h_0 \quad \text{for a tensor } h_0 \text{ over } L_0.$$

2. *A global normalization: choose $D_i \in GO(X)$, and set*

$$A_{i+1} = D_i A_i, \qquad \mathcal{F}_{i+1} = D_i \mathcal{F}_i,$$

   with the value rescaling from each transformed edge given by Equation (F.3).

Then, at every index i,

$$A_i \mathcal{F}_0 \subseteq \mathcal{F}_i, \qquad A_i^{\mathsf{T}} X A_i = \mu_i X \quad (\mu_i \in \overline{\mathbb{Q}}^{\times}), \qquad \text{K-Holant}(\mathcal{F}_i) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}_0), \tag{F.14}$$

and every positive-arity $h \in \mathcal{F}_i$ satisfies

$$h \doteq A_i^{\otimes \mathrm{arity}(h)} h_0 \quad \text{for a tensor } h_0 \text{ over } L_0.$$

At an index where every nonzero directly realizable ordered binary is nonsingular and has finite projective transfer order, let $G(\mathcal{F}_i)$ be the group of its projective transfer classes $[BX]$, and put

$$\overline{G}_i = [A_i]^{-1} G(\mathcal{F}_i) [A_i].$$

Then $\overline{G}_i \leq \mathrm{PGL}_2(L_0)$. On any consecutive sequence of such indices, a global normalization leaves $\overline{G}_i$ unchanged, whereas adding signatures can only enlarge it. Thus all groups compared in the finite-group argument lie in the single fixed field L_0. Any generic interpolation that leaves this field is used only when it completes the complexity classification of the current case.

If $\mathsf{Tract}(K\mathcal{F}_i)$ holds, then $\mathsf{Tract}(K\mathcal{F}_0)$ holds. Separately, if one matrix V satisfies the K-coordinate conditions of Definition F.3.2 for $\mathcal{F}_i$ and $\mathscr{C}$, where $\mathscr{C}$ is closed under nonzero scalar multiplication, then

$$W = A_i^{-1} V$$

satisfies the same $\mathscr{C}$-conditions for $\mathcal{F}_0$. In particular, this applies to $\mathscr{A}, \mathscr{P}, \mathscr{L}$.

Proof. This is [LM26a, Lemma 2.26], applied inductively to the two displayed kinds of step. When signatures are added, restriction supplies the reverse reduction. A normalization conjugates the complete transfer group by $[D_i]$, which cancels when the group is expressed in the original basis using $[A_{i+1}] = [D_i A_i]$. □

A new factor can advance the proof even when the transfer group stays unchanged. To measure this progress, for a factor-closed signature set $\mathcal{F}$ containing an irreducible signature of arity at least three, put

$$\nu(\mathcal{F}) = \min\{\text{arity}(h) : h \in \mathcal{F} \text{ is irreducible and } \text{arity}(h) \geq 3\}.$$

This parameter is used only after the branch in which every irreducible factor has arity at most two has been removed.

Definition F.6.10 (Equivalent updates with strict progress)**.** Let $(A, \mathcal{F})$ satisfy Lemma F.6.9 for the initial set $\mathcal{F}_0$ and its fixed number field L_0. An update to $(A^+, \mathcal{F}^+)$ must satisfy the same invariant and use only the additions of signatures and global changes of basis allowed there. In particular, $\mathcal{F}^+$ is finite, preprocessed, algebraic, and factor-closed, and

$$A^+\mathcal{F}_0 \subseteq \mathcal{F}^+, \qquad (A^+)^{\mathsf{T}} X A^+ = \mu_{A^+} X \quad (\mu_{A^+} \neq 0),$$
$$\text{K-Holant}(\mathcal{F}^+) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}).$$

The transformed height-one mountain and every old irreducible signature remain members up to the same invertible change of basis and their recorded nonzero scalars. Every nonzero directly realizable ordered binary over both sets is nonsingular with finite projective transfer order. Put

$$H = [A]^{-1} G(\mathcal{F})[A], \qquad H^+ = [A^+]^{-1} G(\mathcal{F}^+)[A^+].$$

Both groups are subgroups of $\mathrm{PGL}_2(L_0)$. When both values of ν are defined, the update is required to satisfy

$$H \subseteq H^+, \qquad \nu(\mathcal{F}^+) \leq \nu(\mathcal{F}), \qquad H \neq H^+ \text{ or } \nu(\mathcal{F}^+) \neq \nu(\mathcal{F}).$$

We call such an update a *strict update*. Thus at least one comparison is strict. If no nonbinary irreducible signature remains, the arity-at-most-two alternative gives tractability and no further update is made.

Finiteness of each individual group does not by itself bound the number of updates. The fixed number field gives a uniform bound on group order, which we combine with the decreasing arity parameter.

Lemma F.6.11 (Termination of the strict updates)**.** *Fix the number field L_0 in Lemma F.6.9. There is no infinite sequence*

$$(H_0, \nu_0), (H_1, \nu_1), \ldots$$

such that every H_i is a finite subgroup of $\mathrm{PGL}_2(L_0)$, every ν_i is an integer at least three, and every update satisfies

$$H_i \subseteq H_{i+1}, \qquad \nu_{i+1} \leq \nu_i,$$

with at least one of these two inequalities strict. In the height-one construction these quantities are

$$H_i = [A_i]^{-1} G(\mathcal{F}_i)[A_i], \qquad \nu_i = \min\{\text{arity}(f) : f \in \mathcal{F}_i \text{ is irreducible and } \text{arity}(f) \geq 3\}.$$

If the second set is empty, the arity-at-most-two case is tractable and no further strict update is made.

Proof. Write $\delta = [L_0 : \mathbb{Q}]$. If $[T] \in \mathrm{PGL}_2(L_0)$ has projective order m, the ratio of the two eigenvalues of T is a primitive m-th root of unity ζ_m. The eigenvalues lie in an extension of L_0 of degree at most two, whence

$$\varphi(m) = [\mathbb{Q}(\zeta_m) : \mathbb{Q}] \le [L_0(\zeta_m) : \mathbb{Q}] \le 2\delta.$$

There are therefore only finitely many possible projective element orders. Let M_0 be their maximum. By Proposition F.6.6, every finite subgroup of $\mathrm{PGL}_2(L_0)$ has order at most

$$C_0 = \max\{2M_0, 60\}.$$

For a sequence in the statement, define

$$\Phi_i = (C_0 - |H_i|) + (\nu_i - 3) \in \mathbb{Z}_{\ge 0}.$$

The inclusions and inequalities make Φ_i nonincreasing, and the strict condition decreases it by at least one at every update. Hence the sequence has at most Φ_0 updates. In the application, Lemma F.6.9 places every group in the original basis in $\mathrm{PGL}_2(L_0)$, adding signatures makes the groups nested, and Definition F.6.10 makes ν nonincreasing. Factor closure between updates is finite because every nontrivial tensor split replaces a signature by positive-arity factors of strictly smaller arity. It is not part of the displayed measure. □

When neither a classification nor a strict update occurs, the remaining signatures will share a tractable basis. For either group with odd N, we will show that the single matrix P fixed in Equation (F.10) satisfies

$$(P^{-1})^{\otimes \mathrm{arity}(f)} f \in \mathscr{P} \quad (f \in \mathcal{F}), \qquad P^{\mathsf{T}} X P = 2Z \in \mathscr{P}. \tag{F.15}$$

These conditions show that $K\mathcal{F}$ is $\mathscr{P}$-transformable by Definition F.3.2.

To analyze the remaining branch, we first process contractions and their factors. The condition below describes the case in which these factors are binary and their transfer classes lie in the current group. Fix a signature set $\mathcal{F}$ for which $G(\mathcal{F})$ is defined, a subgroup $H \le G(\mathcal{F})$, and one directly realizable ordered binary B_h with $[B_h X] = h$ for each $h \in H$. We will refer to the following contraction condition for $f \in \mathcal{F}$:

(C) For every ordered pair of distinct variables of f, every chosen B_h, and both kernel orientations XB_hX and $XB_h^{\mathsf{T}}X$, the contraction is zero, or all its irreducible factors are nonsingular binaries B with $[BX] \in H$.

Any nullary factors are absorbed into the overall scalar. The conditions are in the current K-coordinates. After a Walsh change of coordinates the group membership is conjugated by P as well. Applications below take $H = G(\mathcal{F})$, with the complete transfer group from Equation (F.13). Before using condition (C), any factor that gives hardness, tractability, or an update in Definition F.6.10 is handled separately. The next lemma specifies the finite family of outputs and factors retained during this processing. We denote its finite closure under these contractions and factor extractions by $\mathcal{F}^{\mathrm{cl}}$.

Lemma F.6.12 (Finite recursive contraction and factor extraction)**.** *Let $(A, \mathcal{F})$ be a stage of a sequence satisfying Lemma F.6.9, over its fixed number field L_0. Suppose that $\mathcal{F}$ is finite, preprocessed, algebraic, and factor-closed, contains the transformed height-one mountain, and every nonzero directly realizable ordered binary over $\mathcal{F}$ is nonsingular and has finite projective transfer order. Let $H = G(\mathcal{F})$, and fix one directly realizable ordered-binary representative of each class in this finite group.*

Start with every nonbinary irreducible factor currently in $\mathcal{F}$. For each factor f, process every choice of an ordered pair of distinct variables, a fixed representative of a class in H, and either kernel orientation. Add the resulting nonzero contraction of two variables and, through Lemma F.4.3, all of its irreducible factors. Apply the same procedure to each new nonbinary factor. This finite recursive procedure reaches one of the following outcomes:

1. K-Holant($\mathcal{F}$) *is* #P*-hard or* $\mathsf{Tract}(K\mathcal{F})$ *holds.*
2. *there is a strict update* $(A, \mathcal{F}) \mapsto (A^+, \mathcal{F}^+)$ *satisfying Definition F.6.10.*
3. *there is a finite preprocessed algebraic factor-closed set* $\mathcal{F}^{\mathrm{cl}} \supseteq \mathcal{F}$*, with*

$$\text{K-Holant}(\mathcal{F}^{\mathrm{cl}}) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}), \qquad G(\mathcal{F}^{\mathrm{cl}}) = H,$$

for which the coordinate matrix remains A. Every indicated contraction of two variables has been processed, every nonbinary irreducible factor supplied by factor closure is retained in $\mathcal{F}^{\mathrm{cl}}$, and every binary factor it produces is nonsingular with transfer class in H. A nonbinary factor produced from a parent f has arity at most $\mathrm{arity}(f) - 2$.

Outcome 3 is closure under all the indicated contractions and factor extractions, with the same group and coordinate matrix. It may preserve ν, so it need not be a strict update.

Proof. Begin with the finitely many nonbinary irreducible factors in $\mathcal{F}$. For each factor, process its finite set of ordered-variable, representative, and orientation choices. For each choice, add the resulting contraction and all irreducible factors under Equations (F.4) and (F.8). The transformed mountain rules out the common-one-sided factor exception.

An odd-arity factor gives outcome 1 by Theorem F.4.7 and Equation (F.2). A singular binary, or a nonsingular binary whose transfer has infinite projective order, is handled by Lemmas F.4.6 and F.4.10. After each addition, consider all directly realizable nonzero ordered binaries. If any is singular or its transfer has infinite projective order, the preceding argument gives outcome 1. Otherwise the group is defined, and an element outside H strictly enlarges the group in the original basis, and a new nonbinary irreducible factor of arity below $\nu(\mathcal{F})$ strictly decreases ν. These give outcome 2, with both comparisons preserved as in Definition F.6.10. If neither occurs, retain all the binary factors and repeat the procedure on each new nonbinary irreducible factor.

If no classification or strict update occurs, the group and its chosen representatives stay fixed. There are finitely many initial factors and finitely many choices at each step. Every newly produced nonbinary factor has arity at most the arity of its parent minus two. The recursive depth is therefore bounded, so only finitely many factors and contraction choices occur. If neither of the first two outcomes occurs, processing all these choices gives outcome 3. Each addition keeps the transformed mountain in the set and satisfies the fixed-field and scalar conditions of Lemma F.6.9. □

Thus the recursion terminates by arity descent, while Lemma F.6.11 bounds the number of strict updates between such recursions.

F.6.4 Contractions for the odd cyclic and dihedral groups

For tensors A, B of the same arity, a pencil is a family $q_s = A - sB$. Its parameter range is specified at each use.

We now turn the contraction condition into restrictions on the signatures themselves. The next two lemmas constrain four coefficients at a time and then the parity of the whole Walsh support. Both apply in the cyclic and dihedral cases. We use the Walsh coordinates $\widetilde{f}$ defined in Equation (F.10). Each contraction below is realized by an X-edge gadget. The matrix P expresses

the same output in Walsh coordinates, where the dependence on the phase is linear. Representative scalars are retained in the gadget reduction.

F.6.4.1 Contraction kernels and the four-coefficient relation

Fix odd $N \geq 3$. With $R(s)$ as in Theorem F.6.8, the rotation and reflection classes are

$$[R(s)], \qquad [ZR(s)], \qquad s \in \mu_N. \tag{F.16}$$

The Walsh forms are

$$P^{-1}R(s)P = D(s), \qquad P^{-1}ZP = X.$$

For binary chains L, R, with free inputs indexing rows and joined inputs indexing columns, joining their second inputs through one X-edge gives LXR^{T}. Fix a direct binary-chain representative of each projective class in Equation (F.16). Contracting two variables then gives the following Walsh kernels, up to a nonzero scalar:

$$\mathsf{K}^{+}(s) = D(-s) = \begin{pmatrix} 1 & 0 \\ 0 & -s \end{pmatrix}, \qquad \mathsf{K}^{-}(s) = D(-s)X = \begin{pmatrix} 0 & 1 \\ -s & 0 \end{pmatrix}, \qquad s \in \mu_N. \tag{F.17}$$

We call contraction with $\mathsf{K}^{+}(s)$ a *phase contraction* and contraction with $\mathsf{K}^{-}(s)$ a *reflection contraction*. Their nonzero entries select equal and unequal bits, respectively. They correspond to the rotation and reflection classes above. Phase contractions occur in both groups, while reflection contractions occur only in the dihedral group. In Walsh coordinates, swapping the two inputs acts on a transfer matrix T by

$$P^{-1}(PTP^{-1})^{\#}P = ZT^{\mathsf{T}}Z.$$

This map fixes $D(s)$ and sends $XD(s)$ to $-sXD(s^{-1})$. It therefore preserves the projective rotation and reflection classes. Attaching binary representatives to the variables adds no kernels, so Equation (F.17) lists every oriented kernel needed below.

The next coefficient relation and lemma hold for every integer $N \geq 3$. For such an N and $a, b, c, d \in \overline{\mathbb{Q}}$, let $\mathcal{R}_N(a, b, c, d)$ mean that, for some $\alpha \in \mu_N$, one of the following three systems holds:

$$c = -\alpha a, \qquad d = -\alpha b, \qquad \text{(proportional)}, \tag{F.18}$$

$$b = c = 0, \qquad d = \alpha a, \qquad \text{(diagonal monomial)}, \tag{F.19}$$

$$a = d = 0, \qquad c = \alpha b, \qquad \text{(anti-diagonal monomial)}. \tag{F.20}$$

Lemma F.6.13 (Phase-pencil relation)**.** *For every integer $N \geq 3$, the relation $\mathcal{R}_N(a, b, c, d)$ is equivalent to the following condition: for every $s \in \mu_N$,*

$$\operatorname{diag}(a - sb, c - sd)Z \quad \textit{is zero or is proportional to } D(\eta) \textit{ for some } \eta \in \mu_N.$$

Proof. This is [LM26a, Lemma 7.1]. □

F.6.4.2 Support parity

Lemma F.6.14 (Walsh parity from contractions)**.** *Let $N \geq 3$ be odd, and let H be the cyclic or dihedral group in Walsh coordinates, respectively*

$$H = \{[D(s)] : s \in \mu_N\} \quad \text{or} \quad H = \{[D(s)], [XD(s)] : s \in \mu_N\}.$$

Let $f \neq 0$ have positive even arity. At arity two, suppose the Walsh transfer class $[P^{-1}(fX)P]$ belongs to H. At arity at least four, suppose that every irreducible factor of every nonzero contraction of f using an ordered pair of distinct variables, a chosen binary representative with Walsh transfer class in H, and either kernel orientation is supported on one Walsh parity class. In the cyclic case suppose that class is even. Then $\widetilde{f}$ is supported on one parity class, which is even in the cyclic case.

Proof. At arity two, a Walsh binary is, up to a nonzero scalar, a diagonal or anti-diagonal transfer times Z, and hence has one parity. Only the diagonal alternative occurs in the cyclic group. At larger arity, the tensor product of the irreducible factors of a nonzero contraction has one parity. In the cyclic case it is even.

At arity four, each $\mathsf{K}^+(s)$-contraction is zero, diagonal, or anti-diagonal, and every entry is affine in s. Among three phases, either two contractions vanish, annihilating the pencil, or two nonzero contractions have the same support parity, which forces the opposite entries to vanish identically. In the only remaining pattern—one zero, one diagonal, and one anti-diagonal contraction—each opposite pair acquires two roots, with the same conclusion. Hence every equal-bit slice has one parity. For two supported strings $\mathbf{x}, \mathbf{y}$ of opposite parity, there are positions $i \neq j$ with $(x_i, y_i) = (x_j, y_j)$, unless the four pairs (x_k, y_k), $k \in [4]$, exhaust $\mathbb{F}_2^2$. In the latter case both strings have weight two, contradicting their opposite parities. In the former case, deleting positions i, j contradicts the parity of the equal-bit slices. In the cyclic case, odd Walsh parity would realize a reflection and is therefore excluded.

For arity at least six, fix distinct variables i, j. The phase contraction is

$$\widetilde{h}_s(\mathbf{y}) = \lambda_s(\widetilde{f}(00, \mathbf{y}) - s\widetilde{f}(11, \mathbf{y})), \qquad \lambda_s \neq 0.$$

A zero contraction makes all others proportional. Otherwise two phases have the same contraction parity. Every coordinate in the opposite parity then vanishes at two distinct phases, so its affine coefficient vanishes identically. The two nonzero phases also recover the slices A, B with the common parity.

In the cyclic case, delete two equal coordinates of any supported string and choose a phase other than the at most one value cancelling its residual coefficient. The resulting Walsh contraction has even-parity support. In the dihedral case, two supported strings $\mathbf{x}, \mathbf{y}$ of opposite parity at arity at least six have positions $i \neq j$ with $(x_i, y_i) = (x_j, y_j)$. At most one phase cancels each residual coefficient, so $|\mu_N| \geq 3$ leaves a phase preserving both strings. Deleting the chosen pair preserves their opposite parities, contradicting the common slice parity. □

For every f covered by Lemma F.6.14, Walsh duality gives the equivalent complement relation in the K-coordinates,

$$f(\bar{\mathbf{x}}) = f(\mathbf{x}) \quad \text{or} \quad f(\bar{\mathbf{x}}) = -f(\mathbf{x}),$$

with only the plus sign in the cyclic subcase.

F.6.4.3 Quaternary classification under phase contractions

Fix $N \geq 3$ and a quaternary signature q in the current set, and put $\widetilde{q} = (P^{-1})^{\otimes 4} q$. When $\widetilde{q}$ is supported on the even Hamming-weight positions in Walsh coordinates, write

$$p = \widetilde{q}_{0000}, \qquad t = \widetilde{q}_{1111}, \qquad x_{ab} = \widetilde{q}_{\mathbf{1}_{\{a,b\}}}. \tag{F.21}$$

For every ordering (i, j, k, l) of $[4]$, the phase-contraction matrix is

$$H^{+}_{ij}(s) = \begin{pmatrix} p - s x_{ij} & 0 \\ 0 & x_{kl} - st \end{pmatrix}.$$

When its nonzero Walsh transfer classes belong to C_N, Lemma F.6.13 gives $\mathcal{R}_N(p, x_{ij}, x_{kl}, t)$.

Lemma F.6.15 (Quaternary classification under phase contractions)**.** *Let $N \geq 3$, let $\mathcal{F}$ be a finite algebraic signature set in K-coordinates, and, for every $s \in \mu_N$, fix a gadget-realizable ordered binary representative C_s over $\mathcal{F}$, in a specified orientation, whose Walsh matrix is*

$$\widetilde{C}_s = \kappa_s D(s) Z, \qquad \kappa_s \in \overline{\mathbb{Q}}^{\times}.$$

Let $q \in \mathcal{F}$ be quaternary, and suppose $\widetilde{q} = (P^{-1})^{\otimes 4} q$ is supported on the even Hamming-weight positions in Walsh coordinates. Define p, t, x_{ab} by Equation (F.21)*. Assume that, for every contracted variable pair $\{i, j\}$, both orders of the residual pair $\{k, l\} = [4] \setminus \{i, j\}$, and every $s \in \mu_N$, contraction with C_s gives either the zero contraction or a nonsingular binary contraction whose Walsh transfer class belongs to*

$$C_N = \{[D(\eta)] : \eta \in \mu_N\}.$$

Organize the eight coordinates into four complementary pairs. Put $B_0 = (p, t)$ and $\mathfrak{I} = \{\{1,2\},\{1,3\},\{1,4\}\}$. For $I = \{a, b\} \in \mathfrak{I}$, set $x_I = x_{ab}$, put $I^c = [4] \setminus I$, and define the other three complementary pairs $B_I = (x_I, x_{I^c})$, $I \in \mathfrak{I}$. Then exactly one of the following holds:

1. *If $p, t \neq 0$, then $\lambda = t/p \in \mu_N$, and each pair B_I is zero or*

$$B_I = (-p\lambda\alpha_I^{-1}, -p\alpha_I), \qquad \alpha_I \in \mu_N. \tag{F.22}$$

2. *If $p = t = 0$, each pair B_I is zero or*

$$B_I = \rho_I(1, \alpha_I), \qquad \rho_I \neq 0, \quad \alpha_I \in \mu_N.$$

In particular, exactly one of p, t cannot be nonzero. Conversely, every even Walsh coefficient array in these alternatives satisfies the stated contraction condition under every phase contraction.

Proof. This is [LM26a, Lemma 7.2]. □

F.6.5 The odd-dihedral case

Theorem F.6.16 (The odd-dihedral case)**.** *Let $(A, \mathcal{F})$ be a stage of the sequence in Lemma F.6.9 at which $\mathcal{F}$ is finite, preprocessed, algebraic, and factor-closed and contains the transformed height-one mountain. Suppose that, after the normalization of Theorem F.6.8, every nonzero directly realizable binary is nonsingular with finite projective transfer order, and $G(\mathcal{F})$ is*

$$D_{2N} = \{[R(s)], [ZR(s)] : s \in \mu_N\}$$

for an odd $N \geq 3$. Then at least one of the following holds.

1. K-Holant($\mathcal{F}$) *is* #P*-hard or* $\mathsf{Tract}(K\mathcal{F})$ *holds.*

2. *There is a strict update* $(A, \mathcal{F}) \mapsto (A^+, \mathcal{F}^+)$ *satisfying Definition F.6.10.*

3. *There is an equivalent finite preprocessed algebraic factor-closed set* $\mathcal{F}^{\mathrm{cl}} \supseteq \mathcal{F}$*, in the same coordinates and with the same matrix* A*, for which the fixed Walsh matrix* P *satisfies*

$$(P^{-1})^{\otimes \operatorname{arity}(f)} f \in \mathscr{P} \quad (f \in \mathcal{F}^{\mathrm{cl}}), \qquad P^{\mathsf{T}} X P = 2Z \in \mathscr{P}.$$

Hence $K\mathcal{F}$ *is* $\mathscr{P}$*-transformable.*

In the remaining dihedral branch, choose a nonbinary irreducible signature of minimum arity after the finite contraction procedure. Its nonzero contractions factor into binary signatures whose transfer classes lie in the group. We show that this forces a known classification at arity four or contradicts irreducibility at larger arities. The four-variable analysis controls support and coefficient ratios, the six-variable argument uses the finite classification in Theorem F.A.2, and matching pencils supply a common binary factor at higher arities.

F.6.5.1 Support and coefficients on even blocks

For a partition π of a finite variable set V into even blocks, using $\mathbf{1}_C$ for the incidence vector of $C \subseteq V$, define

$$L_\pi = \operatorname{span}_{\mathbb{F}_2}\{\mathbf{1}_C : C \in \pi\}.$$

Its cosets are precisely the sets obtained by toggling each block as one unit.

The class $\mathcal{P}_N(V)$ consists of the zero tensor and all tensors g for which there exist such a partition π, an offset $\mathbf{a} \in \mathbb{F}_2^V$, a scalar $c \in \overline{\mathbb{Q}} \setminus \{0\}$, and block ratios $r_C \in (-1)^{|C|/2}\mu_N$ such that

$$\operatorname{supp}(g) = \mathbf{a} + L_\pi,$$
$$g\left(\mathbf{a} + \sum_{C\in\pi} t_C \mathbf{1}_C\right) = c \prod_{C\in\pi} r_C^{t_C} \qquad ((t_C)_{C\in\pi} \in \mathbb{F}_2^\pi).$$

For $C \subseteq V$, write $\mathbf{a}|_C$ for the restriction of $\mathbf{a}$ to C, and $\overline{\mathbf{a}|_C}$ for its bitwise complement. Equivalently, with the tensor factors placed on their indicated variable sets,

$$g = c \bigotimes_{C\in\pi} (\delta_{\mathbf{a}|_C} + r_C \delta_{\overline{\mathbf{a}|_C}}).$$

The block variables (t_C) therefore have logical coefficient tensor $c\prod_C r_C^{t_C}$. The block ratios will let us track coefficients as well as support through contractions. For nonzero g, its support determines π: the inclusion-minimal nonzero vectors of L_π are precisely the block indicators. We omit V when clear. The class $\mathcal{P}_N$ differs from the standard product class $\mathscr{P}$ of Section F.3.2. Its even-block and coefficient-ratio restrictions remain part of its definition.

Lemma F.6.17 (Closure of $\mathcal{P}_N$ under contractions)**.** *The class* $\mathcal{P}_N$ *is closed under nonzero scalar multiplication, variable permutation, tensor product on disjoint variable sets, and contraction of two variables against any kernel* $\mathsf{K}^+(s)$*,* $\mathsf{K}^-(s)$*, or its transpose from Equation* (F.17)*, where* $s \in \mu_N$*. Consequently, if every irreducible factor of a contraction has Walsh tensor in* $\mathcal{P}_N$*, then the contraction itself has Walsh tensor in* $\mathcal{P}_N$*.*

Proof. For a tensor product, take the disjoint union of the two even-block partitions and concatenate their offsets and block ratios. Scalars and variable permutations merely change the leading coefficient or relabel that data.

Consider a contraction of two variables of a nonzero tensor in $\mathcal{P}_N$. If the variables lie in the same block C, the kernel with the wrong xor gives zero. For the supported xor, delete the two variables. When variables remain in C, its new block ratio is

$$-sr_C \quad \text{or} \quad -r_C/s,$$

and therefore belongs to $(-1)^{(|C|-2)/2}\mu_N$. If $|C| = 2$, the block instead contributes a scalar, possibly zero, which is absorbed into the leading coefficient.

If the variables lie in distinct blocks C, D, the kernel affinely identifies their two block variables. After complementing one residual offset when needed, replace those blocks by

$$(C \setminus \{i\}) \dot\cup (D \setminus \{j\}).$$

Its ratio is one of

$$-s^{\pm 1} r_C r_D^{\pm 1} \in (-1)^{(|C|+|D|-2)/2}\mu_N.$$

All other blocks are unchanged. Transposing $\mathsf{K}^-(s)$ only supplies a nonzero scalar and replaces s by s^{-1}, so it is covered as well. Thus every nonzero contraction is again in $\mathcal{P}_N$. The last assertion follows from factorization into irreducibles. Nullary factors are absorbed into the scalar. □

F.6.5.2 Quaternary support and coefficients

Use the even-parity coordinates in Equation (F.21). For every ordering (i, j, k, l) of $[4]$, the reflection-contraction matrix is

$$H^-_{ij}(s) = \begin{pmatrix} 0 & x_{jl} - sx_{il} \\ x_{jk} - sx_{ik} & 0 \end{pmatrix}.$$

Condition (C) for the dihedral group gives $\mathcal{R}_N(p, x_{ij}, x_{kl}, t)$ from the phase contraction and $\mathcal{R}_N(x_{jk}, x_{ik}, x_{jl}, x_{il})$ from the reflection contraction.

When $\widetilde{q}$ is supported on the odd Hamming-weight positions in Walsh coordinates, write

$$u_i = \widetilde{q}(\mathbf{e}_i), \qquad v_i = \widetilde{q}(\mathbf{1}^4 \oplus \mathbf{e}_i).$$

For every such variable ordering, condition (C) for the dihedral group gives

$$\mathcal{R}_N(u_k, v_l, u_l, v_k), \qquad \mathcal{R}_N(u_j, u_i, v_i, v_j).$$

Proposition F.6.18 (Dihedral quaternary classification)**.** *Let $N \geq 3$ be odd, and let $q \in \mathcal{F}$ be a nonzero quaternary satisfying condition* (C) *with $H = G(\mathcal{F}) = \{[R(s)], [ZR(s)] : s \in \mu_N\}$, with $R(s)$ as in Theorem F.6.8. Put $\widetilde{q} = (P^{-1})^{\otimes 4} q$. Then $\widetilde{q} \in \mathcal{P}_N([4])$. More precisely, $\widetilde{q}$ has one Walsh parity and exactly one or two complementary pairs whose coefficient pairs are nonzero. After choosing an orientation, there is a scalar $\rho \in \overline{\mathbb{Q}} \setminus \{0\}$ such that, for one nonzero complementary pair $\{\mathbf{x}, \bar{\mathbf{x}}\}$, its nonzero coefficients in the order $(\widetilde{q}_{\mathbf{x}}, \widetilde{q}_{\bar{\mathbf{x}}})$ are*

$$\rho(1, \delta), \qquad \delta \in \mu_N, \tag{F.23}$$

or, for two nonzero complementary pairs $\{\mathbf{x}, \bar{\mathbf{x}}\}$ and $\{\mathbf{y}, \bar{\mathbf{y}}\}$, its nonzero coefficients in the order $(\widetilde{q}_{\mathbf{x}}, \widetilde{q}_{\bar{\mathbf{x}}}, \widetilde{q}_{\mathbf{y}}, \widetilde{q}_{\bar{\mathbf{y}}})$ are

$$\rho(1, \delta, -\gamma, -\delta\gamma^{-1}), \qquad \delta, \gamma \in \mu_N. \tag{F.24}$$

Conversely, every Walsh coefficient array in Equations (F.23) *and* (F.24) *satisfies the stated contraction condition under all phase and reflection contractions.*

Proof. Phase contractions give a μ_N-ratio on every nonzero complementary pair. If $p, t \neq 0$, Equation (F.22) puts all nonzero central coefficients in $-p\mu_N$. For two nonzero central coefficient pairs, the 2-by-2 coefficient matrix of the reflection contraction has all four entries nonzero, and its only possible proportional alternative in Lemma F.6.13 requires a ratio in $-\mu_N$, not μ_N. The cosets are disjoint for odd N.

With zero endpoints, three nonzero central coefficient pairs would require three pairwise ratios in $-\mu_N$, although the product of two lies in μ_N. Thus at most two complementary pairs have nonzero coefficients. Applying the same argument to both orientations of the reflection kernels $\mathsf{K}^-(s)$ gives the identical conclusion on the odd Hamming-weight positions.

Substitution gives Equations (F.23) and (F.24) and proves their converse, including zeros. The one-pair case uses $\pi = \{[4]\}$. The two-pair case forms a $2 + 2$ partition coset with ratios $-\gamma, -\delta\gamma^{-1} \in -\mu_N$, whose product is δ. Thus both parities lie in $\mathcal{P}_N([4])$, with odd parity absorbed by the coset offset.

The one-pair case is a CP signature. Up to flips and permutation, a support consisting of two complementary pairs is $\{0000, 1111, 1100, 0011\}$, and Equation (F.24) gives

$$\widetilde{q}_{\mathbf{x}}\widetilde{q}_{\bar{\mathbf{x}}} = \widetilde{q}_{\mathbf{y}}\widetilde{q}_{\bar{\mathbf{y}}},$$

for the two complementary pairs $\{\mathbf{x}, \bar{\mathbf{x}}\}$ and $\{\mathbf{y}, \bar{\mathbf{y}}\}$ in the support. Hence it is a product of two nonsingular weighted equality/disequality binaries. Thus $\widetilde{q} \in \mathscr{P}$, as required by Equation (F.15). □

F.6.5.3 Products of binaries and the arity-six case

Given a finite variable set V, a perfect matching M of V, and $\boldsymbol{\epsilon} \in \mathbb{F}_2^M$, the corresponding *matching coset* is

$$C(M, \boldsymbol{\epsilon}) = \{\mathbf{x} \in \mathbb{F}_2^V : x_u \oplus x_v = \epsilon_{\{u,v\}} \text{ for every } \{u, v\} \in M\}.$$

Equivalently, $C(M, \boldsymbol{\epsilon}) = \mathbf{a} + L_M$, where $\mathbf{a} \in \mathbb{F}_2^V$ satisfies $a_u \oplus a_v = \epsilon_{\{u,v\}}$ for every matched pair. Choose an order (u, v) for each pair and use x_u as its independent bit, so $x_v = x_u \oplus \epsilon_{\{u,v\}}$. The coefficients indexed by these bits form the *logical coefficient tensor*. The all-zero and all-one choices of $\boldsymbol{\epsilon}$ impose equal and unequal bits on every matched pair, respectively. The matching support alone does not impose a rank-one condition on this tensor.

For the minimum-arity argument, contractions must be products of binaries. Write $\mathcal{B}_N(V) \subseteq \mathcal{P}_N(V)$ for zero together with the tensors whose partition consists entirely of pairs. Every nonzero member is a product of nonsingular Walsh equality/disequality binaries, with ratio in $-\mu_N$ on each pair. For the resulting perfect matching M, its support is $\mathbf{a} + L_M$, is complement-closed, and has cardinality $2^{|V|/2}$. The contraction formulas in Lemma F.6.17 show that $\mathcal{B}_N$ is itself closed under all phase and reflection contractions: joining different pairs makes one pair, and closing a pair gives a scalar.

Theorem F.6.19 (Arity-six factorization from contractions)**.** *Let $N \geq 3$ be odd. If every nonzero oriented phase or reflection contraction of an arity-six signature f has Walsh tensor in $\mathcal{B}_N([4])$, then $f = 0$ or f is a tensor product of three nonsingular binaries.*

Proof. Suppose $f \neq 0$, and write $F = \widetilde{f}$. By Lemma F.6.14, F has one parity. Fix a variable pair and fix $x_i \oplus x_j = \xi$, with the two corresponding slices A, B, so the normalized contractions are $q_s = A - sB$. Put $U = \operatorname{supp}(A) \cup \operatorname{supp}(B)$. Each nonzero coordinate remains nonzero at at least $N - 1$ phases, while every nonzero contraction has four support points. Thus

$$(N-1)|U| \leq 4N.$$

Moreover $U = \bigcup_s \operatorname{supp}(q_s)$ is complement-closed. For $N \geq 5$ it is therefore empty or consists of at most two complementary pairs. For $N = 3$, the same conclusion is exactly Theorem F.A.2(i). Part (ii) of that theorem now applies for every odd N: the support of F is a partition coset of type (6), $(4+2)$, or $(2+2+2)$, or belongs to the orbit of

$$S_\star = \{000000, 000011, 001100, 110011, 111100, 111111\}. \tag{F.25}$$

Here and below, bit relabelings describe support coordinates only. They permute the available xor kernels projectively, with possible phase inversion, so all contraction arguments cover the full indicated orbits.

A (6) support has a supported-xor contraction with exactly two nonzero entries, each with a unique lift, meaning a unique extension of the residual assignment to the original support consistent with the kernel. This contradicts the four-point support required by $\mathcal{B}_N([4])$. On a $(4+2)$ support, use block variables and denote the four nonzero coefficients by c_{uv}. Closing the binary block gives, after normalizing its orientation,

$$(c_{00} - sc_{01})\delta_{0000} + (c_{10} - sc_{11})\delta_{1111}.$$

At most two phases are forbidden by cancellation. Since $N \geq 3$, some contraction again has exactly two nonzero entries. On $S_\star$, contract variables $3, 5$ with $\mathsf{K}^+(s)$. Only the two endpoints have equal bits there, giving

$$F_{000000}\delta_{0000} - sF_{111111}\delta_{1111},$$

again with two nonzero entries and unique lifts.

Only the $(2+2+2)$ support remains. Its eight coefficients c_{uvw} are nonzero. Closing any one of its binary blocks leaves a 2-by-2 table whose determinant vanishes at every $s \in \mu_N$: a nonzero contraction has the same four-point matching support, and a zero contraction also has zero determinant. For example,

$$\det\big(c_{uv0} - sc_{uv1}\big)_{u,v\in\mathbb{F}_2} = 0.$$

This polynomial has degree at most two and at least three roots. Its constant and quadratic coefficients show that the two faces obtained by fixing that coordinate have rank one. Doing this for all three axes makes every axis face minor zero. Since all eight entries are nonzero, the ratios along each axis are independent of the other two bits, so $c_{uvw} = cr_1^u r_2^v r_3^w$. This is the required three-binary factorization, which pulls back through the Walsh change of coordinates. □

F.6.5.4 Matching pencils

At higher arities, binary factorizations of separate contractions must be compared as the phase varies. We call a pencil $A - sB$ *projectively nonconstant* when A, B are linearly independent. Otherwise it is zero or has constant projective class wherever it is nonzero. A projective line consists of the classes in a two-dimensional linear span. Here rank one is taken over the matching blocks: it means that the logical coefficient tensor is a product of one vector for each block. A *one-factor Segre ruling* is a linear family of such tensors in which all but one block factor stay fixed. A matrix flattening

uses one block bit as the row index and the remaining block bits as the column index. The next two results use this fixed-factor structure to pass from individual contraction outputs to a common binary factor of the original tensor.

Lemma F.6.20 (Three matching products on a line)**.** *Let three pairwise nonproportional tensor products of nonsingular diagonal or anti-diagonal binaries on the same m variables satisfy a linear relation with all coefficients nonzero. Either their supports coincide and the three tensors lie in a single one-factor Segre ruling, or their matchings differ only on four variables, where the three different pairings occur. In the latter case all binary factors outside those four variables are common up to nonzero scalars.*

Proof. Let the supports be S_0, S_1, S_2, matching cosets of dimension $m/2$. No point can belong to just one support. If two supports coincide, the third is contained in them and has the same cardinality, so all coincide. On their common block coordinates, every 2×2 minor of a matrix flattening has degree at most two along the line. Three rank-one points make these minors vanish identically. A line of rank-one tensors varies in at most one factor: if two factors of two of its points were nonproportional, flattening across one of them would give rank two at a generic third point. This is a single one-factor Segre ruling.

Otherwise each S_i is the union of its intersections with the other two supports. Two proper affine subspaces cover an affine space only when they are disjoint parallel hyperplanes: each has at most half its points, so both must attain that bound. Thus the pairwise intersections are hyperplanes and the triple intersection is empty. The union of two perfect matchings is a disjoint union of alternating even cycles, counting a common edge as a two-cycle. If there are $k = m/2$ edges and b components, the intersection of their translation spaces has dimension b. Here $b = k - 1$, so exactly one component is a four-cycle and all other edges coincide. The offsets on those common edges agree because the supports intersect. After removing their free binary coordinates, S_2 is the four-point symmetric difference of the other two local supports. Its matching is the third pairing on the same four variables. Finally, on each pairwise intersection only two tensors contribute to the dependence. Fixing one assignment to four variables with nonzero coefficient leaves the full remaining matching coset, on which their products are proportional. Uniqueness of the factors on disjoint pairs gives the asserted common factors. □

Proposition F.6.21 (Phase-independent matching and common factors)**.** *Let $N \geq 3$ be odd, and let f have even arity $n \geq 8$. Suppose every nonzero result of one or more successive contractions of two variables using the kernels in Equation* (F.17) *has Walsh tensor in $\mathcal{B}_N$. Then each contraction pencil can be written as $q_s = H \otimes h_s$, where H is a fixed product of nonsingular binaries on disjoint pairs and h_s is a binary tensor on one fixed remaining variable pair. Zero members and proportional pencils are included.*

Proof. Write a pencil as $q_s = A - sB$. A zero member makes all other members proportional, which settles that case. If the pencil is not projectively constant, any three distinct phases give a linear dependence with all three coefficients nonzero, so Lemma F.6.20 applies to these members of $\mathcal{B}_N$. In its common-support case, the 2×2 flattening minors vanish identically, giving the fixed one-factor Segre ruling with that matching for the whole pencil.

In the second case of Lemma F.6.20, the matchings differ on four variables and all other binary factors are common. Close every such common pair, on the original gadget, with its supported-xor kernel at phase 1. For a binary ratio $r \in -\mu_N$, this gives a nonzero multiple of $1 - r$, nonzero because N is odd. These operations multiply the three contractions by the same nonzero scalar. At least one pair is closed since $n \geq 8$. Before closing the original pencil variables, the resulting

arity-six tensor is a nonzero result of contractions of f, hence is itself a binary matching product. Its contractions cannot have three different matchings: closing variables of a fixed matching deletes an edge or merges two edges, independently of the phase. This contradiction proves that the matching is independent of phase. □

F.6.5.5 A common binary factor suffices

Lemma F.6.22 (A binary factor from contractions)**.** *Under the hypotheses of Proposition F.6.21, every nonzero f is reducible.*

Proof. Put $F = \widetilde{f}$, fix variables i, j, and for each $\xi \in \{0, 1\}$ consider the restriction to $x_i \oplus x_j = \xi$, viewed as an n-ary tensor that is zero outside that constraint. Its two slices are $A = F_{ij}^{0,\xi}$, $B = F_{ij}^{1,1-\xi}$. Exactly one of A, B cannot be nonzero. Indeed, if $A \neq 0$ and $B = 0$, take a supported string of A, close any two dangling edges with its supported xor, and choose a phase avoiding cancellation of the projected string. The resulting nonzero contraction has complement-closed support. Its complementary string would require an assignment in B, a contradiction. The case $A = 0$, $B \neq 0$ is identical.

For each nonzero restriction, Proposition F.6.21 writes the pencil as $H \otimes h_s$, where H is a fixed product of external binaries and

$$h_s = (a - sb)\delta_{\mathbf{e}} + (c - sd)\delta_{\bar{\mathbf{e}}}$$

on one external pair, with $\mathbf{e} \in \mathbb{F}_2^2$. Projectively constant pencils are included by choosing any pair. Every nonzero h_s has ratio in $-\mu_N$, so Lemma F.6.13 applies. Restoring the variables i, j, its proportional alternative gives a binary block $\{i, j\}$. Either monomial alternative gives one quaternary CP block containing i, j. All other blocks are nonsingular binaries. Both slices A, B are nonzero, so the restored block has both coefficients nonzero. This factors the restriction in Walsh coordinates.

If only one of the two restrictions is nonzero, this factorization of F already has an external binary factor, since $n \geq 8$. Otherwise denote the restrictions for $\xi = 0, 1$ by U, V. We use the following observation repeatedly. Close an external pair with an xor supported by both restrictions, and choose a phase at which both resulting contractions are nonzero. Such a phase exists: a contraction of either displayed block product with a supported xor has unique lifts, except when an entire binary block is closed, in which case only one phase can cancel its scalar. The nonzero contraction G is a binary matching product with both ij xors present. Its two ij-xor restrictions have:

(O1) the same block of four variables containing i, j.

(O2) the same external matching and the same offset on each external pair.

(O3) proportional binary factors on each external pair, including equality of their ordered coefficient ratios.

Indeed, i, j belong to different matching edges in G. Either xor restriction joins those edges into one block of four variables and leaves every other binary factor unchanged. In particular, this observation identifies coefficients as well as supports.

First, neither U nor V can have $\{i, j\}$ as a binary block. If U did, close variables in two distinct external binary blocks of U. Both xors occur in U, so one is also supported by V. The observation would require a block of four variables through i, j, whereas this contraction leaves their binary block unchanged.

Write the blocks of four variables of U, V as $K = \{i, j, a, b\}$, $L = \{i, j, c, d\}$. Close a, b with its fixed xor in U. That restriction now has $\{i, j\}$ as a binary block, so the observation implies that V does not support this xor. Therefore a, b lie in the same block of V, with the opposite xor. Either $L = K$, or $\{a, b\}$ is an external binary block of V. By symmetry, in the latter case $\{c, d\}$ is an external binary block of U, and the two pairs are disjoint. Choose a variable e outside $\{i, j, a, b, c, d\}$, possible since $n \geq 8$, and close a, e. Both restrictions have unique lifts for either xor. The block of four variables through i, j in the contracted U contains b, whereas that in the contracted V remains $\{i, j, c, d\}$. This contradicts the observation. Thus $K = L$.

Let M_U, M_V be the external matchings on $E = [n] \setminus K$, and write $M_U(e), M_V(e)$ for the respective mates of e. For $e \in E$, closing a, e makes the block of four variables through i, j equal to $\{i, j, b, M_U(e)\}$ in one restriction and $\{i, j, b, M_V(e)\}$ in the other. The observation gives $M_U(e) = M_V(e)$, so the external matchings coincide. There are at least two external pairs. For any one of them, R, choose e on another pair and close a, e. The pair R remains external. The observation now identifies its binary factor in the two restrictions, hence in U, V, since the contraction avoids R. Therefore $U + V = F$ has this common factor. Invertible Walsh coordinates preserve tensor factorization, proving the lemma. □

Proof of Theorem F.6.16. Apply Lemma F.6.12. Its first two outcomes are the first two conclusions of the theorem. In outcome 3, work in the equivalent set $\mathcal{F}^{\mathrm{cl}}$, with the same matrix and group. If a nonbinary irreducible signature remains, choose one, f, of minimum arity n. Odd-arity factors have already been handled by Theorem F.4.7. Every irreducible factor of a contraction of two variables of f has smaller arity, and hence is binary by minimality. Its transfer class is in D_{2N}, so every nonzero contraction of two variables is in $\mathcal{B}_N$ in Walsh coordinates. Closure of $\mathcal{B}_N$ gives the same assertion for all further contractions involving at least one variable pair.

If $n = 4$, Proposition F.6.18 gives either a quaternary CP signature, to which Lemma F.4.19 applies, or a product of two binaries, contradicting irreducibility. If $n = 6$, Theorem F.6.19 contradicts irreducibility. If $n \geq 8$, Lemma F.6.22 gives the same contradiction. Thus either hardness or tractability has been proved, or every irreducible in $\mathcal{F}^{\mathrm{cl}}$ is binary. In the latter case one Walsh basis puts all signatures in $\mathscr{P}$, and $P^{\mathsf{T}} X P = 2Z \in \mathscr{P}$, giving outcome 3 with the stated change of basis. □

F.6.6 The odd-cyclic case

Only the phase-contraction kernels are available in the cyclic case. We first show that a minimum-arity nonbinary irreducible signature must have arity four. Binary attachments then turn that quaternary into a signature covered by an existing dichotomy or force a factorization. This gives the same classification-or-update alternatives as in the dihedral case.

Theorem F.6.23 (The odd-cyclic case)**.** *Let $(A, \mathcal{F})$ be a stage of the sequence in Lemma F.6.9 at which $\mathcal{F}$ is finite, preprocessed, algebraic, and factor-closed and contains the transformed height-one mountain. Let $N \geq 3$ be odd, and suppose every nonzero directly realizable binary is nonsingular with finite projective transfer order, and*

$$[P]^{-1} G(\mathcal{F})[P] = C_N = \{[\mathrm{diag}(1, s)] : s \in \mu_N\}.$$

Then at least one of the following holds.

1. $\mathrm{K\text{-}Holant}(\mathcal{F})$ *is* #P*-hard or* $\mathsf{Tract}(K\mathcal{F})$ *holds.*
2. *There is a strict update $(A, \mathcal{F}) \mapsto (A^+, \mathcal{F}^+)$ satisfying Definition F.6.10.*

3. *There is an equivalent finite preprocessed algebraic factor-closed set $\mathcal{F}^{\mathrm{cl}} \supseteq \mathcal{F}$, in the same coordinates and with the same matrix A, for which the fixed Walsh matrix P satisfies*

$$(P^{-1})^{\otimes \operatorname{arity}(f)} f \in \mathscr{P} \quad (f \in \mathcal{F}^{\mathrm{cl}}), \qquad P^{\mathsf{T}} X P = 2Z \in \mathscr{P}.$$

Hence $K\mathcal{F}$ is $\mathscr{P}$-transformable.

F.6.6.1 Phase-independent matching

Throughout this odd-cyclic proof, $\widetilde{f}$ denotes the fixed Walsh coordinates, and $[P]^{-1}G(\mathcal{F})[P] = C_N = \{[D(s)] : s \in \mu_N\}$, with one direct representative fixed for each class and $N \geq 3$ odd.

For the conditional analysis below, work in outcome 3 of Lemma F.6.12, retaining its finite, preprocessed, algebraic, factor-closed set, still denoted by $\mathcal{F}$, the matrix A, and the transformed height-one mountain. Odd-arity irreducible signatures have already been handled by Theorem F.4.7. If only binary irreducible factors remain, the common Walsh basis gives the third conclusion of the theorem. Otherwise choose a nonbinary irreducible $f \in \mathcal{F}$ of minimum arity n, which is even. For distinct variables i, j, use the phase kernel $\mathsf{K}^+(s)$ from Equation (F.17), with $s \in \mu_N$. After division by the nonzero representative scalar, the contraction has Walsh coefficients

$$(\partial_{ij}^{\mathsf{K}^+(s)} \widetilde{f})(\mathbf{y}) = A(\mathbf{y}) - sB(\mathbf{y}), \qquad A(\mathbf{y}) = \widetilde{f}(00, \mathbf{y}), \quad B(\mathbf{y}) = \widetilde{f}(11, \mathbf{y}). \tag{F.26}$$

By Lemma F.6.12(3), each indicated nonzero contraction and all its irreducible factors belong to $\mathcal{F}$. A nonbinary factor would have arity at most $n-2$, contradicting minimality. Every binary factor is nonsingular and its Walsh transfer class belongs to C_N. Thus condition (C) holds for f. Each nonzero contraction is a product of such binaries on a perfect matching. A binary with Walsh transfer class $[D(t)]$ has Walsh tensor proportional to

$$D(t)Z = \operatorname{diag}(1, -t).$$

Thus the support is the all-zero matching coset on the residual variables: for a perfect matching M,

$$C(M, \mathbf{0}) = \{\mathbf{y} \in \mathbb{F}_2^{[n]\setminus\{i,j\}} : y_a = y_b \text{ for every } \{a, b\} \in M\}. \tag{F.27}$$

The logical coefficient tensor on these matching variables is nowhere zero and has tensor rank one. By Lemma F.6.14, $\widetilde{f}$ has even Walsh parity.

Lemma F.6.24 (Phase-independent matching)**.** *Let $N \geq 3$ be odd, and let $f \in \mathcal{F}$ be a nonbinary irreducible signature of minimum arity under the assumptions of this subsection, including condition* (C) *with $[P]^{-1}G(\mathcal{F})[P] = C_N$. Use the contractions and matching cosets in Equations* (F.26) *and* (F.27)*. For each pair i, j, either every contraction vanishes, or $\partial_{ij}^{\mathsf{K}^+(s)} \widetilde{f} = 0$ for at most one phase $s \in \mu_N$ and there is a fixed perfect matching M_{ij} such that*

$$\partial_{ij}^{\mathsf{K}^+(s)} \widetilde{f} \neq 0 \quad \Longrightarrow \quad \operatorname{supp}(\partial_{ij}^{\mathsf{K}^+(s)} \widetilde{f}) = C(M_{ij}, \mathbf{0}) \qquad (s \in \mu_N).$$

Proof. If $A - sB$ and $A - tB$ vanish at distinct phases, then $A = B = 0$, so the first alternative holds. Otherwise at most one phase gives a zero contraction. Any two distinct nonzero contractions satisfy

$$\operatorname{supp}(A) \cup \operatorname{supp}(B) = \operatorname{supp}(A - sB) \cup \operatorname{supp}(A - tB), \tag{F.28}$$

because a scalar affine polynomial cannot vanish at two phases unless both coefficients vanish.

If no contraction vanishes, choose three phases. By Equation (F.28), the third matching coset lies in the union of the first two. A vector space contained in $U \cup V$ lies in one of them (otherwise sum vectors from its two exclusive parts), and all matching cosets have the same dimension. Thus the third coset equals one of the first two. Two phases with one coset and a third with another would make a string in the latter difference vanish at two phases, contradicting the affine-pencil identity. Hence all cosets coincide.

If $A - s_0 B = 0$, then every other contraction is $(s_0 - s)B$, and the conclusion is immediate. □

F.6.6.2 Pairs with every phase contraction zero

The preceding lemma leaves open the possibility that every phase contraction on a pair is zero. We encode these pairs by a graph and show that none occur above arity four. Define

$$Z_f^{\mathrm{cyc}} = \{\{i,j\} : \partial_{ij}^{\mathsf{K}^+(s)} \widetilde{f} = 0 \text{ for every } s \in \mu_N\}.$$

Two phases in Equation (F.26) show that an edge is exactly the support equation

$$\widetilde{f}(\mathbf{x}) \neq 0 \quad \Longrightarrow \quad x_i + x_j = 1.$$

It follows that every connected component of Z_f^{cyc} is complete bipartite. An odd path forces unequal endpoint bits and hence supplies the corresponding edge. An even path forces equal bits and cannot also be a zero edge.

Lemma F.6.25 (Every variable pair has a nonzero phase contraction above arity four)**.** *Under the assumptions of Lemma F.6.24, if* $\mathrm{arity}(f) = n \geq 6$*, then*

$$Z_f^{\mathrm{cyc}} = \varnothing.$$

Proof. Each color class has size at most two: if one contained p, q, r and the other b, some phase contraction on p, q would be nonzero. Its matching support contains the all-zero residual string, contradicting the forced disequality $x_r \neq x_b$.

An isolated $\{1\} \sqcup \{2\}$ is impossible. For every $k \geq 3$, the all-zero string of a nonzero $(1,k)$-contraction has the unique lift $x_1 = x_k = 1$, whence

$$\widetilde{f}(\mathbf{e}_1 + \mathbf{e}_k) \neq 0 \qquad (k = 3, \ldots, n).$$

A nonzero $(2,3)$-contraction then has each $\mathbf{e}_1 + \mathbf{e}_k$, $k \geq 4$, by the unique lift $x_2 = x_3 = 0$. Its support matching would contain every $\{1,k\}$, impossible.

Thus only $2+1$ and $2+2$ remain. Label the size-two color $\{1,3\}$ and the other $\{2\}$ or $\{2,4\}$. In a $(1,3)$ contraction the forced equations give the unique lift

$$x_1 = x_3 = 1 - x_2.$$

Put

$$h(x_2, \mathbf{y}) = \widetilde{f}(1 - x_2, x_2, 1 - x_2, \mathbf{y}),$$

where $\mathbf{y}$ denotes the remaining coordinates. In the $2+2$ case, also impose the support equation $x_4 = x_2$. Then

$$(\partial_{13}^{\mathsf{K}^+(s)} \widetilde{f})(x_2, \mathbf{y}) = u_s(x_2) h(x_2, \mathbf{y}), \qquad u_s(0) = -s, \quad u_s(1) = 1.$$

Every contraction is nonzero and equals a nowhere-zero unary u_s times h. Dividing this contraction by the nowhere-zero unary u_s only modifies the matching factor incident with variable 2, so h factors over that matching. In the $2+1$ case, let k be the mate of variable 2 in this matching. Reversing the unique lift gives a quaternary CP factor on $\{1,2,3,4\}$ in the $2+2$ case, or on $\{1,2,3,k\}$ in the $2+1$ case, tensor the remaining binaries. For $n \geq 6$ this contradicts irreducibility. □

F.6.6.3 Weight-two coefficients and the arity bound

We next use the weight-two coefficients to reconstruct a matching of the original variables. Once that matching is fixed, contractions impose rank-one restrictions on its logical coefficient tensor. The following lemma recovers the tensor from those restrictions.

Lemma F.6.26 (Rank-one diagonal restrictions)**.** *Let $r \geq 3$, and let $h : \mathbb{F}_2^r \to \mathbb{C}^\times$ satisfy $h(\mathbf{0}^r) = 1$. Suppose that, for every $i \neq j$, the restriction $h|_{t_i=t_j}$, viewed as a function of the common bit and the other $r-2$ bits, has tensor rank one. If $r \geq 4$, then*

$$h(\mathbf{t}) = \prod_{i=1}^{r} \rho_i^{t_i} \qquad (\rho_i \neq 0).$$

If $r = 3$, there are $\rho_1, \rho_2, \rho_3, \kappa \neq 0$ such that

$$h(\mathbf{e}_i) = \rho_i, \qquad h(\mathbf{e}_i + \mathbf{e}_j) = \kappa\rho_i\rho_j, \qquad h(111) = \kappa\rho_1\rho_2\rho_3. \tag{F.29}$$

Proof. This is [LM26a, Lemma 7.8]. □

Theorem F.6.27 (The minimum nonbinary arity is four in the cyclic case)**.** *Let $\mathcal{F}$ be a finite preprocessed algebraic factor-closed signature set containing the transformed height-one mountain. Suppose every nonzero directly realizable binary is nonsingular and has finite projective transfer order, and, in the fixed Walsh coordinates,*

$$[P]^{-1}G(\mathcal{F})[P] = C_N = \{[\mathrm{diag}(1, s)] : s \in \mu_N\}, \qquad N \geq 3 \text{ odd.}$$

Let $f \in \mathcal{F}$ be an even-arity nonbinary irreducible signature of minimum arity among the nonbinary irreducible signatures in $\mathcal{F}$. For every ordered pair of distinct variables of f, every fixed direct representative of a class in $G(\mathcal{F})$, and either kernel orientation, assume that the resulting contraction of two variables is zero or all of its irreducible factors are nonsingular binaries B satisfying

$$[P^{-1}(BX)P] \in C_N.$$

Then $\mathrm{arity}(f) \leq 4$*, and hence* $\mathrm{arity}(f) = 4$*.*

Proof. The contraction hypothesis is condition (C) for the stated group, so Lemmas F.6.24 and F.6.25 apply. Assume that f has arity $n \geq 6$. By Lemma F.6.25, every pair has a nonzero phase contraction. For $S \subseteq [n]$, write $a_S = \widetilde{f}(\mathbf{1}_S)$. We abbreviate $a_{i_1\cdots i_k} = a_{\{i_1,\ldots,i_k\}}$.

Step 1: the constant coefficient is nonzero. If $a_\varnothing = 0$, the all-zero coefficient $(\partial_{ij}^{\mathsf{K}^+(s)}\widetilde{f})(0) = -sa_{ij}$ of a nonzero matching contraction forces $a_{ij} \neq 0$ for every pair. Hence all contractions are nonzero. For disjoint pairs ij, pq,

$$(\partial_{ij}^{\mathsf{K}^+(s)}\widetilde{f})(\mathbf{e}_p + \mathbf{e}_q) = a_{pq} - sa_{ijpq}$$

is nonzero at some phase. The common matching from Lemma F.6.24 would therefore contain every edge on its at least four residual variables, impossible. Normalize henceforth $a_\varnothing = 1$, and put

$$W_f = \{\{p, q\} : a_{pq} \neq 0\}.$$

Step 2: a nonempty W_f factors f. For deleted i, j and a disjoint edge $pq \in W_f$, the coefficients $1 - sa_{ij}$ and $a_{pq} - sa_{ijpq}$ exclude at most one phase each. Thus some contraction contains $\mathbf{e}_p + \mathbf{e}_q$, and its common support matching contains pq. Deleting two variables outside any two adjacent W_f-edges would put both edges in one matching, so W_f is a matching.

Moreover, every $pq \in W_f$ is a global equality constraint. Indeed, if some supported $\mathbf{z}$ had $z_p \neq z_q$, choose equal coordinates i, j outside p, q, possible by even parity and $n \geq 6$. The residual string violates the pq-edge in every nonzero (i, j)-contraction and therefore vanishes at all phases. Two phases of Equation (F.26) then make both lifts vanish, including $\mathbf{z}$. Consequently

$$pq \in W_f \quad \Longrightarrow \quad \operatorname{supp}(\widetilde{f}) \subseteq \{\mathbf{z} : z_p = z_q\}. \tag{F.30}$$

Fix $E_1 = \{u, v\} \in W_f$ and $R = [n] \setminus E_1$. In an (u, k)-contraction, the all-zero string has the unique lift $x_u = x_k = x_v = 0$. Hence, for $p, q \in R \setminus \{k\}$, its coefficient at $\mathbf{e}_p + \mathbf{e}_q$ is exactly a_{pq}. Its matching therefore consists of the W_f-edges on $R \setminus \{k\}$ and one edge incident with v. Since that last edge cannot cover all $n - 2 \geq 4$ variables, the induced matching on R is nonempty. Choose k on one of its edges, with partner ℓ. In the (u, k)-contraction, ℓ has no incident W_f-edge left, so it must be the partner of v. Any vertex unmatched in $W_f|_R$ would also have to be matched to v, which is impossible. Thus $W_f = \{E_1, \ldots, E_r\}$ is perfect, where $r = n/2$.

Equation (F.30) identifies $\widetilde{f}$ with a logical coefficient tensor $h(t_1, \ldots, t_r)$ on the common bits of the E_k's. Contracting one variable from E_i and one from E_j imposes $t_i = t_j$, and the remaining partners determine the deleted bits uniquely. The contraction is nonzero since its all-zero coefficient is one. Its support is contained in the matching coset obtained by merging E_i, E_j. Both this coset and the contraction's matching support have 2^{r-1} points, so they coincide. Removing the nowhere-zero unary phase on the merged variable therefore gives a full-support rank-one diagonal slice of h. Every assignment in $\mathbb{F}_2^r$ has two equal coordinates, so these restrictions cover $\mathbb{F}_2^r$ and h has full support. With $h(\mathbf{0}^r) = 1$, Lemma F.6.26 factors h when $r \geq 4$. If $r = 3$, use Equation (F.29). Let $H_i(s)$ be the two-bit logical coefficient matrix obtained by closing E_i at phase s. For $\{i, j, k\} = [3]$,

$$\det H_i(s) = \rho_j \rho_k (\kappa - 1)(1 - \kappa s^2 \rho_i^2).$$

If $\kappa \neq 1$, the contraction is nonzero and hence rank one for every phase, whereas $s \mapsto s^2$ permutes the odd-order group μ_N. The last factor cannot vanish at two phases. Thus $\kappa = 1$, and the arity-six tensor also factors. Both cases contradict irreducibility.

Step 3: an empty W_f is impossible. Fix i, j. Every contraction now has constant coefficient one and is nonzero. Choose two edges e, e' of its common residual matching. Full matching support gives $a_{\{i,j\} \cup e} a_{\{i,j\} \cup e'} \neq 0$, while its two-bit rank-one identity is

$$a_{e \cup e'} - s a_{\{i,j\} \cup e \cup e'} = s^2 a_{\{i,j\} \cup e} a_{\{i,j\} \cup e'} \qquad (s \in \mu_N).$$

This nonzero quadratic cannot vanish at the $N \geq 3$ distinct phases. The contradiction completes the proof. □

F.6.6.4 The quaternary cyclic case

The minimum nonbinary arity is now four, but the resulting quaternary need not be a mountain. We first record the eight-vertex classification used to finish this branch, then adjust the Walsh coefficients by binary attachments.

An eight-vertex signature has arity four and is zero on inputs of odd Hamming weight.

Theorem F.6.28 (Irreducible quaternary K-Holant dichotomy)**.** *Let $\mathcal{F}$ be a finite algebraic signature set containing an irreducible eight-vertex signature M with $M(0000)M(1111) \neq 0$. If* $\mathsf{Tract}(K\mathcal{F})$ *holds, then* K-Holant$(\mathcal{F}) \in \mathsf{FP}$*. Otherwise* K-Holant$(\mathcal{F})$ *is* #P*-hard. The predicate* $\mathsf{Tract}(K\mathcal{F})$ *is decidable.*

Proof. This is [LM26a, Theorem 1.3]. □

Corollary F.6.29 (Eight-vertex K-Holant dichotomy)**.** *Let $\mathcal{F}$ be a finite algebraic signature set containing an eight-vertex signature M with $M(0000)M(1111) \neq 0$. If $\mathsf{Tract}(K\mathcal{F})$ holds, then* K-Holant$(\mathcal{F}) \in \mathsf{FP}$. *Otherwise* K-Holant$(\mathcal{F})$ *is* $\#\mathsf{P}$*-hard.*

Proof. If $\mathsf{Tract}(K\mathcal{F})$ holds, Theorem F.3.5 and Equation (F.2) give the polynomial-time conclusion. Assume $\neg\mathsf{Tract}(K\mathcal{F})$. If M is irreducible, Theorem F.6.28 gives hardness.

Suppose instead that M is reducible. Its nonzero endpoint values exclude a $1+3$ factorization: setting the unary block to 1 and the other block to 000 would give a nonzero odd-weight entry. Thus every nontrivial factorization has type $2+2$. In such a factorization, the nonzero endpoints and the odd-weight zeros force each binary factor to be diagonal with both diagonal entries nonzero. Up to a variable permutation and a nonzero scalar, one factor is therefore

$$E = \mathrm{diag}(p, q), \qquad p, q \in \overline{\mathbb{Q}}^{\times}.$$

Because M contains both charge signs, the common-one-sided exception in Lemma F.4.3 is impossible. Factor extraction gives K-Holant$(\mathcal{F}, E) \leq_{\mathrm{T}}$ K-Holant$(\mathcal{F})$. Choose algebraic $a, b \neq 0$ with $a^2 p = b^2 q = 1$, and put $D = \mathrm{diag}(a, b)$. Then

$$D^{\mathsf{T}} X D = abX, \qquad DED^{\mathsf{T}} = I,$$

so $D \in GO(X)$ and $D(\mathcal{F} \cup \{E\})$ contains I. By Lemmas F.4.4 and F.4.5, this transformed set still fails Tract. Hence Theorem F.6.2 gives hardness, which the change of basis and the displayed reduction transfer to K-Holant$(\mathcal{F})$. □

The next lemma specifies the output and phase choices of these attachments.

Lemma F.6.30 (Attaching cyclic binary representatives to four variables)**.** *Let $N \geq 3$, let $\mathcal{F}$ be a finite algebraic signature set in K-coordinates, and, for every $s \in \mu_N$, fix a gadget-realizable ordered binary representative C_s over $\mathcal{F}$, in a specified orientation, whose Walsh matrix is*

$$\widetilde{C}_s = \kappa_s D(s) Z, \qquad \kappa_s \in \overline{\mathbb{Q}}^{\times}.$$

Let $q \in \mathcal{F}$ be quaternary. For $r_1, r_2, r_3, r_4 \in \mu_N$, attach C_{r_i} to variable i of q through one X-edge and leave the other variable of each representative exposed. After division by $2^4 \prod_{i=1}^4 \kappa_{r_i}$, the resulting output h satisfies

$$\widetilde{h}_{\mathbf{y}} = \widetilde{q}_{\mathbf{y}} \prod_{i=1}^{4} r_i^{y_i}.$$

Consider the map

$$(r_1, r_2, r_3, r_4) \longmapsto (R, r_{12}, r_{13}, r_{14}), \qquad R = \prod_{i=1}^{4} r_i, \quad r_I = \prod_{i \in I} r_i,$$

Write $(\mu_N)^2 := \{z^2 : z \in \mu_N\}$. A target $(R, u, v, w) \in \mu_N^4$ occurs exactly when

$$\frac{uvw}{R} \in (\mu_N)^2. \tag{F.31}$$

For odd N, this map is bijective.

Proof. In Walsh coordinates an attachment acts by $\widetilde{C}_s(2Z) = 2\kappa_s D(s)$. Thus four attachments give the normalization $2^4 \prod_{i=1}^4 \kappa_{r_i}$ in the statement. The phase-map calculation is [LM26a, Lemma 7.3]. □

Theorem F.6.31 (The quaternary odd-cyclic case)**.** *Let $\mathcal{F}$ be a finite preprocessed algebraic factor-closed signature set containing the transformed height-one mountain. Suppose every nonzero directly realizable ordered binary is nonsingular and has finite projective transfer order, and, in the fixed Walsh coordinates,*

$$[P]^{-1}G(\mathcal{F})[P] = C_N = \{[\mathrm{diag}(1,s)] : s \in \mu_N\}, \qquad N \geq 3 \text{ odd.}$$

Let $0 \neq q \in \mathcal{F}$ be quaternary. For every ordered pair of distinct variables of q, every fixed direct representative of a class in $G(\mathcal{F})$, and either kernel orientation, assume that the resulting contraction of two variables is zero or all of its irreducible factors are nonsingular binaries B with

$$[P^{-1}(BX)P] \in C_N.$$

Then at least one of the following holds.

1. *K-Holant($\mathcal{F}$) is #P-hard or $\mathsf{Tract}(K\mathcal{F})$ holds.*
2. *Up to a permutation of its variables,*

$$q \doteq a \otimes b,$$

where a, b are nonzero positive-arity signatures of arity strictly below four, and

$$\text{K-Holant}(\mathcal{F}, a, b) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}).$$

Proof. The stated contractions satisfy condition (C) for the cyclic group. In the Walsh tensor $\widetilde{q}$, put

$$p = \widetilde{q}_{0000}, \qquad t = \widetilde{q}_{1111}, \qquad x_I = \widetilde{q}_{\mathbf{1}_I} \quad (|I| = 2).$$

For each such I, put $B_I = (x_I, x_{I^c})$, so B_{I^c} has the same two coefficients in reverse order. By Lemma F.6.14, $\widetilde{q}$ has even-parity support. Lemma F.6.15 excludes exactly one nonzero Walsh endpoint. Suppose first that $p, t \neq 0$, and put $\lambda = t/p$. For each $B_I \neq (0,0)$, it gives $(x_I, x_{I^c}) = (-p\lambda\alpha_I^{-1}, -p\alpha_I)$. For each zero pair choose $\alpha_I \in \mu_N$ arbitrarily, using one index from each complementary pair as in Lemma F.6.15. By Lemma F.6.30, prescribe $R = \lambda^{-1}$ and $r_I = \alpha_I\lambda^{-1}$. Let h be the signature obtained with the normalization in that lemma. Its Walsh coefficient array $\widetilde{h}$ has endpoints (p,p), value $(-p,-p)$ on each complementary pair with $B_I \neq (0,0)$, and zero on every pair with $B_I = (0,0)$.

Here the eight even-parity Walsh positions form the endpoint pair and three central pairs $\{\mathbf{1}_I, \mathbf{1}_{I^c}\}$, where $|I| = 2$. In the inverse Walsh formula of Equation (F.10), a pair $\{\mathbf{y}, \bar{\mathbf{y}}\}$ with common coefficient b contributes

$$b((-1)^{\mathbf{x}\cdot\mathbf{y}} + (-1)^{\mathbf{x}\cdot\bar{\mathbf{y}}}).$$

This vanishes when $\mathrm{wt}(\mathbf{x})$ is odd. At either endpoint all signs are positive. At $\mathbf{x} = \mathbf{1}_J$, $|J| = 2$, the endpoint pair and the pair $\{\mathbf{1}_J, \mathbf{1}_{J^c}\}$ have positive signs, and the other two central pairs have negative signs. Thus, if exactly k of the three pairs B_I are nonzero,

$$h(\mathbf{0}^4) = h(\mathbf{1}^4) = 2p(1-k),$$
$$h(\mathbf{1}_J) = \begin{cases} 2p(k-1), & \text{if } B_J \neq (0,0), \\ 2p(k+1), & \text{if } B_J = (0,0). \end{cases}$$

Walsh inversion gives the K-coordinate entries of the normalized gadget output. Thus $k = 1$ gives a nonzero multiple of $X \otimes X$, up to a variable permutation, whose factorization pulls back through the invertible binary attachments just constructed. For $k = 0$, the original Walsh tensor is

already a higher-arity CP signature, so Lemma F.4.19 gives outcome 1. For $k = 2, 3$, the output in K-coordinates has even-parity support and two nonzero endpoints. Add this quaternary and apply Corollary F.6.29. Its conclusion transfers to $\mathcal{F}$ by the gadget reduction and preservation under taking subsets, again giving outcome 1.

It remains that $p = t = 0$. If just one complementary pair has nonzero coefficients, Lemma F.4.19 gives outcome 1 directly. Otherwise label the three complementary pairs $\{\mathbf{1}_{I_\ell}, \mathbf{1}_{I_\ell^c}\}$, $\ell = 1, 2, 3$. Write each nonzero pair B_I as $\rho_I(1, \alpha_I)$, choose the unique $\beta_I \in \mu_N$ with $\beta_I^2 = \alpha_I$. For a zero pair choose $\beta_I \in \mu_N$ arbitrarily. Apply Lemma F.6.30 with $R = 1, r_I = \beta_I$. Write h for the resulting signature, rescaled according to Lemma F.6.30. Its nonzero Walsh coefficient pairs become (σ_I, σ_I), where $\sigma_I = \rho_I \beta_I$. Set $\sigma_I = 0$ when $B_I = (0, 0)$, and write $\sigma_\ell = \sigma_{I_\ell}$. These attachments again give an invertible operation on each variable. Put $c = \sigma_1 + \sigma_2 + \sigma_3$. The same sign calculation, now with zero Walsh endpoints, gives

$$\begin{aligned} h(\mathbf{0}^4) = h(\mathbf{1}^4) &= 2c, \\ h(\mathbf{1}_{I_\ell}) = h(\mathbf{1}_{I_\ell^c}) &= 2(2\sigma_\ell - c) \qquad (\ell = 1, 2, 3), \end{aligned}$$

and all odd-weight entries vanish. If $c \neq 0$, this is a complement-invariant mountain. After division by $2c$, its endpoints are one and its three central coefficient-pair values are

$$r_\ell = 2\sigma_\ell / c - 1, \qquad 1 + r_1 + r_2 + r_3 = 1 + 2c/c - 3 = 0.$$

Up to the nonzero scalar, this mountain is the output of invertible binary attachments to q, so it may be added by Equation (F.4). Corollary F.6.29 makes the enlarged problem either tractable or #P-hard. Preservation under taking subsets in the first case and the gadget reduction in the second transfer that conclusion to $\mathcal{F}$, giving outcome 1.

Assume $c = 0$. Exactly one nonzero pair is impossible. Two nonzero Walsh pairs have values $\sigma, -\sigma$. After permuting variables, $\widetilde{h}$ factors as

$$\sigma \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}_{14} \otimes \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}_{23},$$

so the factor-reduction equivalence applies. Finally suppose that all three σ_i are nonzero. Choose distinct i, j, k with $\{i, j, k\} = \{1, 2, 3\}$ and $\sigma_j^2 + \sigma_k^2 \neq 0$, which exists since otherwise subtraction would give $2\sigma_i^2 = 0$. Put pair i at $12 \mid 34$, and join variables $3, 4$ of one copy to variables $1, 2$ of another by edges. The output is

$$\frac{1}{4}\widetilde{G}_i = \begin{pmatrix} \sigma_i^2 & 0 & 0 & 0 \\ 0 & -(\sigma_j^2 + \sigma_k^2) & -2\sigma_j\sigma_k & 0 \\ 0 & -2\sigma_j\sigma_k & -(\sigma_j^2 + \sigma_k^2) & 0 \\ 0 & 0 & 0 & \sigma_i^2 \end{pmatrix}.$$

If every phase contraction of G_i were zero or had transfer class in C_N in Walsh coordinates, its equal nonzero endpoints and Lemma F.6.15 would give $\lambda = 1$. Complement symmetry and odd N then force every nonzero central coefficient to be $-\sigma_i^2$. In particular,

$$\sigma_j^2 + \sigma_k^2 = \sigma_i^2 = (\sigma_j + \sigma_k)^2,$$

because $\sigma_i = -(\sigma_j + \sigma_k)$, contradicting $\sigma_j\sigma_k \neq 0$. Hence a phase contraction of G_i is a nonzero binary which is singular or has Walsh transfer class outside C_N. A singular binary gives unary factors and the odd-arity theorem. A nonsingular binary of infinite projective transfer order is handled by

Lemma F.4.6. Otherwise its transfer has finite projective order, contradicting the definition of $G(\mathcal{F})$. Every hardness or tractability conclusion above gives outcome 1 through the cited theorem and the equivalent-enlargement reduction. In a factor case, the attached binaries act invertibly, so the displayed factorization pulls back to q. The transformed mountain excludes the common-one-sided exception, and the factor-reduction equivalence gives the reduction in outcome 2. □

Completion of Theorem F.6.23. Apply Lemma F.6.12. Its first two outcomes are the first two conclusions of the theorem. In outcome 3, the listed two-variable contractions and all their irreducible factors are retained in the equivalent set supplied by outcome 3. Other fixed gadget outputs and any required factors are added separately by Equation (F.4) and Theorem F.4.2, as in the quaternary closure proof. If a nonbinary irreducible signature remains, choose one of minimum arity. The argument preceding Lemma F.6.24 proves condition (C) for this signature. Theorem F.6.27 bounds its arity by four, and Theorem F.6.31 then gives hardness, tractability, or a tensor factorization. The last possibility contradicts the choice of an irreducible signature, so outcome 1 holds.

In the remaining case every irreducible signature is binary. By Lemma F.6.12(3), each has transfer class in $G(\mathcal{F})$ and Walsh tensor proportional to $D(r)Z$. Hence Equation (F.15) holds, which proves the third conclusion of Theorem F.6.23. □

F.6.7 Completion of the height-one case

We now combine the binary reduction and the two group cases into one step of the iteration. Its alternatives distinguish a completed classification from a strict update to which the same argument applies.

Theorem F.6.32 (Hardness, tractability, or a strict update)**.** *Let $\mathcal{F}_0$ be the initial signature set, and let $(A, \mathcal{F})$ be a stage of the sequence in Lemma F.6.9. In particular,*

$$A\mathcal{F}_0 \subseteq \mathcal{F}, \qquad A^{\mathsf{T}} X A = \mu_A X \quad (\mu_A \neq 0), \qquad \text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}_0).$$

Suppose $\mathcal{F}$ is finite, preprocessed, algebraic, and factor-closed, and contains the height-one mountain

$$B = \begin{pmatrix} c_0 & u \\ v & c_1 \end{pmatrix}, \qquad c_0 c_1 \neq 0.$$

Then one of the following occurs:

1. K-Holant($\mathcal{F}$) *is* #P*-hard, or* $\mathsf{Tract}(K\mathcal{F})$ *holds.*
2. *there are $A^+ \in GO(X)$ and a finite preprocessed algebraic factor-closed set $\mathcal{F}^+$, continuing the same sequence of additions and changes of basis, such that*

$$A^+\mathcal{F}_0 \subseteq \mathcal{F}^+, \qquad (A^+)^{\mathsf{T}} X A^+ = \mu_{A^+} X \quad (\mu_{A^+} \neq 0), \qquad \text{K-Holant}(\mathcal{F}^+) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}),$$

and, for $D = A^+A^{-1}$, the transformed binary DBD^{T} remains in $\mathcal{F}^+$ up to its known nonzero scalar. In this alternative every nonzero directly realizable ordered binary over both $\mathcal{F}$ and $\mathcal{F}^+$ is nonsingular with finite projective transfer order, so $G(\mathcal{F})$ and $G(\mathcal{F}^+)$ are defined. Moreover, either

$$[A]^{-1} G(\mathcal{F})[A] \subsetneq [A^+]^{-1} G(\mathcal{F}^+)[A^+],$$

or both values of ν are defined and

$$\nu(\mathcal{F}^+) < \nu(\mathcal{F}).$$

Both quantities satisfy the non-strict comparisons and fixed-field conditions of Definition F.6.10.

Proof. Maintain factor closure using the factor-reduction equivalence. The retained mountain excludes the common-one-sided exception. Each nontrivial factorization produces positive-arity factors of strictly smaller arity, so this process terminates. An odd-arity member is handled by Theorem F.4.7 and Equation (F.2). A rank-one binary supplies unaries and hence the same conclusion. If no nonbinary irreducible signature remains, the arity-at-most-two alternative gives tractability.

All additions of signatures and global changes of basis satisfy Lemma F.6.9. Whenever the groups are defined, conjugating them back to the original basis gives nested groups. The old irreducible signatures, up to one common invertible transformation, remain in the set, so ν cannot increase. A strict group inclusion or decrease of ν gives outcome 2. If adding signatures preserves both quantities, the matrix and group remain unchanged. Lemma F.6.12 proves separately that the required recursive contraction and factor extraction is finite.

If B is singular, Lemma F.4.6 supplies its unary factors. If BX is nonsingular of infinite projective order, it first supplies a rank-one binary and then unaries. The same argument applies to every other directly realizable binary that is singular or whose nonsingular transfer has infinite projective order. We may therefore assume that every nonzero directly realizable binary is nonsingular and of finite projective transfer order. Then Lemma F.6.5 and Theorem F.6.8 give either a directly realizable nonsingular weighted equality or, after one $GO(X)$-normalization, an odd cyclic or odd dihedral residual form. In the equality case, write $E = \mathrm{diag}(p, q)$, where $pq \neq 0$. The gadget gives K-Holant$(\mathcal{F}, E) \leq_{\mathrm{T}}$ K-Holant$(\mathcal{F})$. Choose a diagonal $D \in GO(X)$ with $DED^{\mathsf{T}} \doteq I$. The set $D(\mathcal{F} \cup \{E\})$ satisfies the hypotheses of Theorem F.6.2. Its hard conclusion composes back to $\mathcal{F}$, while its tractable conclusion pulls back through the problem equivalence and Lemma F.6.9. Hence this is outcome 1.

Apply Theorems F.6.16 and F.6.23 in the two residual forms. Their first conclusions give outcome 1, and their updates satisfy outcome 2 by Definition F.6.10. Their remaining conclusions give Equation (F.15), which Lemma F.6.9 implies $\mathsf{Tract}(K\mathcal{F})$. This again gives outcome 1 and completes the case analysis. □

Proof of Theorem F.6.1. Maintain $(A, \mathcal{F})$ and the invariant of Lemma F.6.9 over the fixed initial number field, and apply Theorem F.6.32. Its first outcome is the required hardness or tractability conclusion. The second outcome gives an update satisfying Definition F.6.10. By Lemma F.6.11, only finitely many such updates are possible. Between these updates, factor closure terminates by strict arity descent, and the recursive contractions and factor extraction terminate by Lemma F.6.12. Repeated application therefore reaches the first outcome. Finally, Lemma F.6.9 gives the conclusion for the initial signature set. □

F.7 Mountain Height Two

Theorem F.7.1 (Dichotomy with a height-two mountain)**.** *Let $\mathcal{F}$ be a finite algebraic signature set containing a height-two mountain. Then*

$$\mathsf{Tract}(K\mathcal{F}) \qquad \textit{or} \qquad \text{K-Holant}(\mathcal{F}) \textit{ is } \#\mathsf{P}\textit{-hard.}$$

Proof. A height-two mountain has arity four, is supported on Hamming weights $0, 2, 4$, and has two nonzero endpoint values. It is therefore an eight-vertex signature satisfying the hypotheses of Corollary F.6.29, which gives the conclusion. □

F.8 Mountain Height Three

Theorem F.8.1 (Height-three mountain dichotomy)**.** *Let $\mathcal{F}$ be a finite algebraic set of even-arity signatures such that $\chi_r(\mathbf{x}) \in 3\mathbb{Z}$ for every arity-r member $f \in \mathcal{F}$ and every $\mathbf{x} \in \mathrm{supp}(f)$. Suppose that $\mathcal{F}$ contains a mountain*

$$M = a\delta_{\mathbf{0}^6} + b\delta_{\mathbf{1}^6} + Q, \qquad ab \neq 0, \qquad Q \text{ EO}.$$

Then $\mathsf{Tract}(K\mathcal{F})$ *holds or* K-Holant$(\mathcal{F})$ *is* #P*-hard.*

Write $Q = M|_{\mathrm{EO}}$ for the central restriction. We construct a nonzero quaternary EO signature $h \notin \mathcal{B}_\times$, where $\mathcal{B}_\times$ is the class of products of nonsingular anti-diagonal binary signatures defined below, and obtain K-Holant$(\mathcal{F}, h) \leq_{\mathrm{T}}$ K-Holant$(\mathcal{F})$. Since $\mathcal{B}_{\mathrm{ARS}} \subseteq \mathcal{B}_\times$, Theorem F.4.13 then gives hardness in the remaining case $\neg\mathsf{Tract}(K\mathcal{F})$. If $Q \in \mathcal{B}_\times$, a two-copy construction supplies h. Otherwise, we use self-loops and two-copy contractions, together with the computer-assisted exhaustive classification in Appendix F.A.2.2.

F.8.1 Matching Products and Two-Copy Constructions

Let V be an ordered set of even cardinality. For distinct $i, j \in V$, write $\partial_{ij} f$ for the contraction with an X-edge. In the following formula, the first two displayed arguments assign values to variables i, j, respectively, and all other variables retain their order:

$$(\partial_{ij} f)(\mathbf{x}_{V\setminus\{i,j\}}) = f(0, 1, \mathbf{x}_{V\setminus\{i,j\}}) + f(1, 0, \mathbf{x}_{V\setminus\{i,j\}}).$$

Define the matching-product class

$$\mathcal{B}_\times(V) = \left\{ \lambda \bigotimes_{\{u,v\}\in P} \begin{pmatrix} 0 & p_{uv} \\ q_{uv} & 0 \end{pmatrix}_{u,v} : \begin{array}{l} P \text{ is a perfect matching of } V, \\ \lambda \prod_{\{u,v\}\in P} p_{uv} q_{uv} \neq 0 \end{array} \right\},$$

and put $\mathcal{B}_\times^0 = \mathcal{B}_\times \cup \{0\}$. Each factor has a specified orientation. Swapping its two variables exchanges its two nonzero entries. In particular, $\mathcal{B}_{\mathrm{ARS}} \subseteq \mathcal{B}_\times$.

Lemma F.8.2 (A matching-product central restriction yields a quaternary signature)**.** *Let $d \geq 3$, and let $\mathcal{F}$ be a finite algebraic signature set containing*

$$M = a\delta_{\mathbf{0}^{2d}} + b\delta_{\mathbf{1}^{2d}} + Q, \qquad ab \neq 0,$$

where, for a perfect matching $P = \{\{u_1, v_1\}, \ldots, \{u_d, v_d\}\}$ of the ordered variables,

$$Q = \lambda \bigotimes_{k=1}^{d} \begin{pmatrix} 0 & p_k \\ q_k & 0 \end{pmatrix}_{u_k, v_k}, \qquad \lambda \prod_{k=1}^{d} p_k q_k \neq 0.$$

There is a fixed two-copy gadget over $\{M\}$, using only X-edges, whose ordered boundary signature h is EO and has the six weight-two entries

$$(h_{0011}, h_{0101}, h_{0110}, h_{1001}, h_{1010}, h_{1100}) = (ab, c_1, c_2, c_3, c_4, ab), \qquad c_1c_2c_3c_4 \neq 0. \tag{F.32}$$

In particular,

$$\text{K-Holant}(\mathcal{F}, h) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}).$$

The signature h is not a tensor product of two nonsingular anti-diagonal binaries. Moreover, its supported triples include

$$1100 \oplus 1010 \oplus 0110 = 0000, \qquad 0011 \oplus 0101 \oplus 1001 = 1111.$$

Proof. We connect two copies so that their central matchings and the connecting edges form two odd-length alternating components. These supply four boundary patterns with one bit equal to one on each external pair, while the two endpoint choices supply the other two patterns. Write the matching of Q as $P = \{\{w_0, w_1\}, \ldots, \{w_{2d-2}, w_{2d-1}\}\}$. On two copies of the mountain, define a variable bijection p by fixing w_0, w_{2d-1} and transposing

$$(w_1, w_2), (w_3, w_4), \ldots, (w_{2d-3}, w_{2d-2}).$$

Leave (w_0, w_1) external on the first copy and (w_0, w_2) external on the second, in the external order $(w_0^{(1)}, w_1^{(1)}, w_0^{(2)}, w_2^{(2)})$. For every $r \notin \{0, 1\}$, join the first-copy variable w_r to the second-copy variable $p(w_r)$ by one X-edge. With both copies central, the matching edges and the connecting X-edges form two alternating components: the edge between the first external pair and a path of length $4d - 3$ between the second. Both lengths are odd, so each weight-one/weight-one boundary pattern has a unique extension and a nonzero coefficient c_i.

If both copies use endpoints, the edges force the two constant strings to be complementary. This gives exactly the other two EO patterns, with unique extensions of value ab. A mixed endpoint–central extension is impossible: for $d \geq 3$, some central matching edge has both variables connected to the constant copy and is forced equal. Thus each nonzero coefficient in (F.32) has exactly one nonzero extension.

A quaternary member of $\mathcal{B}_\times$ has exactly four supported strings, so the six-string support excludes membership in this class. It also contains the two triples displayed in the statement, whose xor identities follow directly. □

The first identity rules out the positive sign and the second rules out the negative sign in the support clause of $\mathsf{EOTract}$. Hence $\neg\mathsf{EOTract}(\{h\})$, and Theorem F.4.1 makes K-Holant($\{h\}$) #P-hard. Since

$$\text{K-Holant}(\{h\}) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}, h) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}),$$

the problem K-Holant($\mathcal{F}$) is #P-hard.

F.8.2 Arity-Six Loop and Two-Copy Contractions

The fifteen self-loop contractions alone do not force the required restriction on Q, as the counterexample in Remark F.A.5 shows. We therefore also use two-copy contractions. If every output belongs to $\mathcal{B}_\times^0$, the finite classification in Theorem F.A.4 forces the support of Q into one complementary pair. The calculation and its solver dependence are given in Proposition F.A.3.

Definition F.8.3 (Loop and two-copy contractions of an arity-six mountain)**.** Let $U = [6]$, and let

$$M = a\delta_{\mathbf{0}^6} + b\delta_{\mathbf{1}^6} + Q, \qquad Q = \sum_{S \in \binom{U}{3}} q_S \delta_{\mathbf{1}_S}, \qquad ab \neq 0.$$

For each unordered pair $\{i, j\} \in \binom{U}{2}$, let $\partial_{ij} M$ be the quaternary EO signature obtained by one X-self-loop, with the residual variables in increasing order. For $R \in \binom{U \setminus \{i,j\}}{2}$,

$$(\partial_{ij} M)(\mathbf{1}_R) = q_{R \cup \{i\}} + q_{R \cup \{j\}}. \tag{F.33}$$

These are the fifteen one-copy contractions.

For a two-copy contraction, choose unordered two-element external sets $A, B \in \binom{U}{2}$, list them in increasing order

$$A = (a_0, a_1), \qquad B = (b_0, b_1),$$

and retain the first/second labels of the two copies of M. Put $I_A = U \setminus A$ and $I_B = U \setminus B$, and choose a bijection $\pi : I_A \to I_B$. Leave A external on the first copy and B external on the second copy, in the order (a_0, a_1, b_0, b_1), and join every $r \in I_A$ to $\pi(r) \in I_B$ by an X-edge. For $\mathbf{x}, \mathbf{y} \in \{0,1\}^2$, put

$$A_{\mathbf{x}} = \{a_r : x_r = 1\}, \qquad B_{\mathbf{y}} = \{b_r : y_r = 1\}.$$

The resulting quaternary EO signature $C_{A,B,\pi}$ has, when $\mathrm{wt}(\mathbf{x}) + \mathrm{wt}(\mathbf{y}) = 2$,

$$C_{A,B,\pi}(\mathbf{x}, \mathbf{y}) = ab\,\mathbf{1}[(\mathbf{x}, \mathbf{y}) \in \{(00, 11), (11, 00)\}] + \sum_{\substack{T \subseteq I_A \\ |T| = 3 - \mathrm{wt}(\mathbf{x})}} q_{A_{\mathbf{x}} \cup T}\, q_{B_{\mathbf{y}} \cup (I_B \setminus \pi(T))}. \tag{F.34}$$

The first term comes from complementary endpoint choices on the two copies, and the sum comes from two central assignments. A mixed choice would force four variables of the central copy to be equal, which is incompatible with its weight-three support. The endpoint choices leave two external ones, as do two central assignments after the four internal X-edges each account for one bit equal to one. Thus the output is EO. There are $15^2 \cdot 4! = 5400$ labelled two-copy contractions.

Lemma F.8.4 (Support forced by arity-six contractions)**.** *Let M and its contractions $\partial_{ij} M$, $C_{A,B,\pi}$ be as in Definition F.8.3. Suppose that every one of the fifteen signatures $\partial_{ij} M$ and every one of the* 5400 *labelled signatures $C_{A,B,\pi}$ belongs to $\mathcal{B}^0_\times([4])$ on its four ordered dangling edges. Then*

$$\mathrm{supp}(Q) \subseteq \{\mathbf{1}_S, \mathbf{1}_{U \setminus S}\} \qquad \text{for some } S \in \binom{U}{3}. \tag{F.35}$$

Proof. With $t = ab$, the signatures $\partial_{ij} M$ and $C_{A,B,\pi}$ in Equations (F.33) and (F.34) are precisely the loop and labelled two-copy arrays covered by Theorem F.A.4. It applies to arbitrary complex q_S and every $ab \neq 0$, and gives (F.35). □

Theorem F.8.5 (A quaternary EO output outside $\mathcal{B}_\times$ from an arity-six mountain)**.** *Let $\mathcal{F}$ be a finite algebraic signature set containing*

$$M = a\delta_{\mathbf{0}^6} + b\delta_{\mathbf{1}^6} + Q, \qquad ab \neq 0, \qquad Q \text{ EO}.$$

There is a nonzero quaternary EO signature $h \notin \mathcal{B}_\times$ such that

$$\text{K-Holant}(\mathcal{F}, h) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}).$$

The signature h can be chosen as one of the fifteen one-copy contractions $\partial_{ij} M$, one of the 5400 *labelled two-copy contractions $C_{A,B,\pi}$ from Definition F.8.3, or the explicit two-copy output of Lemma F.8.2.*

Proof. Suppose first that $Q \in \mathcal{B}_\times$. Write

$$Q = \lambda \bigotimes_{r=0}^{2} \begin{pmatrix} 0 & p_r \\ q_r & 0 \end{pmatrix}_{w_{2r}, w_{2r+1}}, \qquad \lambda p_0 q_0 p_1 q_1 p_2 q_2 \neq 0.$$

The case $d = 3$ of Lemma F.8.2 gives the six-string output (F.32), with one nonzero extension per supported string. Choose this output as h.

Now suppose $Q \notin \mathcal{B}_\times$. If $Q = 0$, a contraction $C_{A,B,\pi}$ has exactly the two endpoint–endpoint coefficients ab. Its support has size two, whereas a quaternary member of $\mathcal{B}_\times$ has support size four. Choose this contraction as h. It remains to consider $Q \neq 0$. If a loop or two-copy contraction lies

outside $\mathcal{B}_\times^0$, choose it as h. Otherwise Lemma F.8.4 gives $\mathrm{supp}(Q) \subseteq \{\mathbf{1}_S, \mathbf{1}_{U\setminus S}\}$. Choose $i \in S$ and $j \notin S$. The two possible nonzero coefficients of $\partial_{ij} M$ have the unique extensions $\mathbf{1}_S, \mathbf{1}_{U\setminus S}$. Hence this contraction is nonzero with support at most two, contradicting its membership in $\mathcal{B}_\times^0$.

In each case, the selected h is the output of a direct gadget over $\{M\}$, so Equation (F.5) gives the required reduction. □

Proof of Theorem F.8.1. Apply zero/nullary preprocessing and assume $\neg\mathsf{Tract}(K\mathcal{F})$. By Theorem F.8.5, there is a nonzero quaternary EO signature $h \notin \mathcal{B}_\times$ with K-Holant$(\mathcal{F} \cup \{h\}) \leq_{\mathrm{T}}$ K-Holant$(\mathcal{F})$. The finite algebraic set $\mathcal{F} \cup \{h\}$ retains the mountain, whose endpoints witness both charge signs. The added signature has charge zero, so the charge divisibility condition is also preserved. By Lemma F.4.4, $\mathsf{Tract}(K(\mathcal{F} \cup \{h\}))$ fails. Since $\mathcal{B}_{\mathrm{ARS}} \subseteq \mathcal{B}_\times$, we have $h \notin \mathcal{B}_{\mathrm{ARS}}$. Thus Theorem F.4.13 applies to $\mathcal{F} \cup \{h\}$, and the preceding reduction transfers hardness to the original $\mathcal{F}$. □

F.9 Mountain Height at Least Four

Theorem F.9.1 (Mountain dichotomy at every height at least four)**.** *Let $d \geq 4$, and let $\mathcal{F}$ be a finite algebraic set of even-arity signatures such that*

$$\chi_r(\mathbf{x}) \in d\mathbb{Z} \qquad (f \in \mathcal{F},\ \mathrm{arity}(f) = r,\ \mathbf{x} \in \mathrm{supp}(f)).$$

Suppose that $\mathcal{F}$ contains a mountain

$$M = a\delta_{\mathbf{0}^{2d}} + b\delta_{\mathbf{1}^{2d}} + Q, \qquad ab \neq 0, \qquad Q \text{ EO}.$$

Then $\mathsf{Tract}(K\mathcal{F})$ *holds or* K-Holant$(\mathcal{F})$ *is* #P*-hard.*

An *X-matching product* is a nonzero scalar multiple of $\bigotimes_{e\in P} X_e$, where P is a perfect matching of its variables. Such binary factors are irreducible, so uniqueness of irreducible factorization identifies both their variable pairs and their projective values.

We first normalize the complete mountain so that all pair contractions of its central restriction are X-matching products. A common scalar then makes that restriction real-valued and complement-invariant, allowing the reconstruction theorem of Shao and Cai to apply. For $d \geq 5$, it gives a matching-product central restriction, and the two-copy construction of Lemma F.8.2 finishes the case. At height four, the same conclusion holds when the contraction matchings satisfy the overlap condition below. Otherwise we recover the Reed–Muller central restriction, normalize its endpoint coefficients, and obtain equality by a Turing reduction.

Remark F.9.2 (Relation to the real-valued Holant argument)*.* The normalization by attaching binary signatures below follows the contraction and binary-attachment strategy of [SC20, full version, Lemma 8.1]. Their transformed signature set satisfies ARS globally. Here the signature set is complex-valued: charge conservation makes the self-loop outputs EO, and Theorem F.4.13 forces their binary ARS factorization unless hardness has already been proved. This supplies the factorization needed for the binary attachments.

The constructions in Lemma F.9.5 use the complete mountain and its contractions. Together with the phase recovery in Lemma F.9.6, they establish the hypotheses of Lemma F.9.7 for the central restriction.

F.9.1 Common Normalization and Phase Recovery

For the proof of Theorem F.9.1, apply zero/nullary preprocessing and assume $\neg\mathsf{Tract}(K\mathcal{F})$. If $Q = 0$, Lemma F.4.18 already gives hardness. Hence it suffices to consider $Q \neq 0$. Every self-loop on M kills its endpoints:

$$\partial_{ij} M = \partial_{ij} Q. \tag{F.36}$$

More generally, charge conservation keeps every supported half-charge in $d\mathbb{Z}$. For an output of positive arity less than $2d$, its absolute value is less than d, so it must be zero. Such outputs are EO. By Theorem F.4.13, any such nonzero output outside $\mathcal{B}_{\mathrm{ARS}}$ gives hardness. Otherwise its factors are nonsingular binaries

$$B(z) = \begin{pmatrix} 0 & z \\ \bar{z} & 0 \end{pmatrix}, \qquad z \in \overline{\mathbb{Q}}^{\times},$$

up to nonzero scalars. These factors are available by Theorem F.4.2: the mountain is two-sided, excluding the common-one-sided exception. The constructions below use finitely many such factors and compose their reductions. The extracted binaries have charge zero, and Lemma F.4.4 excludes tractability for the corresponding oracle problems.

To normalize a selected binary factor to X, we attach a copy of that factor to one of its variables. Attaching the first variable of an ordered $B(z)$ to a mountain variable through X acts on that variable by

$$\begin{aligned} T_{B(z)^{\mathsf{T}}} &= B(z)^{\mathsf{T}} X = \mathrm{diag}(\bar{z}, z) = \bar{z} D(z/\bar{z}), \\ T_{B(z)^{\mathsf{T}}} B(z) &= |z|^2 X. \end{aligned} \tag{F.37}$$

This invertible diagonal map preserves support and both nonzero endpoints. To preserve a previously normalized contraction, we use the following criterion for consistently oriented factors:

$$B(z)^{\mathsf{T}} X B(w) \doteq X \quad \Longleftrightarrow \quad w/z \in \mathbb{R}^{\times} \quad \Longleftrightarrow \quad B(w) \doteq B(z). \tag{F.38}$$

Indeed the two off-diagonal entries are $\bar{z}w$ and $z\bar{w}$. In this case $T_{B(z)^{\mathsf{T}}}^{\mathsf{T}} X T_{B(w)^{\mathsf{T}}}$ is also a nonzero scalar multiple of X. All these operations use the extracted ordered binaries on the complete mountain.

Lemma F.9.3 (Vanishing pair contractions)**.** *Let h have arity at least three. If $\partial_{ij} h = 0$ for every pair $i \neq j$, then $h(\mathbf{x}) = 0$ whenever $0 < \mathrm{wt}(\mathbf{x}) < \mathrm{arity}(h)$. In particular, an EO signature with all pair contractions zero is zero.*

Proof. This is [CFS20a, Lemma 2.7]. Its three-equation argument is purely linear and uses neither ARS nor nonvanishing contractions. □

We also write ∂_e when $e = \{i, j\}$.

The next lemma propagates the normalized form from selected contractions to all pair contractions of the same EO tensor.

Lemma F.9.4 (Propagation of X-matching products)**.** *Let F be EO on an even variable set V, with $|V| \geq 8$, and suppose every nonzero $\partial_e F$ lies in $\mathcal{B}_{\mathrm{ARS}}$. Either of the following conditions implies that every $\partial_e F$ is an X-matching product.*

1. *Every $\partial_e F$ with $e \cap \{1, 2\} = \varnothing$ is an X-matching product.*
2. *Both $\partial_{12} F$ and $\partial_{ij} F$ are X-matching products, $\{1, 2\} \cap \{i, j\} = \varnothing$, and the matching of $\partial_{ij} F$ pairs $1, 2$ with distinct mates u, v.*

The remaining pair contractions are allowed to be zero in the hypotheses.

Proof. Contracting any pair in an X-matching product gives another nonzero X-matching product: it either removes a matching edge with scalar 2, or splices two edges with scalar 1. Thus, whenever a deleted pair e is disjoint from a pair r for which $\partial_r F$ is an X-matching product,

$$0 \neq \partial_e \partial_r F = \partial_r \partial_e F. \tag{F.39}$$

In particular $\partial_e F \neq 0$. A factor whose variables avoid r survives in this double contraction, so uniqueness of irreducible factorization makes it proportional to X.

For condition 1, only contractions with $e \cap \{1,2\} \neq \varnothing$ remain. A pair r disjoint from e, with $\partial_r F$ an X-matching product, first proves nonvanishing. For each binary factor on variables a, b, choose a pair r avoiding $\{1,2\} \cup e \cup \{a,b\}$. This excludes at most five variables, so such an r exists and $\partial_r F$ is an X-matching product. The factor survives in (F.39), proving condition 1.

For condition 2, put $W = V \setminus \{1,2,i,j\}$, where $u, v \in W$ and $|W| \geq 4$. For $e \in \binom{W}{2} \setminus \{\{u,v\}\}$, nonvanishing follows using the 12-contraction. The variables $1,2$ cannot be paired in $\partial_e F$: that pair would survive deletion of ij, whereas deleting e from the X-matching product $\partial_{ij} F$ pairs $1,2$ only if $e = \{u,v\}$. Consequently write

$$\partial_e F = C(x_1, x_r) D(x_2, x_s) H.$$

The 12-double contraction makes H an X-matching product and the splice of C, D proportional to X. By (F.38), $C \doteq D$ in these orientations. Both factors cannot be affected by deletion of ij, since this would splice $1,2$ together, contradicting the other contraction order. A surviving factor is proportional to X by comparison with the X-matching product $\partial_{ij} F$. Hence $C, D \doteq X$, and the whole contraction is an X-matching product.

For $e = \{u,v\}$, choose two distinct variables $r, s \in W \setminus \{u,v\}$. The contractions at the three disjoint pairs $12, ij, rs$ are now X-matching products. The uv-contraction is nonzero by the 12-comparison. Each of its binary factors avoids at least one of these three pairs, and hence is proportional to X by (F.39). Thus every $\partial_e F$ with $e \subseteq W$ is an X-matching product.

Next let $e = \{i,k\}$ or $\{j,k\}$, with $k \in W$. It is nonzero by the 12-comparison. For a factor on a, b, use $r = 12$ if $\{a,b\}$ avoids 12. Otherwise choose

$$r \in \binom{W \setminus (\{k\} \cup \{a,b\})}{2}.$$

At most two variables of W are excluded, so the choice exists. Here r is disjoint from both e and the factor's variables, and $\partial_r F$ is an X-matching product. The factor remains in (F.39), so it is proportional to X. Along with the already established X-matching product $\partial_{ij} F$, this covers all deleted pairs avoiding 12. Condition 1 completes the proof. □

We now apply this propagation to M. We first normalize a nonzero contraction, then adjust the remaining factors while preserving that normalized contraction. The two cases below correspond to the two conditions of Lemma F.9.4.

Lemma F.9.5 (Normalization by binary attachments at every arity at least eight)**.** *Under the hypotheses of Theorem F.9.1, assume* $\neg\mathsf{Tract}(K\mathcal{F})$ *and* $Q \neq 0$*. Then either* K-Holant($\mathcal{F}$) *is* #P*-hard, or a finite equivalent enlargement of* $\mathcal{F}$*, still supported on half-charges in* $d\mathbb{Z}$*, contains a mountain*

$$M^\circ = A\delta_{\mathbf{0}^V} + B\delta_{\mathbf{1}^V} + F, \qquad |V| = 2d, \quad AB \neq 0, \quad 0 \neq F \ EO,$$

such that

$$\partial_e F = \lambda_e \bigotimes_{t \in P_e} X_t, \qquad \lambda_e \in \overline{\mathbb{Q}}^{\times} \qquad \left(e \in \binom{V}{2} \right). \tag{F.40}$$

Here P_e is a perfect matching of $V \setminus e$.

Proof. At every stage retain the mountain, add the finitely many contractions or factors used, and apply Theorem F.4.13 again. The charge condition and $\neg\mathsf{Tract}(K\mathcal{F})$ are preserved. Thus every current nonzero contraction lies in $\mathcal{B}_{\mathrm{ARS}}$, or hardness has already been proved.

By Lemma F.9.3, some contraction of Q is nonzero. Relabel it as $\partial_{12}Q$. Extract its $d-1$ binary factors and attach a copy of each to one of its variables on M, with the orientation in (F.37). The resulting mountain M_0 has nonzero endpoints and central restriction F_0, with $\partial_{12}F_0$ an X-matching product. For every pair e avoiding 12, $\partial_{12}\partial_e F_0 = \partial_e \partial_{12} F_0 \neq 0$. Hence all these contractions are nonzero.

Suppose first that $1, 2$ form a matching edge in every such contraction. Write $\partial_e F_0 = C_e(x_1, x_2) H_e$. The 12-comparison shows that the contraction of C_e is nonzero and H_e is an X-matching product. If e, f are disjoint pairs avoiding 12, their double contraction is nonzero: in the e-order, deletion of f acts only on the X-matching product H_e. The factor on 12 survives in both orders, so $C_e \doteq C_f$. The graph of disjoint pairs on $V \setminus \{1,2\}$ is connected. Indeed, intersecting distinct pairs exclude only three of its at least six variables, leaving a pair disjoint from both. Thus all C_e are proportional to one ordered factor C. Extract C from one contraction and attach a copy to variable 1 of M_0 with the orientation in (F.37). Every contraction whose deleted pair avoids 12 is now an X-matching product. After reapplying Theorem F.4.13, condition 1 of Lemma F.9.4 makes every pair contraction an X-matching product.

Otherwise choose a pair ij avoiding 12 whose contraction has distinct mates u, v for $1, 2$, and write

$$\partial_{ij} F_0 = C(x_1, x_u) D(x_2, x_v) H.$$

The 12-comparison makes H an X-matching product and gives $C \doteq D$ by (F.38). Extract both factors from this contraction and attach their copies to variables $1, 2$ of M_0 in the corresponding orientations. The new ij-contraction is an X-matching product. The attachments act by $T_{C^\intercal} = C^{\mathsf{T}} X$ and $T_{D^\intercal} = D^{\mathsf{T}} X$, as in (F.37). The effective 12-kernel is $T_{C^\intercal}^{\mathsf{T}} X T_{D^\intercal} \doteq X$, so the 12-contraction remains an X-matching product as well. The diagonal maps preserve the distinct mates of variables $1, 2$. Reapply Theorem F.4.13 to the current contractions, then use condition 2 of Lemma F.9.4.

All extensions are invertible and preserve the central support and the nonzero endpoints. Their number depends only on the fixed source signature. Factor extraction and gadget replacement give the claimed finite equivalence. Every contraction added here is a contraction of the mountain and agrees with the corresponding contraction of its central restriction. □

The normalized contractions may still carry different complex scalars. To apply the reconstruction theorem, we next show that one common scalar makes all these coefficients positive and the central tensor satisfy ARS.

Lemma F.9.6 (A common phase for EO contractions)**.** *Let F be algebraic and EO on an even variable set V, with $|V| \geq 6$, and suppose* (F.40) *holds with every $\lambda_e \neq 0$. There is $c \in \overline{\mathbb{Q}}^{\times}$ such that $G = c^{-1}F$ is real-valued and complement-invariant, and every coefficient λ_e / c in* (F.40) *is positive real. In particular G satisfies ARS.*

Proof. For disjoint pairs e, f, the two contraction orders give

$$\partial_f \partial_e F = m_{ef} \lambda_e X_{P'}, \qquad \partial_e \partial_f F = m_{fe} \lambda_f X_{P''}, \qquad m_{ef}, m_{fe} \in \{1, 2\}.$$

Here $X_P = \bigotimes_{t\in P} X_t$ has only $0, 1$ entries. Both tensors are nonzero, so commutation identifies their supports and gives

$$\lambda_e/\lambda_f = m_{fe}/m_{ef} \in \{1, 2, 1/2\}.$$

The graph of disjoint pairs on V is connected: two intersecting pairs have a common disjoint pair since $|V| \geq 6$. Fix one pair e_0 and take $c = \lambda_{e_0}$. Every contraction of $G = c^{-1}F$ is therefore real and complement-invariant, with a positive coefficient.

Let $\overline{G}$ denote coefficientwise conjugation and $G^{\mathrm{rev}}(\mathbf{x}) = G(\bar{\mathbf{x}})$. Contraction commutes with both operations. Hence every pair contraction of $G - \overline{G}$ and of $G - G^{\mathrm{rev}}$ vanishes. Both differences are EO, so Lemma F.9.3 makes them zero. This is the algebraic normalization used in the reconstruction theorem. □

We use the following algebraic part of the real-valued classification.

Lemma F.9.7 (Shao–Cai matching-product reconstruction [SC20, full version, Lemma 2.11])**.** *Let G be an EO tensor of even arity $n \geq 8$ satisfying ARS. Suppose*

$$\partial_e G = \mu_e \bigotimes_{t\in P_e} X_t, \qquad \mu_e \in \mathbb{R}^{\times} \qquad \left(e \in \binom{[n]}{2}\right),$$

where each P_e is a perfect matching of the residual variables. If $n \geq 10$, then G is an X-matching product. The same conclusion holds for $n = 8$ if $P_A \cap P_B \neq \varnothing$ for some $A \neq B$ with $A \cap B \neq \varnothing$. In that case the X-factor on that shared edge is also a factor of G.

F.9.2 Height at Least Five

Proof of Theorem F.9.1 for $d \geq 5$. As above, assume $\neg\mathsf{Tract}(K\mathcal{F})$ and $Q \neq 0$. Apply Lemma F.9.5. Outside its hard outcome there is a mountain $M^{\circ} = A\delta_{\mathbf{0}^{2d}} + B\delta_{\mathbf{1}^{2d}} + F$ with $AB \neq 0$ and every pair contraction of F an X-matching product. By Lemma F.9.6, $G = c^{-1}F$ is EO with ARS and nonzero real X-matching product contractions. Since $2d \geq 10$, Lemma F.9.7 makes G, and hence F, an X-matching product. Apply Lemma F.8.2 to M°. Its two-copy output is hard, and the finite equivalence transfers hardness to the original $\mathcal{F}$. □

F.9.3 Height Four

At arity eight, the reconstruction theorem needs an overlap between contraction matchings. Without that overlap, the remaining matching family leads to the Reed–Muller structure described below. We recover its central coefficients first, then compare them with the two endpoint coefficients.

Lemma F.9.8 (A shared edge forces a matching product)**.** *Let F be an arity-eight EO tensor satisfying* (F.40)*. If $P_A \cap P_B \neq \varnothing$ for some $A \neq B$ with $A \cap B \neq \varnothing$, then F is an X-matching product.*

Proof. Apply Lemma F.9.6 and then the arity-eight clause of Lemma F.9.7. □

Under the hypotheses of Lemma F.9.8, any finite algebraic signature set containing a mountain $M^{\circ} = A\delta_{\mathbf{0}^8} + B\delta_{\mathbf{1}^8} + F$, with $AB \neq 0$, has a $\#\mathsf{P}$-hard K-Holant problem, by Lemma F.8.2.

We now consider the remaining case: $P_A \cap P_B = \varnothing$ for distinct intersecting deleted pairs A, B.

F.9.3.1 Translation Form of the Matching Family

Lemma F.9.9 (Translation form of the matching family)**.** *Let F be an arity-eight EO tensor on a variable set V, and suppose that for every $u \neq v$ one has*

$$\partial_{uv} F = \lambda_{uv} \bigotimes_{e \in P_{uv}} X_e, \qquad \lambda_{uv} \neq 0,$$

where P_{uv} is a perfect matching of $V \setminus \{u, v\}$. Suppose $P_A \cap P_B = \varnothing$ whenever $A \neq B$ and $A \cap B \neq \varnothing$. Then there is a bijection $\iota : V \to \mathbb{F}_2^3$ such that, after relabelling each variable v by $\iota(v)$, for $p \neq q$ and $h = p + q$ one has

$$P_{pq} = \big\{\{t, t+h\} : t \in \mathbb{F}_2^3 \setminus \{p, q\}\big\}, \tag{F.41}$$

where duplicate unordered pairs are suppressed.

Proof. We apply uniqueness of irreducible factorization and the computer-assisted exhaustive matching classification in Proposition F.A.6. For disjoint deleted pairs e, r, let $\kappa_r(P_e)$ be the two-edge matching left after contracting r in the X-matching product on P_e: delete r when $r \in P_e$, and otherwise splice the mates of the two ends of r and retain the untouched edge. Both operations are nonzero for X factors. Commutativity of the two loops and uniqueness of irreducible factorization give the exact constraint

$$\kappa_r(P_e) = \kappa_e(P_r) \qquad (e \cap r = \varnothing). \tag{F.42}$$

The condition on $P_A \cap P_B$ and (F.42) are exactly the hypotheses of Theorem F.A.7, which supplies the required bijection and (F.41). □

Index the eight variables by the points of $\mathbb{F}_2^3$ in lexicographic order. The first-order Reed–Muller code is

$$RM(1,3) = \{(\ell(t))_{t \in \mathbb{F}_2^3} \mid \ell \colon \mathbb{F}_2^3 \to \mathbb{F}_2 \text{ is affine}\}.$$

Explicitly, its elements are the truth tables of

$$\ell(t_1, t_2, t_3) = a + u_1 t_1 + u_2 t_2 + u_3 t_3, \qquad a, u_1, u_2, u_3 \in \mathbb{F}_2,$$

with all arithmetic over $\mathbb{F}_2$. Evaluating such an ℓ on the eight inputs $t \in \mathbb{F}_2^3$, in the fixed order above, gives one binary string of length eight. The $2^4 = 16$ choices of (a, u_1, u_2, u_3) give exactly the sixteen strings of $RM(1,3)$. For $\mathbf{x} \in \mathbb{F}_2^8$, we write

$$\mathrm{RM}_8(\mathbf{x}) = \mathbf{1}[\mathbf{x} \in RM(1,3)]. \tag{F.43}$$

The arity-eight exception is the central restriction of the Reed–Muller signature

$$R_8^{\mathrm{cen}}(\mathbf{x}) = \mathbf{1}\left[\mathbf{x} = (a + \mathbf{u} \cdot t)_{t \in \mathbb{F}_2^3} \text{ for some } a \in \mathbb{F}_2,\ 0 \neq \mathbf{u} \in \mathbb{F}_2^3\right]. \tag{F.44}$$

This tensor has fourteen supported strings and is not a matching product, but each contraction is an X-matching product. For deleted variables p, q, the surviving pairs are $\{t, t+p+q\}$: each choice of one endpoint from every remaining pair has a unique extension with $\mathbf{u} \cdot (p+q) = 1$. A matching edge can therefore belong to both P_A and P_B, for distinct A, B, only when $A \cap B = \varnothing$, so the arity-eight overlap clause of Lemma F.9.7 does not apply.

For a set of eight variables V and a bijection $\iota : V \to \mathbb{F}_2^3$, write

$$(\iota_* F)((x_t)_{t \in \mathbb{F}_2^3}) = F((x_{\iota(v)})_{v \in V}).$$

The matching family is now fixed, but the coefficients of the original tensor are not yet determined. The next lemma recovers them from the contraction equations, using the symbolic coefficient argument in Proposition F.A.8.

Lemma F.9.10 (The matching family determines the Reed–Muller central restriction)**.** *Let F be an arity-eight EO tensor on a variable set V such that*

$$\partial_{uv} F = \lambda_{uv} \bigotimes_{e \in P_{uv}} X_e, \qquad \lambda_{uv} \neq 0,$$

for every $u \neq v$, where P_{uv} is a perfect matching of $V \setminus \{u, v\}$. Suppose $P_A \cap P_B = \varnothing$ whenever $A \neq B$ and $A \cap B \neq \varnothing$. Choose a bijection $\iota : V \to \mathbb{F}_2^3$ supplied by Lemma F.9.9, put $F^\iota = \iota_* F$, and write

$$c_S = F^\iota(\mathbf{1}_S), \qquad S \in \binom{\mathbb{F}_2^3}{4}.$$

Then

$$\operatorname{supp}(F^\iota) = \left\{ \mathbf{1}_S : S \in \binom{\mathbb{F}_2^3}{4},\ \bigoplus_{s \in S} s = 0 \right\}, \tag{F.45}$$

the indicator vectors of the fourteen affine planes. Moreover all fourteen nonzero coefficients in (F.45) *are equal.*

Proof. Relabel the variables through ι, continue to denote the resulting tensor by $F^\iota = \iota_* F$, and write $V = \mathbb{F}_2^3$. For $S \in \binom{V}{4}$, $p \in S$, and $q \notin S$, put

$$S' = S \triangle \{p, q\}, \qquad T = S \setminus \{p\}, \qquad \sigma(S) = \bigoplus_{s \in S} s, \qquad h = p + q.$$

The contraction coefficient at $\mathbf{1}_T$ is

$$(\partial_{pq} F^\iota)(\mathbf{1}_T) = c_S + c_{S'}.$$

The string $\mathbf{1}_T$ selects one endpoint from each edge of the matching (F.41) if and only if

$$\sigma(S) \in \{0, h\}. \tag{F.46}$$

Indeed, in $V / \operatorname{span}_{\mathbb{F}_2}\{h\}$ the fibre $\{p, q\}$ contains one point of S, while the other occupancies are $1, 1, 1$ or a permutation of $2, 1, 0$. Their quotient xor is respectively zero or the sum of the two odd fibres, proving (F.46). Coefficients at these strings equal λ_{pq} and all others vanish. Thus the c_S and nonzero λ_{pq} satisfy the swap rule in Theorem F.A.9. That theorem eliminates all coefficients on four-sets that are not affine planes and identifies the fourteen plane coefficients with one $c \neq 0$, proving (F.45). □

Equivalently, the central restriction of the Reed–Muller signature from (F.44) is

$$R_8^{\mathrm{cen}}(\mathbf{x}) = \mathbf{1}\left[\operatorname{wt}(\mathbf{x}) = 4,\ \bigoplus_{t : x_t = 1} t = 0 \right].$$

Together with the two constant strings, this gives

$$\mathrm{RM}_8 = \delta_{\mathbf{0}^8} + \delta_{\mathbf{1}^8} + R_8^{\mathrm{cen}}.$$

F.9.3.2 Normalizing the Endpoint Coefficients

An information set of a binary code is a set of coordinates on which restriction is a bijection onto the corresponding binary coordinate space. By Lemma F.9.10, all central coefficients have the same value c. The remaining question is whether the endpoint product AB equals c^2. Restriction to an information set determines an affine string uniquely, so joining two copies along such a set makes the endpoint product visible in an arity-six output. This is the construction in the next lemma.

Lemma F.9.11 (Normalizing the endpoint coefficients)**.** *Let*

$$M^\circ = A\delta_{\mathbf{0}^8} + B\delta_{\mathbf{1}^8} + cR_8^{\mathrm{cen}}, \qquad ABc \neq 0.$$

Then either the two-copy X construction below yields a nonzero arity-six EO signature outside $\mathcal{B}_\times$, or

$$AB = c^2 \tag{F.47}$$

and, for any algebraic s with $s^8 = c/A$,

$$\begin{gathered} \big(sD(s^{-2})\big)^{\otimes 8} M^\circ = c\mathrm{RM}_8, \\ sD(s^{-2}) = \mathrm{diag}(s, s^{-1}), \qquad \big(sD(s^{-2})\big)^{\mathsf{T}} X \big(sD(s^{-2})\big) = X. \end{gathered} \tag{F.48}$$

The first outcome uses two source vertices, four information-set edges, and one final X-closure.

Proof. Index the variables by $V = \mathbb{F}_2^3$ and take the complementary affine information sets

$$I_0 = \{000, 100, 010, 001\}, \qquad I_1 = V \setminus I_0.$$

Join corresponding I_0-variables of two copies of M° by four X-edges. An affine string is determined by its values on either information set. Hence the internal equations force the two $RM(1,3)$ strings to be complementary on all eight coordinates. The remaining variables form four opposite pairs indexed by I_1, and their logical coefficient tensor on the four independent pair-choice bits is

$$H(\mathbf{y}) = \begin{cases} AB, & \mathbf{y} = 0^4 \text{ or } 1^4, \\ c^2, & \text{otherwise.} \end{cases}$$

Close one of these four pairs by an X-edge. On the three remaining matching bits the two coefficients at complementary strings are $AB + c^2$, whereas the other six coefficients are $2c^2$. If $AB = -c^2$, the output is nonzero but its support omits two of the eight choices of one endpoint from each remaining pair. Otherwise that support uniquely determines its three opposite pairs, and a rank-one logical coefficient tensor would have all 2×2 minors vanish after fixing any one of the three pair-choice bits. Choosing a minor through one of these complementary strings instead gives determinant

$$\det \begin{pmatrix} AB + c^2 & 2c^2 \\ 2c^2 & 2c^2 \end{pmatrix} = 2c^2(AB - c^2).$$

Thus $AB \neq c^2$ gives the first outcome, and the remaining case is (F.47).

Finally, $\big(sD(s^{-2})\big)^{\otimes 8}$ multiplies a weight-k coefficient by s^{8-2k}. It fixes the central restriction, sends A to $As^8 = c$, and sends B to $Bs^{-8} = AB/c = c$. This proves (F.48). Direct multiplication gives the displayed $GO(X)$ identity. □

In the current height-four branch, the first outcome of Lemma F.9.11 gives hardness by Theorem F.4.13. In the remaining case $AB = c^2$, the global transformation in Equation (F.48) supplies $c\mathrm{RM}_8$. We next obtain binary equality by reduction.

F.9.3.3 Obtaining Equality from the Reed–Muller Signature

Remark F.9.12 (The Reed–Muller signature in Shao–Cai). With the variable order in Equation (F.43), RM_8 is the signature $f_8 = \widehat{f_8}$ of [SC20, full version, Section 8]. Its Bell contraction identities are established in [CFS20b, conference version, Theorem 13 and the construction following it].

We use the Bell contraction identity and occurrence reduction of [SC20, full version, Definition 8.7 and Lemma 8.9]. They reduce the number of added Bell vertices to at most two. The half-charge condition then lets us evaluate the remaining equality vertices by closures over $\mathcal{F}$, giving the reduction needed for the binary-equality dichotomy. We then add the remaining Bell signatures. Recall $\mathcal{B} = \{I, X, Y, Z\}$, with the ordered matrices of Equation (F.11).

Lemma F.9.13 (The Reed–Muller Bell reduction)**.** *Let $\mathcal{F}$ be a finite algebraic set of even-arity signatures such that*

$$\chi_r(\mathbf{x}) \in 4\mathbb{Z} \qquad (f \in \mathcal{F},\ \mathrm{arity}(f) = r,\ \mathbf{x} \in \mathrm{supp}(f)).$$

Suppose $\mathcal{F}$ contains $c\mathrm{RM}_8$ for a known $c \neq 0$. Then

$$\text{K-Holant}(\mathcal{F}, \mathcal{B}) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}). \tag{F.49}$$

Proof. Boundary signatures below are realized by direct X-gadgets over the indicated signature set, with ordered dangling edges. Disconnected gadgets are allowed, with closed components contributing scalars.

Occurrence reduction. For each ordered $b \in \mathcal{B}$, attaching b to two variables of $c\mathrm{RM}_8$ leaves, up to a known nonzero scalar and a fixed ordering of the six remaining variables, $b^{\otimes 3}$. This is the identity used in [SC20, full version, Lemma 8.9]. It remains valid in K-coordinates, since $XbX = \pm b$. Thus any three b-vertices can be replaced by one b-vertex and one $c\mathrm{RM}_8$-vertex, reducing their number by two. Repeating this operation leaves at most two occurrences, with linear growth in the instance size and a tracked nonzero scalar. This is a tensor identity, so arbitrary complex labels in the remaining instance do not affect the reduction.

For $b = Y$ or I, a self-loop on a b-vertex has value zero, while two such vertices sharing an edge collapse to an X-edge, up to their recorded orientation signs. If they share both edges, they form a component of known scalar value. We may therefore assume that the remaining b-vertices are disjoint. Removing $k \leq 2$ of them leaves a direct gadget with $2k$ ordered dangling edges.

Adding I. All supported half-charges in $\mathcal{F}$ lie in $4\mathbb{Z}$. Charge conservation across edges therefore makes every binary boundary tensor over $\mathcal{F}$ anti-diagonal and every quaternary boundary tensor EO.

Use occurrence reduction for I. One disjoint occurrence evaluates to zero, since $XIX = I$ is orthogonal to every anti-diagonal binary for the bilinear pairing. For two disjoint occurrences, let H be the quaternary residual tensor, with the removed vertices on 12 and 34. On its EO support,

$$(I_{12} \otimes I_{34})|_{HW_4^=} = \left.\frac{X_{13} \otimes X_{24} + X_{14} \otimes X_{23} - X_{12} \otimes X_{34}}{2}\right|_{HW_4^=}. \tag{F.50}$$

Thus three X closures, with respective coefficients $1/2, 1/2, -1/2$, evaluate the desired contraction. These are K-Holant($\mathcal{F}$)-oracle queries, valid for arbitrary complex coefficients of H. We have proved

$$\text{K-Holant}(\mathcal{F}, I) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}). \tag{F.51}$$

Adding Y. The following identity holds on all sixteen four-bit assignments:

$$Y_{12} \otimes Y_{34} = X_{14} \otimes X_{23} - X_{13} \otimes X_{24}. \tag{F.52}$$

Hence two disjoint Y-occurrences can be evaluated by the difference of two X closures of the residual gadget. The effective kernel of a Y-vertex attached through two edges is $XYX = -Y$, so the two signs cancel here.

It remains to evaluate one occurrence. Use the bilinear coefficient pairing of Section F.2.7. If every binary boundary tensor H over $\mathcal{F}$ satisfies $\langle H, Y\rangle = 0$, every such instance has value zero. Otherwise fix one direct binary gadget G over $\mathcal{F}$ with

$$\kappa = \langle G, Y\rangle \neq 0.$$

Its size and the nonzero algebraic number κ depend only on the fixed signature set $\mathcal{F}$, not on the input instance. Its truth table is computed by finite exact summation. For the residual binary H of an input, apply (F.52) to the disconnected gadget $H \otimes G$. Two oracle closures compute

$$\langle H \otimes G, Y \otimes Y\rangle = \kappa\langle H, Y\rangle.$$

Dividing by the known κ, and restoring the sign from $XYX = -Y$, gives the required one-occurrence value. Together with occurrence reduction, this proves

$$\text{K-Holant}(\mathcal{F}, Y) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}). \tag{F.53}$$

Since Y has charge zero, the argument for I also applies over $\mathcal{F} \cup \{Y\}$, giving

$$\text{K-Holant}(\mathcal{F}, Y, I) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}, Y). \tag{F.54}$$

Finally $YXI = Z$ supplies the remaining Bell signature by a direct gadget, and an X-vertex merely subdivides an edge. This proves (F.49). □

Under the hypotheses of Lemma F.9.13, suppose also that $\neg\mathsf{Tract}(K\mathcal{F})$. By Lemma F.4.4, $\mathsf{Tract}(K(\mathcal{F} \cup \{I\}))$ fails. The binary-equality dichotomy Theorem F.6.2 makes this enlarged problem hard, and (F.51) transfers hardness to K-Holant($\mathcal{F}$).

F.9.3.4 Completion of the height-four case

Proof of Theorem F.9.1 for $d = 4$. Assume $\neg\mathsf{Tract}(K\mathcal{F})$ and $Q \neq 0$, as in Section F.9.1. Apply Lemma F.9.5 and suppose hardness has not already been proved. If $P_A \cap P_B \neq \varnothing$ for some $A \neq B$ with $A \cap B \neq \varnothing$, Lemma F.9.8 gives a matching-product central restriction, and Lemma F.8.2 gives hardness. Otherwise Lemmas F.9.9 and F.9.10 give a bijection $\iota : V \to \mathbb{F}_2^3$ under which the mountain is

$$M^\circ = A\delta_{\mathbf{0}^{\mathrm{s}}} + B\delta_{\mathbf{1}^{\mathrm{s}}} + cR_8^{\mathrm{cen}}, \qquad ABc \neq 0.$$

The first outcome of Lemma F.9.11 is a nonzero EO signature outside $\mathcal{B}_\times$, so Theorem F.4.13 gives hardness. Otherwise (F.48) supplies $sD(s^{-2}) = \mathrm{diag}(s, s^{-1}) \in GO(X)$ with

$$\big(sD(s^{-2})\big)^{\otimes 8} M^\circ = c\mathrm{RM}_8.$$

Apply $sD(s^{-2})$ to the entire current signature set and again call it $\mathcal{F}$. This set contains $c\mathrm{RM}_8$, the image of the retained mountain. The whole-set equivalence preserves support and charge, and Lemma F.4.5 preserves $\neg\mathsf{Tract}(K\mathcal{F})$. Apply Lemma F.9.13 and its hardness consequence to this set. Composing its reduction with the preceding gadget and factor reductions and the change-of-basis equivalence gives hardness for the original signature set. □

F.10 Synthesis and Proof of the Dichotomy

F.10.1 Combining the Height Cases

The four height cases apply to the same ambient signature set. For heights at least three, the whole-set divisibility hypothesis is supplied by Lemma F.5.2. We first combine the cases under that hypothesis and then apply the result to an arbitrary input signature set.

Theorem F.10.1 (Dichotomy for sets containing a mountain)**.** *Let $\mathcal{F}$ be a finite algebraic set of even-arity signatures containing a mountain of height d. If $d \geq 3$, assume that $\chi_r(\mathbf{x}) \in d\mathbb{Z}$ for every $f \in \mathcal{F}$ of arity r and every $\mathbf{x} \in \operatorname{supp}(f)$. Then*

$$\mathsf{Tract}(K\mathcal{F}) \qquad \textit{or} \qquad \text{K-Holant}(\mathcal{F}) \textit{ is } \#\mathsf{P}\textit{-hard.}$$

Proof. Apply Theorem F.6.1 for $d = 1$, Theorem F.7.1 for $d = 2$, Theorem F.8.1 for $d = 3$, and Theorem F.9.1 for $d \geq 4$. □

F.10.2 Proof of the Main Theorem

We complete the main proof in K-coordinates and then return to ordinary coordinates. Before constructing a mountain, we handle the case where every signature is supported on one Hamming layer, using the following classification.

Theorem F.10.2 (Single-layer K-Holant [MWX25, Corollary 68])**.** *Let $\mathcal{F}$ be a finite preprocessed algebraic signature set in which every signature is supported on one Hamming layer. Then* K-Holant($\mathcal{F}$) *is $\#\mathsf{P}$-hard unless either $\mathcal{F}$ is common one-sided with $\mathsf{EOTract}(\mathcal{F}|_{\mathrm{EO}})$, or both positive and negative charges occur and $\mathsf{EOTract}(\mathcal{F}_{\to\mathrm{EO}})$. Both exceptions are in FP by Theorem F.4.1.*

Theorem F.10.3 (Structural classification in K-coordinates)**.** *Let $\mathcal{F}$ be a finite algebraic signature set. If $\mathsf{Tract}(K\mathcal{F})$ holds, then* K-Holant($\mathcal{F}$) $\in \mathsf{FP}$*. Otherwise* K-Holant($\mathcal{F}$) *is $\#\mathsf{P}$-hard. The predicate $\mathsf{Tract}(K\mathcal{F})$ is decidable.*

Proof. The tractable implication and its effective recognition follow from Theorem F.3.5 and Equation (F.2). For the converse, assume that K-Holant($\mathcal{F}$) is not $\#\mathsf{P}$-hard. We prove $\mathsf{Tract}(K\mathcal{F})$. By the preprocessing in Section F.2.3, we may assume that $\mathcal{F}$ contains neither zero nor nullary signatures. We use the equivalence convention throughout, retaining $\mathcal{F}$ for the current set.

If there is an odd-arity signature, apply Theorem F.4.7 and Equation (F.2). The hard case is excluded, so $\mathsf{Tract}(K\mathcal{F})$ holds. We may therefore assume that $\mathcal{F}$ has only even-arity signatures. In this remaining case, test common one-sided support before factor extraction. For common one-sided support, charge conservation identifies the closed-instance values with those of the central restrictions, as in Section F.3.3. By Theorem F.4.1, failure of $\mathsf{EOTract}(\mathcal{F}|_{\mathrm{EO}})$ would give hardness. Hence alternative (1) holds. If every member is single-layer, Theorem F.10.2 excludes hardness and gives $\mathsf{EOTract}(\mathcal{F}_{\to\mathrm{EO}})$ in the remaining two-sided case. This is alternative (2).

In the remaining case, use Theorem F.4.2 and Lemma F.4.3 and the tensor-reconstruction equivalence in Section F.4.1.1 to replace each signature by its irreducible factors, still writing $\mathcal{F}$. Both charge signs persist: a supported charge is the sum of the charges on its factor blocks, so a positive or negative sum has a factor of that sign. The set also remains multilayer, since a tensor product of single-layer signatures is single-layer. An odd-arity factor gives $\mathsf{Tract}(K\mathcal{F})$ by Theorem F.4.7 and Equation (F.2), since hardness is excluded. If every remaining factor has arity at most two, alternative (3) holds. We may therefore start the remaining analysis with nonzero irreducible even-arity signatures, at least one of arity greater than two.

A tractability conclusion for this factor set transfers to the original set by reconstruction. Indeed, the first two alternatives cannot hold for this two-sided, multilayer set, and each of the other five alternatives is closed under tensor products.

Let d be the gcd of the nonzero supported half-charges of this preprocessed set $\mathcal{F}$. Apply Lemma F.5.2 to $\mathcal{F}$, which has supported half-charges of both signs. In its pinning outcome, apply Lemma F.4.10 to $\mathcal{F} \cup \{\Delta_0, \Delta_1\}$. Hardness would transfer by the stated reduction and is excluded. The tractability predicate therefore holds on that set, and Lemma F.4.4 gives $\mathsf{Tract}(K\mathcal{F})$. In the remaining outcome, the lemma supplies a mountain M of height d with K-Holant$(\mathcal{F}, M) \leq_{\mathrm{T}}$ K-Holant$(\mathcal{F})$ and

$$\chi_{\mathrm{arity}(f)}(\mathbf{x}) \in d\mathbb{Z} \qquad (f \in \mathcal{F}, \mathbf{x} \in \mathrm{supp}(f)).$$

The same divisibility holds for M, so Theorem F.10.1 applies to $\mathcal{F} \cup \{M\}$. Its hard outcome would transfer to the current problem and is therefore excluded. By Lemma F.4.4, $\mathsf{Tract}(K\mathcal{F})$ holds for the current factor set. Tensor reconstruction then gives the predicate for the original set. □

For the transformability cases, as in Definition F.3.2, if W satisfies the transformability conditions in K-coordinates, the ordinary-coordinate basis matrix is $M = KW$. Since $K^{\mathsf{T}}K = X$,

$$M^{-1}(K\mathcal{F}) = W^{-1}\mathcal{F}, \qquad M^{\mathsf{T}}M = W^{\mathsf{T}}XW,$$

so one basis handles both the edge and the entire signature set.

Proof of Theorem F.1.1. Decidability and polynomial-time evaluation in every tractable case follow from Theorem F.3.5. For hardness, put $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$. By Equation (F.2),

$$\text{Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \text{K-Holant}(\widehat{\mathcal{F}}).$$

Since $K\widehat{\mathcal{F}} = \mathcal{F}$,

$$\mathsf{Tract}(K\widehat{\mathcal{F}}) = \mathsf{Tract}(\mathcal{F}).$$

If $\neg\mathsf{Tract}(\mathcal{F})$, then $\neg\mathsf{Tract}(K\widehat{\mathcal{F}})$, so Theorem F.10.3 implies that K-Holant$(\widehat{\mathcal{F}})$ is #P-hard. The displayed problem equivalence transfers this hardness to Holant$(\mathcal{F})$. □

F.11 Conclusion

We have proved an FP-versus-#P-hard dichotomy for Boolean Holant on general graphs with arbitrary finite sets of algebraic complex-valued signatures, under polynomial-time Turing reductions. The same seven conditions as in the odd-arity dichotomy now apply without an odd-arity assumption. This criterion is decidable from exact algebraic input.

For the all-even cases, charge conservation and tensor-factor extraction either lead to an existing classification or produce a mountain signature of controlled support. Classifying these mountains by height completes the remaining cases. The gadget and factor reductions transfer the conclusions back to the original signature set.

Code Availability

The verification code for the finite calculations in Appendix F.A is available in the `certificates/` directory of the *GitHub repository: Full-Holant-Dichotomy*. The complete suite is run from the repository root with `python3 certificates/run_all_exact.py`. The `README.md` in that directory gives requirements and reproduction commands. `theorem-to-script-map.md` identifies the statements checked by each program and distinguishes supplementary checks from calculations used in the proof.

F.A Exact Finite Certificates

F.A.1 Exact finite verification

The appendix supplies four results used in the main proof:

1. Theorem F.A.2 classifies the arity-six support candidates used in Theorem F.6.19.
2. Theorem F.A.4 gives the central-support restriction used in Lemma F.8.4 and Theorem F.8.5.
3. Theorem F.A.7 gives the translation form of the matching family in Lemma F.9.9.
4. Theorem F.A.9 determines the central coefficients used in Lemma F.9.10.

The first three use exhaustive finite calculations. The last has a symbolic proof, with supplementary computational checks recorded at the end of the appendix. The arity-six coefficient calculation also uses exact real-polynomial satisfiability queries, whose solver dependence is stated in Proposition F.A.3. Code Availability gives the programs, requirements, and reproduction commands.

All arithmetic is exact over the indicated fields. We normalize a projective vector by making its first nonzero coordinate one, unless a primitive integer representative is specified. For a rational vector, the latter is obtained by clearing denominators, dividing by the gcd of the coordinates, and making the first nonzero coordinate positive.

F.A.2 Odd-dihedral, height-three, and height-four certificates

F.A.2.1 The order-three odd-dihedral arity-six certificate

Write $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$. Fix $\zeta_3 \in \overline{\mathbb{Q}}$ with $\zeta_3^2 + \zeta_3 + 1 = 0$, and identify a residue $r \in \mathbb{F}_3$ with its representative $0, 1, 2$ whenever it appears as an exponent ζ_3^r. For $\epsilon \in \mathbb{F}_2$, let

$$\mathcal{C}_\epsilon = \{\{\mathbf{x}, \bar{\mathbf{x}}\} : \mathbf{x} \in \mathbb{F}_2^4,\ \mathrm{wt}(\mathbf{x}) \equiv \epsilon \pmod 2\}$$

be the four complementary pairs of parity ϵ. Choose an orientation $(\mathbf{x}_p, \bar{\mathbf{x}}_p)$, $0 \le p \le 3$, of these four pairs. Using the indicator signatures of Section F.2.2, for $r, t \in \mathbb{F}_3$ put

$$\mathbf{V}_{p,r} = \zeta_3^r \delta_{\mathbf{x}_p} + \zeta_3^{2r} \delta_{\bar{\mathbf{x}}_p}, \qquad \mathbf{E}_{pq}(r,t) = \mathbf{V}_{p,r} - \mathbf{V}_{q,t}. \tag{F.55}$$

Define the finite projective set

$$\mathcal{Q}_3^\epsilon = \{[\mathbf{V}_{p,r}] : 0 \le p \le 3,\ r \in \mathbb{F}_3\} \cup \{[\mathbf{E}_{pq}(r,t)] : 0 \le p < q \le 3,\ r, t \in \mathbb{F}_3\}.$$

For an arity-six tensor F, use the slices F_{ij}^{ab} from Section F.2.7, with $i < j$ and the residual variables in increasing order. The kernels $D(-s)X^\xi$, for $\xi = 0, 1$, are respectively $\mathsf{K}^+(s)$ and $\mathsf{K}^-(s)$ from Equation (F.17). For $s \in \mu_3$,

$$\partial_{ij}^{D(-s)X^\xi} F = F_{ij}^{0,\xi} - s F_{ij}^{1,1-\xi}. \tag{F.56}$$

For $\eta \in \mathbb{F}_2$, if $\mathrm{supp}(F) \subseteq \{\mathbf{x} \in \mathbb{F}_2^6 : \mathrm{wt}(\mathbf{x}) \equiv \eta \pmod 2\}$, then this contraction has residual parity $\eta \oplus \xi$.

For a support $S \subseteq \mathbb{F}_2^6$, define

$$U_{ij}^\xi(S) = \{\mathbf{x}_{\widehat{\{i,j\}}} : \mathbf{x} \in S,\ x_i \oplus x_j = \xi\}.$$

The (i, j, ξ) *xor sector* consists of the assignments satisfying $x_i \oplus x_j = \xi$. We will use the condition that $U_{ij}^{\xi}(S)$ is empty or is the union of one or two complete complementary pairs from a single parity class $\mathcal{C}_\epsilon$.

A secant is a line through two distinct projective points. In the table below, the middle column lists the numbers of complementary pairs supporting the three projective points. For example, 1/1/2 means that two points each use one complementary pair and the third uses two. For three distinct complementary pairs p, q, u, the points $[\mathbf{E}_{pq}(r, t)]$, $[\mathbf{E}_{pu}(r, v)]$, and $[\mathbf{E}_{qu}(t, v)]$ are collinear because $\mathbf{E}_{pq}(r, t) - \mathbf{E}_{pu}(r, v) + \mathbf{E}_{qu}(t, v) = 0$.

We use the pencil terminology of Section F.6.5.4. Consider labelled pencils $q_s = A - sB$, $s \in \mu_3$, modulo one common nonzero scalar, such that all three q_s are nonzero and their three distinct projective classes are points of $\mathcal{Q}_3^{\epsilon}$ assigned bijectively to the phase labels. Only pencils satisfying these conditions are counted below.

Proposition F.A.1 (Order-three odd-dihedral arity-six classification)**.** *The following statements hold.*

(i) *Each set $\mathcal{Q}_3^{\epsilon}$ has* 66 *points. Its projective lines containing at least three of its points are exactly*

number	*supported pairs*	*geometry*
4	1/1/1	*one fixed complementary pair*
36	2/2/2	*a one-factor Segre ruling*
54	1/1/2	*a secant between two complementary pairs*
108	2/2/2	*the displayed relation on three complementary pairs*

Every such line contains exactly three points of $\mathcal{Q}_3^{\epsilon}$. Hence there are 202 *such lines and* $202 \cdot 6 = 1212$ *labelled pencils satisfying the stated conditions in each parity.*

(ii) *Let $\eta \in \mathbb{F}_2$, let* $\operatorname{supp}(F) \subseteq \{\mathbf{x} \in \mathbb{F}_2^6 : \operatorname{wt}(\mathbf{x}) \equiv \eta \pmod 2\}$*, and suppose that every contraction in Equation* (F.56) *is zero or has projective class in the appropriate set $\mathcal{Q}_3^{\eta \oplus \xi}$. For no fixed $i < j$ and ξ does the projectively nonconstant family $s \mapsto \partial_{ij}^{D(-s)X^{\xi}} F$ lie on a line from the third or fourth row of part* (i)*. Consequently, for every such F, each projectively nonconstant family $s \mapsto \partial_{ij}^{D(-s)X^{\xi}} F$ either is supported on one complementary pair or lies in a one-factor Segre ruling.*

(iii) *Exactly* 873 *supports $S \subseteq \mathbb{F}_2^6$ make every $U_{ij}^{\xi}(S)$ empty or a union of one or two complete complementary pairs of a single parity. They are the empty support, the* 392 *distinct cosets,*

$$\mathbf{a} + L_\pi, \qquad L_\pi = \operatorname{span}_{\mathbb{F}_2}\{\mathbf{1}_C : C \in \pi\},$$

where $\mathbf{a} \in \mathbb{F}_2^6$ and π is a partition of the six variables into even blocks, and the 480*-element bit-flip/variable-permutation orbit of*

$$S_\star = \{000000, 000011, 001100, 110011, 111100, 111111\}.$$

The distinct affine supports occur in the respective numbers

$$32, \qquad 240, \qquad 120$$

for partition types (6)*,* $(4 + 2)$*, and* $(2 + 2 + 2)$*, where a partition type is the multiset of its block sizes. Every nonempty support in the classification has one parity.*

Proof. The vectors in Equation (F.55) give twelve projective points supported on one complementary pair and fifty-four supported on two. Normalize every nonzero vector at its first nonzero coordinate. For every one of the $\binom{66}{2}$ point pairs, take the exact reduced-row-echelon form of its two-dimensional span and group equal row spaces. The groups containing at least three points of $\mathcal{Q}_3^\epsilon$ all have size three and give the table in part (i).

The table also has a direct interpretation in terms of complementary pairs. The three points on each of the four complementary pairs give one line, for four lines in total. On two fixed complementary pairs, the nine points supported on both pairs form a 3-by-3 Segre grid in the transformed phase coordinates $u = r + t$, $v = r - t$ in $\mathbb{F}_3$. Fixing u or v, not the original r or t in Equation (F.55), gives its two rulings and hence $6 \cdot 2 \cdot 3 = 36$ lines. Every $\mathbf{E}_{pq}(r,t)$ lies on the secant through $\mathbf{V}_{p,r}$ and $\mathbf{V}_{q,t}$, giving $6 \cdot 3^2 = 54$ secants. For three distinct complementary pairs,

$$\mathbf{E}_{pq}(r,t) - \mathbf{E}_{pu}(r,v) + \mathbf{E}_{qu}(t,v) = 0$$

gives $\binom{4}{3}3^3 = 108$ lines supported on three complementary pairs. Conversely, in a dependent triple every complementary pair occurring in one vector must occur in another. According as the triple uses one, two, or three complementary pairs, it is one of the displayed types. Four complementary pairs would leave one occurring only once.

Assigning the phases $1, \zeta_3, \zeta_3^2$ to the three points on one of these 202 lines gives a unique affine pencil up to one common nonzero scalar. Indeed, after the unique dependence is scaled, solve

$$q_1 = A - B, \qquad q_{\zeta_3} = A - \zeta_3 B$$

and verify $q_{\zeta_3^2} = A - \zeta_3^2 B$. All six phase assignments are distinct. This proves the last assertion of part (i).

For part (ii), after a bit flip to even parity, fix $\mathbf{x}_0 = 0000$, $\mathbf{x}_1 = 0011$, $\mathbf{x}_2 = 0101$, and $\mathbf{x}_3 = 0110$. For $\mathbf{t} = (t_1, \ldots, t_4) \in \mathbb{F}_3^4$, define $(R_\mathbf{t} q)(\mathbf{x}) = \zeta_3^{\sum_{j=1}^4 t_j x_j} q(\mathbf{x})$. Under $R_\mathbf{t}$, the four oriented complementary-pair labels change by $H\mathbf{t}$, where

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad \det H = -16 \neq 0 \pmod 3.$$

The phase permutations are generated by $s \mapsto \alpha s$ and $s \mapsto \alpha / s$, with $\alpha \in \mu_3$. These maps form $D_6 \cong S_3$ and realize all six permutations of μ_3. Together with variable permutations and bit flips, these actions reduce the secant cases supported on two complementary pairs and the cases supported on three to the following representative pencils for a fixed restriction $x_i \oplus x_j = \xi$. Group coordinates by their four oriented complementary pairs, writing $\mathbf{0}_2 = (0,0)$:

$$\begin{aligned} A_{\text{sec}} &= ((1,1), (\zeta_3, \zeta_3), \mathbf{0}_2, \mathbf{0}_2), \\ B_{\text{sec}} &= ((\zeta_3^2, \zeta_3^2), (\zeta_3, \zeta_3), \mathbf{0}_2, \mathbf{0}_2), \\ A_{\text{tri}} &= ((1,1), (\zeta_3, \zeta_3), (\zeta_3^2, \zeta_3^2), \mathbf{0}_2), \\ B_{\text{tri}} &= ((\zeta_3, \zeta_3), (1,1), (\zeta_3^2, \zeta_3^2), \mathbf{0}_2). \end{aligned}$$

Each displayed pair (A_*, B_*) denotes the pencil $q_s = A_* - sB_*$, $s \in \mu_3$. Since H is invertible over $\mathbb{F}_3$, the maps $R_\mathbf{t}$ shift the complementary-pair labels independently. Variable permutations and bit flips act transitively on the complementary-pair choices and the two total parities. Together

with the phase permutations, this proves that every orbit of the secant cases supported on two complementary pairs and the cases supported on three under these operations is represented.

Fix the selected pair $i < j$ and sector ξ. It remains to exclude arity-six tensors F satisfying the hypotheses of part (ii) and realizing either selected representative pencil. Put $\zeta = \xi \oplus 1$. Under the restriction $x_i \oplus x_j = \zeta$, a projectively nonconstant pencil is one of the 1212 labelled pencils from part (i). Its slices $C = F_{ij}^{0,\zeta}$ and $D = F_{ij}^{1,1-\zeta}$ are fixed up to a common scalar $\lambda \in \overline{\mathbb{Q}}^{\times}$. For each of the fourteen variable pairs other than the selected pair, and for each xor sector and phase, substitution into Equation (F.56) gives a quaternary contraction $\mathbf{q}_0 + \lambda \mathbf{q}_1$, where the computed vectors $\mathbf{q}_0, \mathbf{q}_1 \in \overline{\mathbb{Q}}^{\mathbb{F}_2^4}$ are independent of λ. For a fixed representative $\mathbf{v} \in \overline{\mathbb{Q}}^{\mathbb{F}_2^4} \setminus \{0\}$ of a point of the appropriate set $\mathcal{Q}_3^{\epsilon}$, the set of λ for which this vector belongs to $\operatorname{span}\{\mathbf{v}\}$ is all of the field, empty, or one point. Exact two-column row reduction determines which case occurs. Intersecting the resulting finite unions over all remaining pairs, both xor sectors, and all three phases leaves no nonzero λ for either representative.

For an opposite pencil with constant projective class, write $C = c\mathbf{v}$ and $D = d\mathbf{v}$, where $\mathbf{v} \in \overline{\mathbb{Q}}^{\mathbb{F}_2^4} \setminus \{0\}$ represents one of the 66 points of the appropriate $\mathcal{Q}_3^{\epsilon}$ and $(c, d) \in \overline{\mathbb{Q}}^2 \setminus \{(0, 0)\}$. For each remaining contraction, substitution gives $\mathbf{q}_0 + c\mathbf{q}_1 + d\mathbf{q}_2$, where the computed $\mathbf{q}_0, \mathbf{q}_1, \mathbf{q}_2 \in \overline{\mathbb{Q}}^{\mathbb{F}_2^4}$ are independent of c, d. Its preimage over all target points of $\mathcal{Q}_3^{\epsilon}$ is a finite union of affine linear systems in (c, d). Exact row reduction, followed by intersection over all contractions in the hypothesis of part (ii), leaves no nonzero pair (c, d). The identically zero opposite sector is checked separately and also fails. All matrices have entries in $\mathbb{Q}(\zeta_3)$. Their ranks and consistency are unchanged after extension to $\overline{\mathbb{Q}}$. Swapping the two variables when $\xi = 0$ changes nothing, whereas for $\xi = 1$ reversal at s gives a nonzero scalar multiple of the contraction at s^{-1}, so unordered variable pairs exhaust the ordered conditions.

For part (iii), fix one variable pair and one xor sector. A union satisfying the stated condition is empty, one of the eight unions consisting of one complementary pair across the two parities, or one of the twelve unions consisting of two complementary pairs. Each residual string in a nonempty union occurs only in $F_{ij}^{0,\xi}$, only in $F_{ij}^{1,1-\xi}$, or in both. Hence the exact finite superset of local candidates has

$$1 + 8 \cdot 3^2 + 12 \cdot 3^4 = 1045$$

members. Combine the two disjoint xor sectors on the first variable pair, giving 1045^2 candidates. Retain a candidate precisely when, on each of the other fourteen pairs, each of the two unions is empty or is the union of one or two complete complementary pairs of its own parity. Encode a support by a 64-bit integer whose bit k corresponds to the k-th string in lexicographic order on $\{0, 1\}^6$, starting at $k = 0$. Exact intersection of these encoded support lists leaves 873 distinct supports.

Independently generate every partition into even blocks and every coset of its block-indicator space. This gives

$$32 + 240 + 120 = 392$$

distinct supports. Generating all bit flips and variable permutations of $S_\star$ gives 480 further supports. These two families and the empty support are disjoint, and their union is exactly the 873-support intersection. Since every block indicator has even weight and $S_\star$ has even parity, every nonempty support has one parity.

The programs regenerate the projective point sets, lines, labelled pencils, affine preimages, support candidates, and orbits from these definitions, using exact linear algebra over $\mathbb{Q}(\zeta_3)$ and exhaustive enumeration. □

Part (i) below obtains the local support condition for $N = 3$. Part (ii) depends only on that support condition, so Theorem F.6.19 also uses it for $N \geq 5$ after its counting argument.

Theorem F.A.2 (Arity-six support classification)**.** (i) *Let F be an arity-six tensor of one parity, and suppose that every contraction $\partial_{ij}^{D(-s)X^{\xi}} F$, for every variable pair, xor, and $s \in \mu_3$, belongs to $\mathcal{B}_3([4])$. Then, for every fixed pair and xor, with $A = F_{ij}^{0,\xi}$, $B = F_{ij}^{1,1-\xi}$, $\operatorname{supp}(A) \cup \operatorname{supp}(B)$ is empty or is the union of at most two complementary pairs in its residual parity.*

(ii) *Let F be an arity-six tensor of one parity. If the support union in part* (i) *is empty or consists of at most two complementary pairs for every pair and xor, then* $\operatorname{supp}(F)$ *is empty, a partition coset of type* (6)*,* $(4+2)$*, or* $(2+2+2)$*, or a bit-flip/variable-permutation image of $S_\star$ in Equation* (F.25)*.*

Proof. For part (i), a nonzero member of $\mathcal{B}_3([4])$ has the two-pair coefficient form (F.24) with $N = 3$. Rescaling this form gives one of the vectors $\mathbf{E}_{pq}(r,t)$ in (F.55). Thus every nonzero contraction has projective class in the corresponding $\mathcal{Q}_3^{\epsilon}$. By Proposition F.A.1(ii), a projectively nonconstant pencil lies on a line supported on one complementary pair or a one-factor Segre ruling. Its support union consequently contains at most two complete complementary pairs. Coordinatewise,

$$\operatorname{supp}(A) \cup \operatorname{supp}(B) = \bigcup_{s \in \mu_3} \operatorname{supp}(A - sB),$$

since a nonzero affine coefficient has at most one root. This also handles zero and projectively constant pencils. For part (ii), the hypothesis says precisely that every local union $U_{ij}^{\xi}(\operatorname{supp}(F))$ is empty or consists of one or two complete complementary pairs of a single parity, as required in the enumeration of Proposition F.A.1(iii). That enumeration gives the asserted support alternatives. □

F.A.2.2 Height-three loop and two-copy arrays

The arrays below list the EO coefficients of the contractions in Definition F.8.3. For a mountain with endpoint coefficients a, b, the parameter is $t = ab$, and q_S are the coefficients of its central restriction. We retain the external variable orders from that definition. Let

$$\mathcal{W}_4 = (0011, 0101, 0110, 1001, 1010, 1100).$$

The arrays below use this order. Let $U = [6]$, let $t \in \mathbb{C}^{\times}$, and assign arbitrary complex coefficients q_S to the twenty triples $S \in \binom{U}{3}$, and put $Q = \sum_{S \in \binom{U}{3}} q_S \delta_{\mathbf{1}_S}$. For each deleted pair $i < j$, order the residual variables as $v_0 < v_1 < v_2 < v_3$. For $\mathbf{w} \in \mathcal{W}_4$, write $\mathbf{w} = (w_0, w_1, w_2, w_3)$ and put $R_{\mathbf{w}} = \{v_r : w_r = 1\}$, and define the six EO entries

$$(\partial_{ij} Q)(\mathbf{w}) = q_{R_{\mathbf{w}} \cup \{i\}} + q_{R_{\mathbf{w}} \cup \{j\}}, \qquad \mathbf{w} \in \mathcal{W}_4. \tag{F.57}$$

For unordered two-element external sets $A, B \in \binom{U}{2}$, use the canonical increasing orders $A = (a_0, a_1)$, $B = (b_0, b_1)$ and the external order (a_0, a_1, b_0, b_1). The two copies are distinguished. Put $I_A = U \setminus A$, $I_B = U \setminus B$, and let $\pi : I_A \to I_B$ be a bijection. Define the six EO entries, indexed by the concatenated strings $xy \in \mathcal{W}_4$, where $\mathbf{x} = (x_0, x_1)$ and $\mathbf{y} = (y_0, y_1)$, by

$$C_{A,B,\pi}(\mathbf{x}, \mathbf{y}) = t\, \mathbf{1}[(\mathbf{x}, \mathbf{y}) \in \{(00,11), (11,00)\}] + \sum_{\substack{T \subseteq I_A \\ |T| = 3 - \mathrm{wt}(\mathbf{x})}} q_{A_{\mathbf{x}} \cup T}\, q_{B_{\mathbf{y}} \cup (I_B \setminus \pi(T))}, \tag{F.58}$$

where $A_{\mathbf{x}} = \{a_r : x_r = 1\}$, and similarly for $B_{\mathbf{y}}$.

For any quaternary EO array $\mathbf{z} = (z_0, z_1, z_2, z_3, z_4, z_5)$ indexed in the order $\mathcal{W}_4$, put $P_0 = z_0 z_5$, $P_1 = z_1 z_4$, and $P_2 = z_2 z_3$.

Proposition F.A.3 (Height-three coefficient classification)**.** *Suppose every one of the fifteen arrays in Equation* (F.57) *and all* $15^2 \cdot 4! = 5400$ *labelled arrays in Equation* (F.58) *satisfy*

$$z_0 = 0 \iff z_5 = 0, \qquad z_1 = 0 \iff z_4 = 0, \qquad z_2 = 0 \iff z_3 = 0, \tag{F.59}$$

$$P_0P_1P_2 = 0, \qquad P_iP_j(P_i^2 - P_j^2) = 0 \quad (0 \le i < j \le 2). \tag{F.60}$$

Then

$$\{R \in \tbinom{U}{3} : q_R \neq 0\} \subseteq \{S, U \setminus S\} \qquad \text{for some } S \in \binom{U}{3}.$$

Proof. For each of the fifteen loop arrays, Equations (F.59) and (F.60) force at least one of its three complementary EO-entry pairs to vanish. Selecting such a pair gives, for a suitable $S \in \binom{U}{3}$ containing exactly one of i, j, two signed equations

$$q_S = -q_{S\triangle\{i,j\}}, \qquad q_{U\setminus S} = -q_{U\setminus(S\triangle\{i,j\})}.$$

Thus every coefficient vector lies in at least one of exactly $3^{15} = 14\,348\,907$ branches defined by signed linear equations. Signed union–find deduplicates them to 1868 linear subspaces. Modulo the S_6 variable action and simultaneous complementation, they form exactly 24 orbits.

For each family, substitute independent complex parameters into Equations (F.57) and (F.58). Impose Equations (F.59) and (F.60), $t \neq 0$, and the counterexample condition that two distinct complementary coefficient pairs $\{q_S, q_{U\setminus S}\}$ are nonzero. Split each complex parameter into real and imaginary parts. A selected nonzero parameter may be normalized to 1, since the equations are homogeneous under $(q, t) \mapsto (cq, c^2t)$ for $c \in \mathbb{C}^\times$. Exact satisfiability queries find no solution to these systems.

All generated expressions have integer coefficients, and all reductions performed before the solver use exact rational polynomial arithmetic. The all-zero orbit is immediate. The exact real-polynomial queries for the remaining twenty-three nonzero orbit families were evaluated with Z3 4.14.0. The answers are `UNSAT`, meaning that no satisfying assignment exists. The calculation uses 61 queries. For nineteen of the nonzero orbit families, 51 queries cover the cases obtained by choosing a nonzero parameter and normalizing it to 1. The script checks that parameter sets omitted by these normalizations are supported on at most one complementary pair and hence cannot meet the counterexample condition. For the other four families, 10 queries verify reductions to one of those nineteen families, the zero vector, or one complementary pair.

The wrapper requires Z3 4.14.0 and records compiler and solver versions and source hashes. It regenerates the orbit representatives and queries on each run. The option `-q6-audit-dir` saves inputs, parameter encodings, branch labels, and solver responses.

The runs neither request nor check solver proof objects. The `UNSAT` answers therefore rely on the correctness of Z3. With this solver dependence, the exhaustive calculation proves the proposition for arbitrary complex coefficients. □

Theorem F.A.4 (Consequence for loop and two-copy arrays)**.** *Let* $U = [6]$*, let* $t \in \mathbb{C}^\times$*, and let* $(q_S)_{S\in\binom{U}{3}}$ *be arbitrary complex coefficients. Form the fifteen loop arrays in Equation* (F.57) *and all* 5400 *labelled two-copy arrays in Equation* (F.58)*. If every one of these arrays belongs to* $\mathcal{B}^0_\times([4])$*, then*

$$\{R \in \tbinom{U}{3} : q_R \neq 0\} \subseteq \{S, U \setminus S\} \qquad \text{for some } S \in \binom{U}{3}.$$

Proof. Every member of $\mathcal{B}^0_\times([4])$ satisfies Equations (F.59) and (F.60). Hence Proposition F.A.3 applies and yields the assertion. □

Remark F.A.5 (Why the two-copy arity-six arrays are necessary). Order each triple as $S = \{s_0 < s_1 < s_2\}$, and order the twenty triples by the lexicographic order of (s_0, s_1, s_2). Consider the coefficient vector whose twenty weight-three entries are all nonzero:

$$(1, -1, -1, 1, i, -i, -1, 1, -i, -i, -i, -1, 1, -i, i, -1, 1, 1, -1)$$

This vector does not define a member of $\mathcal{B}_\times([6])$, although all fifteen arrays Equation (F.57) lie in $\mathcal{B}^0_\times([4])$. Take $A = B = \{1, 2\}$ and $\pi(3) = 3$, $\pi(4) = 4$, $\pi(5) = 6$, $\pi(6) = 5$. In the order $\mathcal{W}_4$, Equation (F.58) is

$$(t, 2, -2, -2, 2, t),$$

where $t \neq 0$. Its six nonzero entries put it outside that class. Thus the one-loop conditions alone cannot imply the conclusion of Theorem F.A.4.

F.A.2.3 Arity-eight matching families

Let $V = \{0, 1, \ldots, 7\}$, and for distinct labels a, b write ab for the unordered pair $\{a, b\}$. For every pair $e \in \binom{V}{2}$, let P_e be a perfect matching of $V \setminus e$. For disjoint $e, r \in \binom{V}{2}$, write $r = \{u, v\}$ and define $\kappa_r(P_e)$ as follows. If $r \in P_e$, delete that edge. Otherwise let u', v' be the mates of u, v, remove uu' and vv', add $u'v'$, and retain the unique untouched edge. In both cases $\kappa_r(P_e)$ is a two-edge matching on $V \setminus (e \cup r)$.

Proposition F.A.6 (Arity-eight matching classification)**.** *Suppose*

$$\kappa_r(P_e) = \kappa_e(P_r) \qquad (e \cap r = \varnothing) \tag{F.61}$$

for every pair of disjoint deleted pairs. After relabeling so that $P_{01} = \{23, 45, 67\}$, exactly nine labelled families satisfy these constraints. Seven are induced by a global perfect matching M on eight variables, meaning $P_e = \kappa_e(M)$ under the same deletion-and-splicing rule. Each of these seven families has $P_A \cap P_B \neq \varnothing$ for some distinct pairs A, B with $A \cap B \neq \varnothing$. With labels $0, \ldots, 7$ interpreted as three-bit vectors, one remaining family is given by the matchings below and the other is obtained by swapping $6, 7$:

h	$\{\{t, t+h\} : t \in \mathbb{F}_2^3\}$
001	$01, 23, 45, 67$
010	$02, 13, 46, 57$
011	$03, 12, 47, 56$
100	$04, 15, 26, 37$
101	$05, 14, 27, 36$
110	$06, 17, 24, 35$
111	$07, 16, 25, 34.$

(F.62)

For a deleted pair p, q, removing the edge pq from the row $h = p + q$ gives P_{pq}.

Proof. Once P_{01} is fixed, each of the other twenty-seven variable pairs has the fifteen perfect matchings on its six residual variables as candidates. An exhaustive backtracking search imposes Equation (F.61) whenever both sides become known. Exactly nine complete matching families remain. They give the stated $7 + 2$ split and the table in Equation (F.62).

The programs reproducing these nine families are described in Appendix F.A.3. □

Theorem F.A.7 (Translation form of the matching family)**.** *Let V be an eight-element variable set and, for every $e \in \binom{V}{2}$, let P_e be a perfect matching of $V \setminus e$. Suppose*

$$\kappa_r(P_e) = \kappa_e(P_r) \qquad (e \cap r = \varnothing).$$

If $P_A \cap P_B = \varnothing$ whenever $A \neq B$ and $A \cap B \neq \varnothing$, then there is a bijection $\iota : V \to \mathbb{F}_2^3$ such that, after relabelling, for $p \neq q$ and $h = p + q$,

$$P_{pq} = \{\{t, t+h\} : t \in \mathbb{F}_2^3 \setminus \{p, q\}\},$$

where duplicate unordered pairs are suppressed.

Proof. Relabel V so that $P_{01} = \{23, 45, 67\}$. By Proposition F.A.6, seven of the nine labelled families violate the condition on $P_A \cap P_B$. The remaining two differ only by a relabelling and give the displayed matching family. □

Proposition F.A.8 (Coefficients on affine planes)**.** *Identify $V = \mathbb{F}_2^3$, with the labels $0, \dots, 7$ interpreted as their three-bit binary expansions, and for $S \in \binom{V}{4}$ put $\sigma(S) = \bigoplus_{s \in S} s$. Let $c_S \in \mathbb{C}$, and let $\lambda_{pq} \in \mathbb{C}^\times$ for every unordered pair $\{p, q\} \in \binom{V}{2}$. Suppose that, whenever $p \in S$, $q \notin S$, $h = p + q$, and $S' = S \triangle \{p, q\}$,*

$$c_S + c_{S'} = \begin{cases} \lambda_{pq}, & \sigma(S) \in \{0, h\}, \\ 0, & \sigma(S) \notin \{0, h\}. \end{cases} \tag{F.63}$$

Then the four-sets with nonzero coefficients are exactly the fourteen affine planes of V, and there is one $c \in \mathbb{C}^\times$ such that

$$c_S = \begin{cases} c, & \sigma(S) = 0, \\ 0, & \sigma(S) \neq 0. \end{cases} \tag{F.64}$$

Call a four-subset a nonplane *when it is not an affine plane. Let G be the graph on the fifty-six nonplanes, joining S to $S \triangle \{p, q\}$ whenever $p \in S$, $q \notin S$, and the latter four-set is also a nonplane. Let H be the bipartite incidence graph between planes and unordered variable pairs, with an edge between S and $\{p, q\}$ exactly when $|S \cap \{p, q\}| = 1$. Then G is connected and contains an odd cycle, and H is connected.*

Proof. A four-set has xor zero exactly when it is an affine plane: xor zero makes the fourth point the affine sum of the other three, and the seven nonzero linear functionals on $\mathbb{F}_2^3$ each have two affine level-set planes. Thus there are fourteen such sets and fifty-six nonplanes. For a nonplane S, the exchanged set $S' = S \triangle \{p, q\}$ is again a nonplane exactly when $\sigma(S) \notin \{0, p+q\}$. Lift $t \in V$ to $\widehat{\mathbf{t}} = (1, t) \in \mathbb{F}_2^4$, and let M be the represented matroid on ground set V with representing vectors $\widehat{\mathbf{t}}$. Independence in this matroid means linear independence of those vectors. A four-set is a nonplane exactly when its four lifted points form a basis. Thus G is the basis-exchange graph of the rank-four binary matroid M, whose vertices are bases and whose edges exchange one element while preserving a basis. By basis exchange, a basis different from a target basis admits an exchange that increases their intersection. Repeating such exchanges reaches the target basis, so this graph is connected.

In binary labelling it contains the triangle below, where, for example, 0124 abbreviates the four-set $\{0, 1, 2, 4\}$:

$$\mathtt{0124} \text{ — } \mathtt{0125} \text{ — } \mathtt{0245} \text{ — } \mathtt{0124}. \tag{F.65}$$

The three xor sums are $7, 6, 3$, while the corresponding swap directions are $1, 5, 4$. Hence all three edges belong to G. The second line of Equation (F.63) gives, on every edge of G,

$$c_{S'} = -c_S.$$

Alternating around Equation (F.65) and using characteristic zero forces the three coefficients on the triangle to vanish. Connectivity then forces $c_S = 0$ for every nonplane.

Let S now be a plane and let $p \in S, q \notin S$. Then $S' = S \triangle \{p, q\}$ has xor $p + q \neq 0$, so it is a nonplane. The first line of Equation (F.63) therefore gives

$$c_S = \lambda_{pq} \neq 0. \tag{F.66}$$

It remains only to show that these values agree. Regard a plane as incident with every unordered pair having one endpoint in the plane and one outside it. Two nonparallel affine planes meet in two points and their union has six points. A pair from their intersection to the complement of their union is incident with both planes in H. Two distinct parallel planes are complementary and hence have the same incident variable pairs. Thus H is connected. Every pair $\{p, q\}$ is incident with a plane: choose a linear functional ℓ with $\ell(p + q) = 1$ and take $\{x : \ell(x) = \ell(p)\}$. Equation (F.66) makes all fourteen plane coefficients and all incident λ_{pq}'s equal to one common nonzero c, proving Equation (F.64). □

Theorem F.A.9 (Affine-plane coefficients)**.** *Identify $V = \mathbb{F}_2^3$. Let $c_S \in \mathbb{C}$ for $S \in \binom{V}{4}$, and let $\lambda_{pq} \in \mathbb{C}^\times$ for every unordered pair $\{p, q\} \in \binom{V}{2}$. Suppose that, whenever $p \in S$, $q \notin S$, $h = p + q$, and $S' = S \triangle \{p, q\}$,*

$$c_S + c_{S'} = \begin{cases} \lambda_{pq}, & \bigoplus_{s \in S} s \in \{0, h\}, \\ 0, & \bigoplus_{s \in S} s \notin \{0, h\}. \end{cases}$$

Then there is one $c \in \mathbb{C}^\times$ such that

$$c_S = \begin{cases} c, & \bigoplus_{s \in S} s = 0, \\ 0, & \bigoplus_{s \in S} s \neq 0. \end{cases}$$

Equivalently, the support consists exactly of the fourteen affine planes of $\mathbb{F}_2^3$, and all supported coefficients are equal.

Proof. This follows from Proposition F.A.8. □

F.A.3 Implementation details and supplementary checks

For the arity-six coefficient calculation, the Python loop generator lists the six entries in reverse order. Both (F.59) and (F.60) are invariant under $z_k \mapsto z_{5-k}$. The two-copy generator uses $\mathcal{W}_4$ directly. For the C++ enumeration, variables are numbered from 0 to 5, and triples are ordered by increasing $\sum_{k \in S} 2^k$. In a nonzero component C of the signed equations, write $q_i = \varepsilon_i u_C$, where $\varepsilon_i \in \{1, -1\}$, choose $\varepsilon_{\min C} = 1$, and name its parameter by $\min C$. Choose the lexicographically least encoding under variable permutations and simultaneous complementation as the orbit representative. Before any solver query, the Python stage reindexes the C++ output and verifies that both stages give the same representatives.

For the arity-eight matching classification, the scripts `verify_high_charge_q8_atlas.py` and `verify_high_charge_q8_normalized.py` regenerate the nine families. The search has 236 nodes. The programs share the matching-generation logic and are not independent implementations. The normalized script also checks the linear systems for the original tensor's seventy weight-four coefficients and twenty-eight contraction scales. For each matching family, requiring every pair

contraction to equal its scalar times the corresponding X-matching product gives a 560-by-98 integer matrix, with twenty equations for each of the twenty-eight deleted pairs. A nonzero rational kernel vector gives rank at most 97, and rank 97 modulo 1,000,000,007 gives rank at least 97 over $\mathbb{Q}$. This rank calculation is supplementary to the matching classification.

For Proposition F.A.8, the atlas script checks the $14 + 56$ split, the connectivity of the nonplane exchange graph G, its displayed odd cycle, and the connectivity of the plane–pair incidence graph H. These checks supplement the symbolic proof above. The normalized script separately encodes the multiplicative rank-one relations among the fourteen plane coefficients by an integer matrix, whose rows record the exponents in the relations. An explicit kernel and a nonzero 13-by-13 minor of absolute value 16 give rank 13. Rank 9 modulo two then forces the nonzero Smith invariant factors to be nine 1's and four 2's. These supplementary checks use integer rank and divisibility arguments and are not needed for the symbolic deduction of the affine-plane coefficients.